EIGENFUNCTIONS, FREE BOUNDARIES, AND TIME-FREQUENCY LOCALIZATION

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Abstract. We study Gaussian time–frequency localization operators through their Bargmann–Fock representation and develop an inverse and free-boundary approach to their spectral theory.

We develop a perturbative inverse theory showing that functions sufficiently close to the ground-state Hermite function can be realized as eigenfunctions of localization operators. As a byproduct, we recover the classical disk characterization of Abreu and Dörfler in the simply connected setting by the same free-boundary method. As an application we prove that the stability estimate of Gomez-Guerra, Ramos and Tilli is sharp.

In the second part of the paper we apply these inverse results to Faber–Krahn-type problems for Gaussian time–frequency concentration. We prove that local maximizers of the first localization eigenvalue under a measure constraint are necessarily disks, and we establish existence of global maximizers via a concentration–compactness argument in the Bargmann–Fock space, recovering the Nicola–Tilli theorem from this perspective.

1. Introduction

1.1. **Time–frequency localization and the Gaussian model.** The study of time–frequency concentration lies at the heart of modern harmonic analysis, with deep connections to signal processing, mathematical physics, and phase–space methods; see, for instance, [22, 30, 17]. A central object in this theory is the *short-time Fourier transform* (STFT), defined for a function $f \in L^2(\mathbb{R})$ and a window $\varphi \in L^2(\mathbb{R})$ by

$$V_\varphi f(x,\omega) = \int_{\mathbb{R}} f(t)\,\overline{\varphi(t-x)}\,e^{-2\pi i t\omega}\,dt.$$

This transform encodes the joint time–frequency content of f, and its squared modulus $|V_\varphi f|^2$ provides a phase–space energy density. Understanding how this energy can be concentrated in regions of the time–frequency plane is a fundamental problem, with both theoretical and applied significance.

In applications ranging from signal recovery to uncertainty principles, one is often interested in maximizing the quantity

$$\int_\Omega |V_\varphi f(x,\omega)|^2\,dx\,d\omega$$

over functions f of unit norm and over sets Ω of prescribed measure. This leads naturally to the study of *time–frequency localization operators*, introduced by Daubechies in [15] and further developed in [16, 35, 36, 14, 12, 7, 27, 34, 37]. For general windows, sharp concentration bounds are notoriously elusive, and Lieb's uncertainty principle [29] remains a fundamental benchmark; see also [2] for related large-sieve phenomena in modulation and polyanalytic Fock spaces. In the Gaussian setting, by contrast, these operators acquire a particularly rigid structure: via

the Bargmann transform [4], they can be realized as Toeplitz-type integral operators on the Bargmann–Fock space, and the problem becomes one of complex analysis in weighted spaces; compare [8, 9, 10, 11, 43, 22, 17]. In this picture, the STFT is intertwined with an entire function whose weighted modulus encodes the concentration properties of the signal, and geometric questions translate into analytic ones.

1.2. **Spectral and geometric interplay.** A central theme in this area is the interplay between the *spectral properties* of localization operators and the *geometry* of the underlying domains. In the Gaussian case, Nicola and Tilli proved the sharp Faber–Krahn inequality for the STFT, showing that disks maximize concentration among all measurable sets of a fixed measure [33]; see also [34, 5, 6] for complementary spectral and regularity results on localization operators. The stability theory for this variational problem was subsequently developed in [20], where quantitative control of both the first eigenfunction and the Fraenkel asymmetry is obtained from the spectral deficit. Related concentration questions for other transforms, including wavelet transforms, have been investigated in [38].

From a complementary perspective, one may pose inverse questions: to what extent does spectral information determine the geometry of the localization domain? Conversely, given a function, can one construct a domain for which it is an eigenfunction of the associated localization operator? The first systematic result in this direction is due to Abreu and Dörfler [1], who showed that if a Hermite function is an eigenfunction of a localization operator on a simply connected domain, then the domain must be a disk. This inverse viewpoint sits naturally within the broader circle of ideas surrounding weighted quadrature identities, free-boundary problems, and inverse potential theory: classical results of Aharonov, Schiffer, and Zalcman [3] characterize disks via mean-value identities for harmonic functions; Karp [26] and Cherednichenko [13] address inverse problems for Newtonian potentials; Gustafsson and collaborators [23, 41, 40] develop the theory of quadrature domains and obstacle problems; and Isakov's monograph [25] provides the perturbative framework for inverse source problems that we shall employ. The inverse viewpoint is also tied to the special role of Gaussian decay and Hermite structures in the Bargmann model; see [28, 21, 42].

1.3. **Eigenfunctions as geometric data.** The guiding principle of the present paper is that, for Gaussian localization operators, eigenfunctions should be regarded as *geometric data*. More precisely, the eigenfunction should determine the domain in rigid situations, and the possibility of prescribing the eigenfunction should in turn have strong consequences for nearby variational problems. The three main threads of the paper are organized around this idea:

(1) rigid eigenfunctions force rigid domains;
(2) near the Gaussian, one can prescribe the eigenfunction and construct the corresponding localization domain;
(3) once such an eigenfunction-to-domain mechanism is available, it can be used to analyze sharpness and extremizers in concentration inequalities.

The present work contributes to this line of research by developing a unified inverse and free-boundary framework for Gaussian time–frequency localization operators. Our approach builds

on, and extends, the rigidity result of Abreu and Dörfler [1], the variational theory of Nicola and Tilli [33, 34], and the quantitative stability theorems of Gómez, Guerra, Ramos, and Tilli [20]. At the technical level, our arguments also draw on ideas from inverse potential problems and quadrature-domain theory, in the spirit of [25, 13, 23, 41, 40]. In this way, the paper places localization operators squarely within the broader free-boundary and inverse-potential framework suggested by the Gaussian Bargmann model.

Despite these advances, two fundamental questions have remained open. First, no general construction of localization domains associated with prescribed near-Gaussian functions has been available, beyond a handful of highly symmetric examples. Second, the role of eigenfunction rigidity in variational concentration problems has not been clarified, leaving the sharpness of recent stability inequalities and the structure of local maximizers out of reach. The results of the present paper address these questions in a unified manner.

1.4. **An inverse construction near the Gaussian.** Our first main result is the constructive half of this picture. It shows that, near the Gaussian, one can indeed prescribe the eigenfunction and realize it through a suitable localization domain.

Theorem 1. *Let $\lambda \in (0,1)$ and $N \in \mathbb{N}$. There exists $\varepsilon_0(\lambda, N) > 0$ such that if*

$$f_0 = h_0 + \sum_{k=1}^{N} a_k h_k$$

satisfies $\max_k |a_k| < \varepsilon_0$, then there exists a domain U_λ such that

$$\mathcal{L}_{U_\lambda} f_0 = \lambda f_0.$$

This theorem provides, to the best of our knowledge, the first systematic construction of localization domains associated with nontrivial linear combinations of Hermite functions. In particular, it supplies the inverse counterpart to the Abreu–Dörfler rigidity phenomenon: rather than deducing the domain from a rigid eigenfunction, we construct domains starting from prescribed near-Gaussian eigenfunctions. The dependence of ε_0 on λ and N is explicit in our argument, and reflects the size of the spectral gap of the disk localization operator on the relevant Hermite subspace.

The proof proceeds in three stages, combining tools from free-boundary problems and complex analysis. First, we reformulate the eigenvalue equation in the Bargmann–Fock space as a nonlinear integral identity for the squared modulus of an entire function with polynomial growth. Applying a suitable differential operator, we reduce the problem to that of constructing a domain satisfying a weighted identity that is reminiscent of a quadrature condition.

Next, we produce an approximate solution by solving a free-boundary problem of obstacle type. This is achieved through a perturbative theorem of Isakov [25], which provides a domain whose associated logarithmic potential satisfies a prescribed equation under small perturbations of the source. This step yields a domain for which the desired identity holds up to a controlled error.

In the third stage, we refine this construction using a weighted quadrature-domain theorem in the spirit of [23, 41, 40]. Viewing the problem through the lens of complex analysis, we

interpret the identity as a condition on a weighted Cauchy transform, and a perturbative implicit-function argument allows us to deform the approximate domain so that the identity becomes exact. Finally, we show that the resulting function satisfies the original eigenvalue equation by analyzing an associated differential operator and ruling out nontrivial obstructions through growth estimates. The combination of these steps yields a robust local inverse theory.

A noteworthy aspect of this construction is that it reaches well beyond monomial-type eigenfunctions, the only case previously accessible by direct symmetry arguments. Our theorem shows that a much broader class of near-Gaussian functions can be realized spectrally, opening the door to a richer inverse theory and revealing new connections with nonlinear potential theory and weighted quadrature domains in the spirit of [41, 40].

1.5. **A new proof of the Abreu–Dörfler rigidity theorem.** The same free-boundary framework also yields the rigid half of the correspondence, namely a short recovery of the classical theorem of Abreu and Dörfler.

More precisely, we provide two new proofs of the Abreu–Dörfler rigidity result based on our methods: one through the local radiality mechanism for the logarithmic potential, and a second through a holomorphic auxiliary function attached to the associated Poisson problem.

Theorem 2. *Let U be a simply connected bounded domain. If a Hermite function is an eigenfunction of the localization operator associated with U, then U is a disk.*

The proof transports the eigenvalue equation to the Bargmann–Fock space, where Hermite functions correspond to monomials and the localization operator becomes an integral operator with reproducing-kernel structure. In this setting, the eigenvalue equation translates into a functional identity involving exponential generating functions. Differentiating appropriately and invoking a Paley–Wiener type argument, we reduce the problem to the vanishing of a Fourier transform along complex lines.

This allows us to apply a divisibility lemma of Ehrenpreis–Malgrange type, which produces a logarithmic potential solving a Poisson equation with source supported on the domain. The decisive insight is that the spectral assumption imposes strong constraints on this potential: it must exhibit a form of radial behavior on the boundary. In the simply connected setting, this constraint forces the domain to be a disk, recovering the Abreu–Dörfler rigidity theorem from the same inverse/free-boundary perspective.

1.6. **Sharpness of the GGRT stability inequality.** We now turn to consequences of this inverse point of view. Our first application concerns the quantitative stability theorem of Gómez, Guerra, Ramos, and Tilli for the Faber–Krahn inequality for the Gaussian STFT. In the formulation most relevant here, if $U \subset \mathbb{R}^2$ is a measurable set of finite positive measure, $\lambda_1(U)$ denotes the first eigenvalue of the associated localization operator, and f_U is a corresponding L^2-normalized first eigenfunction, then Corollary 1.4 of [20] gives the bounds

$$\min_{z_0\in\mathbb{C},\ |c|=1} \|f_U - c\phi_{z_0}\|_{L^2(\mathbb{R})} \le Ce^{|U|/2}\left(1-\frac{\lambda_1(U)}{1-e^{-|U|}}\right)^{1/2},$$

and

$$\mathcal{A}(U) \leq K(|U|)\left(1 - \frac{\lambda_1(U)}{1 - e^{-|U|}}\right)^{1/2},$$

where ϕ_{z_0} ranges over the Gaussian optimizers and $\mathcal{A}(U)$ denotes the Fraenkel asymmetry of U.

Our perturbative inverse theorem provides the missing input needed to test this stability estimate on genuine spectral data. Starting from a prescribed polynomial perturbation of the Gaussian, Theorem 1 produces an exact eigenpair $(U_\varepsilon, f_\varepsilon)$ for the localization operator, with U_ε a smooth perturbation of the disk and f_ε a corresponding first eigenfunction. The constructive direction of the paper thus yields explicit near-extremizers, against which one can compare the size of the spectral deficit with the actual geometric displacement of the domain from the class of disks. The resulting one-parameter family shows that the square-root dependence in Corollary 1.4 of [20] is the best possible.

Theorem 3. *The exponent* $1/2$ *in Corollary 1.4 of* [20] *is optimal. More precisely, the powers of the deficit*

$$1 - \frac{\lambda_1(U)}{1 - e^{-|U|}}$$

appearing in both the eigenfunction estimate and the Fraenkel-asymmetry estimate there cannot be replaced by β *for any* $\beta > 1/2$.

The argument rests on constructing a carefully chosen family of perturbations of the disk via the inverse theorem above. Starting from a polynomial perturbation of the ground-state Hermite function, we obtain a family of domains whose geometry deviates from that of a disk at first order. A detailed asymptotic analysis of the eigenvalue equation, including a first-variation computation by a transport formula, then identifies the leading Fourier modes of the boundary deformation.

This analysis shows that the deformation is governed by second-order harmonics, which cannot be eliminated by translations of the disk. Comparing the perturbed domains with arbitrary disks then yields lower bounds on their symmetric difference that match the upper bounds in the stability estimate of [20]. The proof combines spectral perturbation theory, geometric measure estimates, and harmonic analysis on the circle, illustrating how inverse constructions can be used to probe the sharpness of variational inequalities.

1.7. **Local maximizers are disks.** We next connect this eigenfunction-to-domain philosophy to variational problems. Our fourth main result concerns the structure of local maximizers for Gaussian time–frequency concentration.

Theorem 4. *Any local maximizer of* $\lambda_{1,\varphi}$ *under a measure constraint is, up to translation, a disk.*

The proof relies on combining variational arguments with the inverse spectral perspective developed earlier. We first show that any local maximizer must coincide with a superlevel set of the density associated with its first eigenfunction in the Bargmann–Fock space. This reduces the geometric problem to a functional one.

We then analyze the spectral properties of the localization operator at a maximizer, showing that the first eigenvalue is simple and that derivatives of the eigenfunction remain eigenfunctions. This strong rigidity property forces the maximizing eigenfunction into a Gaussian form. Once this is known, the corresponding domain is determined as well, and one concludes that the only admissible configuration is a disk. The argument thus shows that extremizers are governed by the same rigidity mechanism that underlies the inverse theory.

1.8. **A new proof of the Nicola–Tilli theorem.** Finally, we address the global variational problem and recover the Nicola–Tilli theorem from the same perspective.

Theorem 5 (New proof of Nicola–Tilli)**.** *Disks maximize Gaussian time–frequency concentration among all sets of fixed measure.*

Our proof is based on a concentration–compactness argument in the Bargmann–Fock space. We consider maximizing sequences and analyze their possible limiting behavior, ruling out vanishing and dichotomy scenarios using the structure of the Fock space and the invariance properties of the STFT. This yields the existence of a maximizer. Related concentration–compactness methods have recently been used in nearby time–frequency concentration problems, notably by Nicola, Romero, and Trapasso for ambiguity-function type concentration functionals [32]; see also [31, 39] for recent work on Fourier concentration operators, and [18] for closely related developments in the polyanalytic Fock-space setting.

We then combine this existence result with the rigidity of local maximizers established earlier. Since any global maximizer is, in particular, a local maximizer, its eigenfunction is rigid and therefore determines the maximizing domain. This provides a new conceptual route to the Nicola–Tilli theorem, avoiding rearrangement inequalities and instead relying on inverse and free-boundary methods. The argument highlights the power of treating eigenfunctions as geometric data.

1.9. **Organization of the paper.** The paper is organized as follows. In Section 2 we review the Bargmann–Fock representation and establish several auxiliary results, including a key divisibility lemma and perturbative free-boundary theorems. Section 3 is devoted to the perturbative inverse construction underlying Theorem 1. Section 4 explains how the same machinery recovers the classical theorem of Abreu and Dörfler in the simply connected case (Theorem 2). Section 5 contains the sharpness result, Theorem 3, while the remaining sections are devoted to variational applications, culminating in the new proof of the Nicola–Tilli theorem.

In summary, the results of this paper establish a bridge between time–frequency analysis, inverse spectral theory, and free-boundary problems in potential theory. This perspective not only resolves several open problems, but also supplies a flexible and robust framework for studying a wide range of questions in phase–space analysis.

2. Preliminaries

2.1. **Properties of the Bargmann-Fock space.** We briefly recall the connection between the Gaussian short-time Fourier transform (STFT) and the Bargmann–Fock space, which will be

used systematically in what follows. Excellent references for the material in this subsection are [4, 17, 22, 43]; see also [8, 10, 15].

Let $\varphi(x) = 2^{1/4}e^{-\pi x^2}$ be the normalized Gaussian. For $f \in L^2(\mathbb{R})$, its Gaussian STFT is

$$V_\varphi f(x,\omega) = \int_{\mathbb{R}} f(t)\,\varphi(x-t)\,e^{-2\pi i t\omega}\,dt. \tag{2.1}$$

The Bargmann transform $\mathcal{B} : L^2(\mathbb{R}) \to \mathcal{F}^2(\mathbb{C})$ is defined by

$$(\mathcal{B}f)(z) = 2^{1/4}\int_{\mathbb{R}} f(t)\,e^{2\pi t z - \pi t^2 - \frac{\pi}{2}z^2}\,dt, \qquad z \in \mathbb{C}. \tag{2.2}$$

It is well known that $\mathcal{B}$ is a unitary isomorphism onto the Bargmann–Fock space $\mathcal{F}^2(\mathbb{C})$, endowed with the norm

$$\|F\|^2_{\mathcal{F}^2} = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\,dA(z).$$

A direct computation shows that, for $z = x + i\omega$,

$$V_\varphi f(x,-\omega) = e^{\pi i x\omega}\,(\mathcal{B}f)(z)\,e^{-\frac{\pi}{2}|z|^2}. \tag{2.3}$$

In particular, setting $F = \mathcal{B}f$, one obtains

$$\int_{\mathbb{R}^2} |V_\varphi f(x,\omega)|^2\,dx\,d\omega = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\,dA(z) = \|F\|^2_{\mathcal{F}^2}. \tag{2.4}$$

Weyl operators and invariance. For $a \in \mathbb{C}$, define the operator

$$(T_a F)(z) := e^{\pi a z - \frac{\pi}{2}|a|^2} F(z-a). \tag{2.5}$$

Then T_a is unitary on $\mathcal{F}^2(\mathbb{C})$. Indeed,

$$\begin{aligned}\|T_a F\|^2_{\mathcal{F}^2} &= \int_{\mathbb{C}} e^{2\pi\Re(az)-\pi|a|^2}\,|F(z-a)|^2 e^{-\pi|z|^2}\,dA(z)\\ &= \int_{\mathbb{C}} |F(z-a)|^2 e^{-\pi|z-a|^2}\,dA(z) = \|F\|^2_{\mathcal{F}^2},\end{aligned}$$

because

$$2\Re(az) - |a|^2 - |z|^2 = -|z-a|^2.$$

Moreover,

$$u_{T_a F}(z) = u_F(z-a). \tag{2.6}$$

Indeed, since $2\Re(az) - |a|^2 - |z|^2 = -|z-a|^2$,

$$u_{T_a F}(z) = |T_a F(z)|^2 e^{-\pi|z|^2} = e^{2\pi\Re(az)-\pi|a|^2}|F(z-a)|^2 e^{-\pi|z|^2} = |F(z-a)|^2 e^{-\pi|z-a|^2} = u_F(z-a). \tag{2.7}$$

In particular, the functionals I_F defined by, for $s > 0$,

$$I_F(s) := \sup_{|\Omega|=s} \int_\Omega u_F\,dA,$$

satisfy, by the translation-invariance of the Lebesgue measure on $\mathbb{C}$,

$$I_{T_a F}(s) = I_F(s), \qquad a \in \mathbb{C}, \tag{2.8}$$

reflecting the invariance of the Gaussian STFT under time–frequency shifts; see [15, 22, 28, 21].

Connection with localization. If $F = \mathcal{B}f$, then for every measurable $\Omega \subset \mathbb{C}$,

$$\int_\Omega u_F \, dA = \int_{\widetilde{\Omega}} |V_\varphi f(x,\omega)|^2 \, dx \, d\omega,$$

where $\widetilde{\Omega} = \{(x,\omega) : (x,-\omega) \in \Omega\}$. Thus, concentration properties of the STFT are naturally encoded by the function

$$u_F(z) := |F(z)|^2 e^{-\pi|z|^2}, \qquad z \in \mathbb{C}, \tag{2.9}$$

which is the main object of interest in the Bargmann–Fock setting, in terms of the quantities $I_F(s)$.

We next collect a few elementary facts that will be used repeatedly.

Lemma 2.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) The space $\mathcal{F}^2(\mathbb{C})$ is a reproducing kernel Hilbert space with kernel

$$K_w(z) = e^{\pi z \overline{w}}, \qquad z, w \in \mathbb{C}.$$

(ii) (Sharp pointwise bound) For every $z \in \mathbb{C}$,

$$u_F(z) \le \|F\|_{\mathcal{F}^2}^2,$$

and equality holds at some point if and only if F is proportional to F_w for some $w \in \mathbb{C}$.

(iii) $u_F(z) \to 0$ as $|z| \to \infty$.

(iv) One has the estimate

$$\|u_F - u_G\|_{L^1(\mathbb{C})} \le \big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big)\|F - G\|_{\mathcal{F}^2}.$$

Proof. (i) For $w \in \mathbb{C}$, the function $K_w(z) = e^{\pi z \overline{w}}$ is entire and satisfies, after completing the square via $2\Re(z\overline{w}) - |z|^2 = -|z-w|^2 + |w|^2$,

$$\|K_w\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} e^{2\pi\Re(z\overline{w}) - \pi|z|^2} \, dA(z) = e^{\pi|w|^2} \int_{\mathbb{C}} e^{-\pi|z-w|^2} \, dA(z) = e^{\pi|w|^2},$$

since the Gaussian $e^{-\pi|\cdot|^2}$ integrates to 1 over $\mathbb{C}$ (with the Lebesgue measure dA on $\mathbb{C} \simeq \mathbb{R}^2$). In particular $K_w \in \mathcal{F}^2(\mathbb{C})$. Now, expanding $F \in \mathcal{F}^2(\mathbb{C})$ along the orthonormal basis $\{e_k\}_{k\ge0}$ of Lemma 2.2 and using the Taylor expansion $K_w(z) = \sum_{k\ge0} \frac{\pi^k \overline{w}^k}{k!} z^k = \sum_{k\ge0} \overline{e_k(w)}\, e_k(z)$, one obtains the reproducing identity

$$\langle F, K_w \rangle_{\mathcal{F}^2} = \sum_{k\ge0} a_k \, e_k(w) = F(w), \qquad F(z) = \sum_{k\ge0} a_k e_k(z).$$

Point evaluation is therefore continuous, and F is entire as a uniform limit on compact sets of polynomials. This is the standard description of the Bargmann–Fock space, see [4, 17, 22, 43].

(ii) Applying Cauchy–Schwarz to the reproducing identity from (i),

$$|F(z)| = |\langle F, K_z \rangle_{\mathcal{F}^2}| \le \|F\|_{\mathcal{F}^2} \, \|K_z\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2} \, e^{\frac{\pi}{2}|z|^2}, \tag{2.10}$$

which immediately gives $u_F(z) = |F(z)|^2 e^{-\pi|z|^2} \le \|F\|_{\mathcal{F}^2}^2$. The case of equality at a point $z = w$ corresponds to equality in Cauchy–Schwarz, that is, $F = c\, K_w$ for some $c \in \mathbb{C}$; writing

$F_w := K_w$, this proves the rigidity statement. The sharpness of this pointwise bound goes back to [29].

(iii) Fix $\varepsilon > 0$, and choose a polynomial P such that

$$\|F - P\|_{\mathcal{F}^2} < \varepsilon$$

using the density of polynomials in $\mathcal{F}^2(\mathbb{C})$. By the pointwise bound from (ii),

$$|F(z) - P(z)|e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon \qquad \text{for every } z \in \mathbb{C}.$$

Since P is a polynomial,

$$P(z)e^{-\frac{\pi}{2}|z|^2} \to 0 \qquad \text{as } |z| \to \infty.$$

Hence

$$\limsup_{|z|\to\infty} |F(z)|e^{-\frac{\pi}{2}|z|^2} \leq \varepsilon.$$

As $\varepsilon > 0$ is arbitrary, we conclude that

$$|F(z)|e^{-\frac{\pi}{2}|z|^2} \to 0,$$

that is, $u_F(z) \to 0$ as $|z| \to \infty$.

(iv) Writing

$$|u_F - u_G| = \big||F|^2 - |G|^2\big|e^{-\pi|z|^2} \leq |F - G|(|F| + |G|)e^{-\pi|z|^2},$$

the claim follows by Cauchy–Schwarz. □

We now note the following crucial fact about the Bargmann transform: it transforms the basis of normalized Hermite functions $\{h_k\}_{k\geq 0}$ on $L^2(\mathbb{R})$ (see [42]) onto normalized monomials, yielding a natural orthonormal basis on $\mathcal{F}^2(\mathbb{C})$. More precisely, one has $\mathcal{B}h_k = e_k$ for every $k \geq 0$, see [4, 17, 22].

Lemma 2.2. *The functions*

$$e_k(z) := b_k z^k, \qquad k \geq 0,$$

form an orthonormal basis of $\mathcal{F}^2(\mathbb{C})$*, where*

$$b_k = \left(\frac{\pi^k}{k!}\right)^{1/2}.$$

In particular,

$$\langle e_k, e_m\rangle_{\mathcal{F}^2} = \delta_{k,m}.$$

Proof. Writing $z = re^{i\theta}$ and using $dA(z) = r\,dr\,d\theta$ on $\mathbb{C}$, a direct computation in polar coordinates gives

$$\int_{\mathbb{C}} |z|^{2k}e^{-\pi|z|^2}\,dA(z) = \int_0^{2\pi}\int_0^\infty r^{2k+1}e^{-\pi r^2}\,dr\,d\theta = 2\pi\int_0^\infty r^{2k+1}e^{-\pi r^2}\,dr = \frac{k!}{\pi^k},$$

where the last identity follows from the substitution $s = \pi r^2$, giving $2\pi\int_0^\infty r^{2k+1}e^{-\pi r^2}\,dr = \pi^{-k}\int_0^\infty s^k e^{-s}\,ds = \pi^{-k}\Gamma(k+1)$. Thus $\|z^k\|^2_{\mathcal{F}^2} = \frac{k!}{\pi^k}$, which yields the normalization. For

orthogonality, again writing $z = re^{i\theta}$,

$$\langle z^k, z^m \rangle_{\mathcal{F}^2} = \int_0^\infty r^{k+m+1} e^{-\pi r^2}\, dr \int_0^{2\pi} e^{i(k-m)\theta}\, d\theta = 0 \qquad \text{for } k \neq m.$$

For completeness, observe that if $F \in \mathcal{F}^2(\mathbb{C})$ satisfies $\langle F, e_k \rangle_{\mathcal{F}^2} = 0$ for every $k \geq 0$, then in particular all Taylor coefficients of F at the origin vanish: since $F = \mathcal{B}f$ for a unique $f \in L^2(\mathbb{R})$, and $\mathcal{B}h_k = e_k$ (so that $\{e_k\}$ is the image of the orthonormal basis of Hermite functions under the unitary $\mathcal{B}$), the family $\{e_k\}_{k\geq 0}$ is total in $\mathcal{F}^2(\mathbb{C})$. Equivalently, one may invoke directly the fact that any entire function of order at most 2 lying in $\mathcal{F}^2(\mathbb{C})$ is the limit of its Taylor polynomials in the Fock norm; see [4, 43]. □

Lemma 2.3. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) F admits the expansion

$$F(z) = \sum_{k=0}^{\infty} a_k e_k(z), \qquad \sum_{k=0}^{\infty} |a_k|^2 = \|F\|_{\mathcal{F}^2}^2.$$

(ii) (Growth bound) For every $z \in \mathbb{C}$,

$$|F(z)| \leq \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) (Derivative bounds) For every $z \in \mathbb{C}$,

$$|F'(z)| \leq C(1 + |z|)\, \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

Proof. (i) follows from the orthonormal basis in Lemma 2.2.

(ii) follows from the reproducing kernel estimate:

$$|F(z)| = |\langle F, K_z \rangle| \leq \|F\|_{\mathcal{F}^2} \|K_z\|_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

(iii) Differentiating the reproducing identity $F(z) = \langle F, K_z \rangle_{\mathcal{F}^2}$ in the parameter z, and using $\partial_z K_w(z) = \pi \overline{w}\, K_w(z)$ (which is to be read in the antiholomorphic variable w, namely $\overline{\partial_{\overline{w}} K_z(w)} = \pi w K_z(w)$ once one views $z \mapsto K_z$ as an antiholomorphic family), we obtain

$$F'(z) = \pi \langle F, \overline{w} K_z(w) \rangle_{\mathcal{F}^2}.$$

Cauchy–Schwarz then gives

$$|F'(z)| \leq \pi \|F\|_{\mathcal{F}^2}\, \|\overline{w} K_z\|_{\mathcal{F}^2}.$$

Now, by completing the square as in (i),

$$\|\overline{w} K_z\|_{\mathcal{F}^2}^2 = \int_{\mathbb{C}} |w|^2 e^{2\pi \Re(w\overline{z}) - \pi|w|^2}\, dA(w) = e^{\pi|z|^2} \int_{\mathbb{C}} |w + z|^2 e^{-\pi|w|^2}\, dA(w),$$

after the change of variables $w \mapsto w + z$. Expanding $|w + z|^2 = |w|^2 + 2\Re(w\overline{z}) + |z|^2$, and using that $\int_{\mathbb{C}} e^{-\pi|w|^2}\, dA(w) = 1$, $\int_{\mathbb{C}} |w|^2 e^{-\pi|w|^2}\, dA(w) = \frac{1}{\pi}$, and $\int_{\mathbb{C}} w\, e^{-\pi|w|^2}\, dA(w) = 0$ (by rotational invariance), we conclude

$$\int_{\mathbb{C}} |w + z|^2 e^{-\pi|w|^2}\, dA(w) = \frac{1}{\pi} + |z|^2 \leq C(1 + |z|^2),$$

for some absolute constant $C > 0$. Therefore

$$|F'(z)| \le C(1+|z|)\,\|F\|_{\mathcal{F}^2} e^{\frac{\pi}{2}|z|^2}.$$

□

Lemma 2.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$ and set*

$$T := \sup_{z\in\mathbb{C}} u_F(z).$$

Then $0 \le T \le \|F\|_{\mathcal{F}^2}^2$, and $T = \|F\|_{\mathcal{F}^2}^2$ if and only if F is proportional to F_w for some $w \in \mathbb{C}$.

Proof. The bound follows from Lemma 2.1(ii). The characterization of equality is the same as in the reproducing kernel inequality. □

We proceed with a few elementary facts.

Lemma 2.5. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$. Then:*

(i) $I_F(s)$ is attained by a superlevel set of u_F, and

$$0 \le I_F(s) \le \|F\|_{\mathcal{F}^2}^2.$$

(ii) For every $c \in \mathbb{C}$,

$$I_{cF}(s) = |c|^2 I_F(s).$$

(iii) One has the Lipschitz bound

$$|I_F(s) - I_G(s)| \le \|u_F - u_G\|_{L^1(\mathbb{C})} \le \big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big)\,\|F-G\|_{\mathcal{F}^2}.$$

Proof. The identity

$$\int_{\mathbb{C}} u_F\,dA = \int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\,dA(z) = \|F\|_{\mathcal{F}^2}^2$$

is immediate from the definition of the Fock norm. Since $F \in \mathcal{F}^2$, the standard reproducing-kernel estimate implies that u_F is bounded and continuous, and in fact $u_F(z) \to 0$ as $|z| \to \infty$; thus $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$.

The existence of a maximizing set for $I_F(s)$, given by a superlevel set of u_F, is the classical bathtub principle. The bound

$$0 \le I_F(s) \le \int_{\mathbb{C}} u_F\,dA = \|F\|_{\mathcal{F}^2}^2$$

is immediate. The scaling identity in (ii) is obvious.

For (iii), for every measurable $\Omega \subset \mathbb{C}$ with $|\Omega| = s$,

$$\left|\int_\Omega u_F\,dA - \int_\Omega u_G\,dA\right| \le \int_{\mathbb{C}} |u_F - u_G|\,dA.$$

Taking suprema over $|\Omega| = s$ gives

$$|I_F(s) - I_G(s)| \le \|u_F - u_G\|_{L^1(\mathbb{C})}.$$

Finally,

$$\begin{aligned}\|u_F - u_G\|_{L^1(\mathbb{C})} &= \int_{\mathbb{C}} \left||F|^2 - |G|^2\right| e^{-\pi|z|^2}\, dA \\ &\le \int_{\mathbb{C}} |F-G|(|F|+|G|)e^{-\pi|z|^2}\, dA \\ &\le \|F-G\|_{\mathcal{F}^2}\big(\|F\|_{\mathcal{F}^2} + \|G\|_{\mathcal{F}^2}\big)\end{aligned}$$

by Cauchy–Schwarz. □

2.2. **From time-frequency analysis to free boundary problems.** Having translated, in the previous subsection, the time–frequency localization problem into a question about the vanishing of a Fourier transform along the two complex lines $\eta = \pm i\xi$, we now bridge the gap between this Fourier-analytic statement and the free boundary formulation that will dominate the remainder of the paper. The bridge is provided by a divisibility lemma of Ehrenpreis–Malgrange flavour: any compactly supported distribution whose Fourier transform vanishes on the characteristic variety $\{\eta = \pm i\xi\}$ of the Laplacian must be the Laplacian of a compactly supported L^2 function which, in our setting, is moreover continuous off a finite set of points.

In order to state the lemma cleanly, we make precise the class of distributions with which we work. We say that a compactly supported distribution ω is a *singular-continuous free measure* if it defines a (finite, signed) Radon measure on $\mathbb{R}^2$ whose Lebesgue decomposition with respect to planar Lebesgue measure $\mathrm{d}x\,\mathrm{d}y$ takes the form

$$\omega = \omega_{\mathrm{ac}} + \omega_{\mathrm{pp}} = f\,\mathrm{d}x\,\mathrm{d}y + \sum_{j=1}^{N} c_j\,\delta_{p_j},$$

with $f \in L^1(\mathbb{R}^2)$ compactly supported, $c_j \in \mathbb{R}\setminus\{0\}$, and $p_1,\dots,p_N \in \mathbb{R}^2$ pairwise distinct. In other words, the continuous singular part ω_{cs} in the Lebesgue–Radon–Nikodym decomposition $\omega = \omega_{\mathrm{ac}} + \omega_{\mathrm{cs}} + \omega_{\mathrm{pp}}$ is required to vanish identically, and the pure point part is finite. The terminology *free measure* is suggested by the role of ω as the right-hand side of a free boundary problem arising in the next subsection.

Lemma 2.6. *Let $\omega \in \mathcal{S}'(\mathbb{R}^2)$ be a compactly supported singular-continuous free distribution such that*

$$\widehat{\omega}(w, \pm iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Then there exist a finite set $P \subset \mathbb{R}^2$ and a compactly supported function $u \in L^2(\mathbb{R}^2)$ such that

$$\Delta u = \omega$$

in the sense of distributions, and

$$u \in C(\mathbb{R}^2 \setminus P).$$

Proof. By the Paley–Wiener–Schwartz theorem (see, e.g., [24, Theorem 7.3.1] or [17, §9.3]), the Fourier transform $\widehat{\omega}(\xi,\eta)$ extends to an entire function on $\mathbb{C}^2$ of exponential type, satisfying the bound

$$|\widehat{\omega}(\xi,\eta)| \le C(1+|\xi|+|\eta|)^N\, e^{R(|\Im\xi|+|\Im\eta|)}$$

for some $C, R > 0$ and some integer $N \geq 0$ depending on the order of ω, where R is the radius of any disk containing $\operatorname{supp}\omega$. In particular, $\widehat{\omega}$ admits an everywhere convergent power series expansion

$$\widehat{\omega}(\xi,\eta) = \sum_{k,l\geq 0} a_{k,l}\,\xi^k\eta^l, \qquad (\xi,\eta)\in\mathbb{C}^2.$$

Using the vanishing hypothesis on the complex lines $\eta = \pm i\xi$, we may group terms by total homogeneity: setting $\eta = \pm iw$, $\xi = w$, and collecting powers of w,

$$0 = \widehat{\omega}(w,\pm iw) = \sum_{n\geq 0}\Big(\sum_{k=0}^{n} a_{k,n-k}(\pm i)^{\,n-k}\Big)w^n, \qquad \forall w\in\mathbb{C}.$$

Since this holds for all $w \in \mathbb{C}$, comparing coefficients yields

$$\sum_{k=0}^{n} a_{k,n-k}(\pm i)^{\,n-k} = 0, \qquad \forall n \geq 0.$$

For each $n \geq 0$, define the auxiliary one-variable polynomial

$$p_n(t) := \sum_{k=0}^{n} a_{k,n-k}\,t^{\,n-k}.$$

The two scalar equations above precisely say that $p_n(\pm i) = 0$, so $(t-i)(t+i) = t^2+1$ divides $p_n(t)$. Hence we may write

$$p_n(t) = (t^2+1)\,q_n(t),$$

with q_n a polynomial of degree at most $n-2$ (and $q_n \equiv 0$ when $n < 2$).

We now reassemble the homogeneous pieces. Let

$$H_n(\xi,\eta) := \sum_{k=0}^{n} a_{k,n-k}\,\xi^k\eta^{\,n-k}, \qquad n\geq 0,$$

so that $\widehat{\omega} = \sum_{n\geq 0} H_n$ is the decomposition of $\widehat{\omega}$ into homogeneous components (of degree n). The argument above shows that each H_n vanishes on the two complex lines $\eta = \pm i\xi$, hence is divisible, as a homogeneous polynomial in two variables, by $\xi^2+\eta^2$. Concretely,

$$H_n(\xi,\eta) = (\xi^2+\eta^2)\,G_{n-2}(\xi,\eta)$$

for a (uniquely determined) homogeneous polynomial G_{n-2} of degree $n-2$, with $G_{n-2} \equiv 0$ for $n < 2$. Summing the homogeneous pieces, the formal series

$$F(\xi,\eta) := \sum_{n\geq 2} G_{n-2}(\xi,\eta)$$

converges absolutely on every polydisc on which $\widehat{\omega}$ does, and defines an entire function on $\mathbb{C}^2$ satisfying

$$\widehat{\omega}(\xi,\eta) = (\xi^2+\eta^2)\,F(\xi,\eta).$$

Since $\widehat{\omega}$ has a zero of order at least two at the origin (coming from the vanishing of H_0 and H_1, which is forced by $p_0(\pm i) = p_1(\pm i) = 0$), this factorization is consistent at 0 and F is entire there as well.

We next quantify the size of F on $\mathbb{R}^2$. Since ω is a finite (signed) Radon measure with compact support, its Fourier transform is in fact bounded on $\mathbb{R}^2$:

$$|\widehat{\omega}(\xi,\eta)| \le \|\omega\|_{\mathrm{TV}}, \qquad (\xi,\eta)\in\mathbb{R}^2,$$

where $\|\omega\|_{\mathrm{TV}}$ is the total variation; equivalently, the Paley–Wiener bound holds with $N=0$. On $\mathbb{R}^2$, $\xi^2+\eta^2=|\xi|^2+|\eta|^2$, so for $|\xi|^2+|\eta|^2\ge 1$ we have $\xi^2+\eta^2\ge \frac{1}{2}(1+|\xi|^2+|\eta|^2)$, whence

$$|F(\xi,\eta)| = \frac{|\widehat{\omega}(\xi,\eta)|}{\xi^2+\eta^2} \le \frac{2\|\omega\|_{\mathrm{TV}}}{1+|\xi|^2+|\eta|^2}, \qquad |\xi|^2+|\eta|^2\ge 1.$$

Near the origin, F is entire (as established above) and hence bounded; absorbing this region into the constants we obtain

$$|F(\xi,\eta)| \le \frac{C''}{1+|\xi|^2+|\eta|^2}, \qquad (\xi,\eta)\in\mathbb{R}^2,$$

for some $C''>0$. Therefore $F|_{\mathbb{R}^2}\in L^2(\mathbb{R}^2)\cap L^1(\mathbb{R}^2)$. On $\mathbb{C}^2$, the analogous Paley–Wiener-type bound

$$|F(\xi,\eta)| \le C'\,\frac{e^{R(|\Im\xi|+|\Im\eta|)}}{1+|\xi|^2+|\eta|^2}$$

follows from the same argument off a neighbourhood of $\{\xi^2+\eta^2=0\}$ and from boundedness of F on a fixed compact neighbourhood of 0 inside $\mathbb{C}^2$, after enlarging C' if necessary.

By the Paley–Wiener–Schwartz theorem applied in the reverse direction (see again [24, Theorem 7.3.1]), the entire function F satisfies an exponential-type bound on $\mathbb{C}^2$ with the same exponential rate R as $\widehat{\omega}$, and integrable decay along $\mathbb{R}^2$, so it is the Fourier–Laplace transform of a distribution of order $\le N$ compactly supported in the closed disk $\overline{B(0,R)}$. The L^2 bound on the real locus shows that this distribution is, in fact, an $L^2(\mathbb{R}^2)$ function. Let $u\in L^2(\mathbb{R}^2)$ denote this compactly supported representative, characterised by $\widehat{u}=F$ on $\mathbb{R}^2$ and $\operatorname{supp} u\subset\overline{B(0,R)}$.

Computing distributional Laplacians on the Fourier side, with the convention $\widehat{g}(\xi,\eta)=\int g(x,y)e^{-2\pi i(x\xi+y\eta)}\,\mathrm{d}x$ we have

$$\widehat{\Delta u}(\xi,\eta) = -4\pi^2(\xi^2+\eta^2)\,\widehat{u}(\xi,\eta) = -4\pi^2(\xi^2+\eta^2)\,F(\xi,\eta) = -4\pi^2\,\widehat{\omega}(\xi,\eta);$$

with any other standard normalisation, the same identity holds up to a nonzero real multiplicative factor. In all cases, Δu is a constant nonzero real multiple of ω. Replacing u by λu for the appropriate (nonzero, real) λ – this is the harmless rescaling referred to above – we may, and do, assume that

$$\Delta u = \omega$$

in the sense of distributions. The rescaling preserves all the previously established properties of u: compact support, membership in $L^2(\mathbb{R}^2)$, and the Fourier identity $\widehat{u}=\lambda F$.

To pin down u pointwise, observe that since ω is a compactly supported Radon measure on $\mathbb{R}^2$, the convolution

$$v(x,y) := \frac{1}{2\pi}(\log|\cdot| * \omega)(x,y)$$

is a well-defined locally integrable function on $\mathbb{R}^2$ which solves $\Delta v=\omega$ in the sense of distributions; here $G(z)=\log|z|$ is (up to the standard normalisation $\frac{1}{2\pi}$) the fundamental solution of the Laplacian in $\mathbb{R}^2$, see, e.g., [19, Ch. II] or [17, §6.2] for the relevant logarithmic-potential theory,

and [3] for related representations in the planar quadrature-domain context. The difference $u-v$ is then a tempered distribution which is harmonic on $\mathbb{R}^2$ (since $\Delta(u-v) = \omega-\omega = 0$), and hence, being harmonic, real-analytic. Outside the union of the supports of u and ω, the function v is in fact real-analytic and grows at most logarithmically at infinity, with $v(x) = \frac{M}{2\pi}\log|x| + O(|x|^{-1})$ as $|x| \to \infty$ where $M = \omega(\mathbb{R}^2)$, while u vanishes there because of its compact support. Hence $u - v$ is harmonic on $\mathbb{R}^2$ and grows at most logarithmically at infinity; a standard refinement of the harmonic Liouville theorem (any harmonic function on $\mathbb{R}^2$ growing slower than a polynomial of degree 1 is constant) then forces $u - v$ to be a constant, and this constant must vanish to be consistent with $u \in L^2(\mathbb{R}^2)$ and the local integrability of v. Equivalently, and more transparently: by the compact support of ω together with $\widehat{u} = F$ on $\mathbb{R}^2$, the function u is uniquely determined among compactly supported L^2 solutions of $\Delta u = \omega$, since any two such would differ by an entire L^2 function which is harmonic, hence vanishes. Thus

$$u(x,y) = \frac{1}{2\pi}\int_{\mathbb{R}^2} \log|(x,y)-(x',y')|\,\mathrm{d}\omega(x',y'),$$

the logarithmic potential of ω.

We finally analyse the regularity of u. Decomposing $\omega = f\,\mathrm{d}x\,\mathrm{d}y + \sum_{j=1}^N c_j\,\delta_{p_j}$ as in the definition of singular-continuous free measure, we obtain correspondingly

$$u(x,y) = \frac{1}{2\pi}\int_{\mathbb{R}^2} \log|(x,y)-(x',y')|\,f(x',y')\,\mathrm{d}x'\mathrm{d}y' + \frac{1}{2\pi}\sum_{j=1}^N c_j\,\log|(x,y)-p_j|.$$

The first summand is the logarithmic potential of the L^1, compactly supported density f and is therefore continuous on all of $\mathbb{R}^2$ – this is a classical fact, see e.g. [19, Ch. II, Theorem 1.3] or [17, §6.2]. It is here that the absence of a continuous singular part in ω is essential: a generic continuous singular measure (e.g. the Hausdorff measure on a Cantor-type set of positive 1-capacity) gives rise to a logarithmic potential which need not be continuous, only quasi-continuous, and the conclusion of the lemma would fail in that generality. Each term $\log|(x,y)-p_j|$ is continuous on $\mathbb{R}^2\setminus\{p_j\}$, with a logarithmic singularity at p_j. Setting $P := \{p_1,\dots,p_N\}$, we conclude that $u \in C(\mathbb{R}^2\setminus P)$, as claimed. The fact that u is compactly supported has already been established in the construction above and is consistent with the representation as a logarithmic potential, which differs from a compactly supported function only by a globally harmonic correction that, as explained, must vanish identically. □

2.3. **Two perturbative auxiliary results.** In this appendix we prove two perturbative results used in the proof of Theorem 1. The first is a free-boundary stability statement around a strictly negative obstacle-type solution, in the spirit of the perturbative one-phase results of Isakov [25, Chapter 5] and the classical free-boundary literature on quadrature domains [3, 26, 23, 40, 41]. The second is a local realization theorem for weighted quadrature domains near a reference domain. Both statements are tailored to the setting of the present paper.

2.3.1. *A perturbative free-boundary theorem.* We begin with a perturbative result for the one-phase problem

$$-\Delta u = f\,\mathbf{1}_\Omega, \qquad u = 0 \text{ on } \mathbb{C}\setminus\Omega,$$

near a reference pair (Ω_0, u_0). The reader interested in more general statements of this type may consult Chapter 5 of [25].

Let $\Omega_0 \subset \mathbb{C} \simeq \mathbb{R}^2$ be a bounded domain of class C^2. For $\rho > 0$ sufficiently small, every C^2 domain Ω which is ρ-close to Ω_0 (in the Hausdorff sense, or equivalently in the sense of C^2-graphs over $\partial\Omega_0$) may be written uniquely as a normal graph over $\partial\Omega_0$:

$$\partial\Omega = \{x + \varphi(x)\nu(x) : x \in \partial\Omega_0\},$$

where ν denotes the outer unit normal to $\partial\Omega_0$, and $\varphi \in C^2(\partial\Omega_0)$ has small norm. We shall write $\Omega = \Omega_\varphi$. Throughout this subsection we keep a Hölder exponent $\alpha \in (0,1)$ fixed; the choice of C^5 regularity on the right-hand side f is dictated by the requirement that the implicit function theorem close in the space of C^2 (equivalently $C^{2,\alpha}$) boundaries: C^5 data on f and C^2 regularity of $\partial\Omega_0$ ensure that the solution operator $(\varphi, f) \mapsto u_{\varphi,f} \circ \Phi_\varphi$ is of class C^1 into $C^{2,\alpha}(\overline{\Omega_0})$ on a small neighbourhood of $(0, f_0)$.

Proposition 2.1 ([25])**.** *Let $\Omega_0 \subset \mathbb{C}$ be a bounded C^2 domain, and let $f_0 \in C^5(\overline{\Omega'})$ for some open neighborhood $\Omega' \supset \overline{\Omega_0}$, with*

$$f_0 > 0 \qquad \text{on } \partial\Omega_0.$$

Assume there exists $u_0 \in C^{2,\alpha}(\overline{\Omega_0})$ for some $\alpha \in (0,1)$ such that

$$\begin{cases} -\Delta u_0 = f_0 & \text{in } \Omega_0, \\ u_0 = 0 & \text{on } \partial\Omega_0, \\ u_0 < 0 & \text{in } \Omega_0. \end{cases} \tag{2.11}$$

Then there exist $\varepsilon > 0$ and $\rho > 0$ such that the following holds.

Whenever $f \in C^5(\overline{\Omega'})$ satisfies

$$\|f - f_0\|_{C^5(\overline{\Omega'})} < \varepsilon,$$

there exists a unique C^2 function $\varphi \in C^2(\partial\Omega_0)$, with $\|\varphi\|_{C^2} < \rho$, such that for $\Omega_f := \Omega_\varphi$ the Dirichlet problem

$$\begin{cases} -\Delta u_f = f & \text{in } \Omega_f, \\ u_f = 0 & \text{on } \partial\Omega_f \end{cases} \tag{2.12}$$

has a solution $u_f \in C^{2,\alpha}(\overline{\Omega_f})$ satisfying

$$u_f < 0 \qquad \text{in } \Omega_f.$$

Moreover, the map

$$f \mapsto \varphi(f)$$

is C^1 from a neighborhood of f_0 in $C^5(\overline{\Omega'})$ into $C^2(\partial\Omega_0)$; in particular, φ depends continuously on f in the indicated norms.

Proof. We divide the argument into several steps; the structure follows that of Corollary 5.1.4 of [25], adapted to the present formulation.

Step 1: nondegeneracy of the reference solution. By assumption, $f_0 > 0$ on the compact set $\partial\Omega_0$. Since $f_0 \in C^5(\overline{\Omega'})$ is, in particular, continuous on $\overline{\Omega_0}$, there exist constants $c_0 > 0$ and

$\eta > 0$ such that

$$f_0(x) \geq c_0 \qquad \text{for every } x \in \overline{\Omega_0} \text{ with } \operatorname{dist}(x, \partial\Omega_0) < \eta.$$

Combined with $u_0 < 0$ in Ω_0, $u_0 = 0$ on $\partial\Omega_0$, and $-\Delta u_0 = f_0 \geq c_0 > 0$ in this neighborhood of $\partial\Omega_0$, the Hopf boundary point lemma (in the standard application discussed in the perturbative one-phase setting of [25, Chapter 5]) yields

$$\partial_\nu u_0(x) > 0 \qquad \text{for every } x \in \partial\Omega_0,$$

where ν is the outer unit normal. Since $\partial\Omega_0$ is compact and $u_0 \in C^{2,\alpha}(\overline{\Omega_0})$, the function $x \mapsto \partial_\nu u_0(x)$ is continuous on $\partial\Omega_0$, and we obtain a uniform lower bound

$$\partial_\nu u_0 \geq \kappa_0 \qquad \text{on } \partial\Omega_0, \tag{2.13}$$

for some constant $\kappa_0 > 0$.

Step 2: reduction to a boundary equation. Fix a tubular neighborhood

$$\mathcal{N} := \{x + t\nu(x) : x \in \partial\Omega_0,\ |t| < \delta\}$$

on which the normal coordinates are well-defined and the map $(x, t) \mapsto x + t\nu(x)$ is a C^2 diffeomorphism onto its image. For $\varphi \in C^2(\partial\Omega_0)$ sufficiently small in C^2, the perturbed boundary

$$\Gamma_\varphi := \{x + \varphi(x)\nu(x) : x \in \partial\Omega_0\}$$

bounds a C^2 domain $\Omega_\varphi \subset \Omega'$, and the normal graph map

$$\Phi_\varphi : \overline{\Omega_0} \longrightarrow \overline{\Omega_\varphi},$$

constructed by extending φ to a C^2 function $\widetilde{\varphi}$ on a neighbourhood of $\partial\Omega_0$ and setting $\Phi_\varphi(y) = y + \widetilde{\varphi}(y)\nu(y)$ on $\mathcal{N} \cap \Omega_0$ and the identity outside, is a C^2 diffeomorphism with $\Phi_0 = \mathrm{Id}$, depending in a C^1 fashion on φ in the C^2-topology.

For each pair (φ, f) with φ small in $C^2(\partial\Omega_0)$ and f close to f_0 in $C^5(\overline{\Omega'})$, let $u_{\varphi,f}$ denote the unique solution of

$$\begin{cases} -\Delta u_{\varphi,f} = f & \text{in } \Omega_\varphi, \\ u_{\varphi,f} = 0 & \text{on } \Gamma_\varphi. \end{cases}$$

Pulling back via Φ_φ, the function $v_{\varphi,f} := u_{\varphi,f} \circ \Phi_\varphi$ satisfies a transformed boundary-value problem of the form

$$\begin{cases} L_\varphi v_{\varphi,f} = f \circ \Phi_\varphi & \text{in } \Omega_0, \\ v_{\varphi,f} = 0 & \text{on } \partial\Omega_0, \end{cases}$$

where L_φ is a second-order uniformly elliptic operator with $C^{0,\alpha}$ (in fact C^1) coefficients depending in a C^1 fashion on φ, and with $L_0 = -\Delta$. Standard Schauder theory for elliptic boundary value problems (see e.g. [25, Ch. 5]) then implies that the map

$$(\varphi, f) \mapsto v_{\varphi,f} = u_{\varphi,f} \circ \Phi_\varphi$$

is of class C^1 from a neighborhood of $(0, f_0)$ in $C^2(\partial\Omega_0) \times C^5(\overline{\Omega'})$ into $C^{2,\alpha}(\overline{\Omega_0})$; the choice of C^5 data on the right-hand side guarantees the requisite regularity of the coefficients of L_φ

and of the source term $f \circ \Phi_\varphi$ in this Schauder framework, and in particular that the linearised problems are well-posed in $C^{2,\alpha}$.

Now define the nonlinear boundary operator

$$\mathcal{F}(\varphi, f)(x) := u_{\varphi,f}\big(x + \varphi(x)\nu(x)\big), \qquad x \in \partial\Omega_0;$$

here we regard $u_{\varphi,f}$ as a $C^{2,\alpha}$ function defined in a neighbourhood of $\overline{\Omega_\varphi}$ (which contains a fixed neighbourhood of $\overline{\Omega_0}$ for φ small in C^2) and evaluate it at the point $x + \varphi(x)\nu(x)$, which lies on Γ_φ. Since $u_{\varphi,f}$ vanishes on Γ_φ, the condition that Γ_φ be the free boundary of (2.12) is precisely

$$\mathcal{F}(\varphi, f) = 0. \tag{2.14}$$

By the regularity statement above, $\mathcal{F}$ is C^1 from a neighborhood of $(0, f_0)$ into $C^{2,\alpha}(\partial\Omega_0)$.

Step 3: computation of the linearization in φ. For the purpose of the implicit function theorem we replace $\mathcal{F}$ by the equivalent boundary functional

$$\widetilde{\mathcal{F}}(\varphi, f)(x) := u_{0,f}\big(x + \varphi(x)\nu(x)\big), \qquad x \in \partial\Omega_0,$$

where $u_{0,f}$ denotes the solution of $-\Delta u = f$ on the *fixed* reference domain Ω_0, with $u_{0,f} = 0$ on $\partial\Omega_0$, extended to a fixed neighbourhood of $\overline{\Omega_0}$ by a bounded linear extension operator $E\colon C^{2,\alpha}(\overline{\Omega_0}) \to C^{2,\alpha}(\mathbb{R}^2)$ (such an extension exists by standard arguments, e.g. Stein-type extension or local reflection across the C^2 boundary $\partial\Omega_0$). For (φ, f) close to $(0, f_0)$ the equations

$$\widetilde{\mathcal{F}}(\varphi, f) = 0 \qquad \text{and} \qquad \big(\Gamma_\varphi \text{ is the free boundary of (2.12)}\big)$$

cut out the same set of (φ, f): both express that Γ_φ is the zero level set of the solution corresponding to f. The advantage of $\widetilde{\mathcal{F}}$ is that the underlying solution $u_{0,f}$ does not depend on φ, so its derivative in φ reduces to a pure shape derivative of a fixed function. From now on we abuse notation and continue to write $\mathcal{F}$ for $\widetilde{\mathcal{F}}$.

Let $\psi \in C^2(\partial\Omega_0)$, extend ψ to a C^2 function in a neighbourhood of $\partial\Omega_0$, and consider the family $\varphi_\varepsilon = \varepsilon\psi$, $\varepsilon \in \mathbb{R}$. Using $u_0(x) = 0$ for $x \in \partial\Omega_0$ and Taylor-expanding u_0 at first order in the normal direction,

$$\mathcal{F}(\varepsilon\psi, f_0)(x) = u_0\big(x + \varepsilon\psi(x)\nu(x)\big) = u_0(x) + \varepsilon\,\psi(x)\,\partial_\nu u_0(x) + o(\varepsilon) = \varepsilon\,\psi(x)\,\partial_\nu u_0(x) + o(\varepsilon),$$

uniformly in $x \in \partial\Omega_0$. Hence

$$D_\varphi\mathcal{F}(0, f_0)[\psi] = \psi\,\partial_\nu u_0 \qquad \text{on } \partial\Omega_0.$$

This is the standard Hadamard formula in the present Dirichlet setting: moving the boundary in the normal direction by ψ changes the boundary trace of the (fixed) solution at first order by $\psi\,\partial_\nu u_0$.

In view of (2.13), the multiplication operator

$$\psi \mapsto \psi\,\partial_\nu u_0$$

is invertible, with continuous inverse $g \mapsto g/\partial_\nu u_0$. Since $\partial_\nu u_0 \in C^{1,\alpha}(\partial\Omega_0)$ (being the boundary trace of a $C^{1,\alpha}$ vector field on a C^2 curve, derived from the $C^{2,\alpha}$ function u_0) and is bounded below by $\kappa_0 > 0$, this operator is an isomorphism of $C^2(\partial\Omega_0)$ onto itself, and likewise of $C^{2,\alpha}(\partial\Omega_0)$ onto itself.

Step 4: application of the implicit function theorem. We apply the implicit function theorem in Banach spaces to the C^1 map

$$\mathcal{F} : U \subset C^2(\partial\Omega_0) \times C^5(\overline{\Omega'}) \longrightarrow C^{2,\alpha}(\partial\Omega_0)$$

at the point $(0, f_0)$. We have $\mathcal{F}(0, f_0) = 0$, and by Step 3 the partial derivative $D_\varphi\mathcal{F}(0, f_0)$ is an isomorphism onto its image. The implicit function theorem therefore yields $\varepsilon > 0$, $\rho > 0$, and a unique C^1 map

$$f \mapsto \varphi(f) \in C^2(\partial\Omega_0), \qquad \|\varphi(f)\|_{C^2} < \rho,$$

defined for $\|f - f_0\|_{C^5} < \varepsilon$, such that

$$\mathcal{F}(\varphi(f), f) = 0, \qquad \varphi(f_0) = 0.$$

Equivalently, the corresponding domain $\Omega_f := \Omega_{\varphi(f)}$ has the property that the solution $u_f := u_{\varphi(f),f}$ of (2.12) vanishes on $\partial\Omega_f$. This already gives existence and uniqueness of $\varphi(f)$ inside the small C^2-ball of radius ρ about 0.

Step 5: preservation of the sign. It remains to show that, possibly after shrinking ε and ρ, the solution u_f is strictly negative on the whole of Ω_f.

By Step 2, the map $(\varphi, f) \mapsto u_{\varphi,f} \circ \Phi_\varphi$ is C^1 (in particular continuous) from a neighbourhood of $(0, f_0)$ into $C^{2,\alpha}(\overline{\Omega_0})$. By the embedding $C^{2,\alpha} \hookrightarrow C^1$,

$$\left\|u_f \circ \Phi_{\varphi(f)} - u_0\right\|_{C^1(\overline{\Omega_0})} \longrightarrow 0 \qquad \text{as } f \to f_0 \text{ in } C^5(\overline{\Omega'}). \tag{2.15}$$

To deduce strict negativity from this C^1-closeness, we split $\overline{\Omega_0}$ into a fixed interior region and a thin collar around the boundary, where additional care is needed because $u_0 = 0$ on $\partial\Omega_0$.

Fix $\delta_0 > 0$ so small that the inner collar

$$\mathcal{N}_{\delta_0} := \{x - t\nu(x) : x \in \partial\Omega_0,\ 0 < t < \delta_0\}$$

is contained in Ω_0 and the map $(x, t) \mapsto x - t\nu(x)$ is a diffeomorphism of $\partial\Omega_0 \times (0, \delta_0)$ onto $\mathcal{N}_{\delta_0}$. By (2.13) and the C^1-regularity of u_0 up to the boundary, after possibly shrinking δ_0 we obtain the quantitative collar estimate

$$u_0(x - t\nu(x)) \le -\frac{\kappa_0}{2}\, t \qquad \text{for } x \in \partial\Omega_0,\ 0 < t < \delta_0.$$

Set

$$K_{\delta_0} := \big\{y \in \Omega_0 : \operatorname{dist}(y, \partial\Omega_0) \ge \delta_0/2\big\},$$

which is a compact subset of Ω_0 on which, by continuity and strict negativity of u_0 in Ω_0, one has $u_0 \le -m_0$ for some $m_0 > 0$.

By (2.15), for f sufficiently close to f_0 in $C^5(\overline{\Omega'})$ the function $v_f := u_f \circ \Phi_{\varphi(f)}$ satisfies

$$\|v_f - u_0\|_{C^1(\overline{\Omega_0})} < \min\Big\{\frac{\kappa_0}{4}, \frac{m_0}{2}\Big\}.$$

On K_{δ_0} this immediately gives $v_f \le -m_0 + m_0/2 = -m_0/2 < 0$. On the closed collar $\overline{\mathcal{N}_{\delta_0}}$, the C^1-closeness combined with (2.13) gives, on $\partial\Omega_0$,

$$\partial_\nu v_f \ge \partial_\nu u_0 - \|v_f - u_0\|_{C^1} \ge \kappa_0 - \frac{\kappa_0}{4} = \frac{3\kappa_0}{4}.$$

By uniform continuity of ∇v_f and ∇u_0 on $\overline{\Omega_0}$ — recall both lie in a fixed bounded set of $C^{1,\alpha}(\overline{\Omega_0})$ by the Schauder estimate, hence in a precompact subset of $C^1(\overline{\Omega_0})$ — after a further shrinking of δ_0 (independent of f) we have, for every $x \in \partial\Omega_0$ and $0 < t < \delta_0$,

$$\partial_\nu v_f(x - t\nu(x)) \ge \frac{\kappa_0}{2}.$$

Since $v_f = 0$ on $\partial\Omega_0$, the fundamental theorem of calculus applied along inward normal segments yields

$$v_f(x - t\nu(x)) = v_f(x) - \int_0^t \partial_\nu v_f(x - s\nu(x))\, ds \le -\frac{\kappa_0}{2}\, t < 0 \qquad \text{for } x \in \partial\Omega_0,\ 0 < t < \delta_0.$$

Hence $v_f < 0$ on $\mathcal{N}_{\delta_0}$ as well, and combining with the estimate on K_{δ_0} we conclude that $u_f \circ \Phi_{\varphi(f)} < 0$ throughout Ω_0. Since $\Phi_{\varphi(f)}$ is a diffeomorphism from Ω_0 onto Ω_f, this is equivalent to $u_f < 0$ throughout Ω_f.

This proves existence. Uniqueness among sufficiently small C^2 normal graphs follows from the uniqueness statement in the implicit function theorem applied in Step 4. □

2.3.2. *A perturbative weighted quadrature-domain theorem.* We now prove a local realization statement for weighted quadrature domains near a reference C^2 domain. The result is a tailored, perturbative version of the classical setup of Cherednichenko [13], adapted to the form needed in the proof of Theorem 1. Conceptually, the statement may also be viewed as a quantitative implicit-function counterpart of the variational characterization of quadrature domains studied by Gustafsson [23], Gustafsson–Shahgholian [41], Shahgholian [40], Karp [26], and Aharonov–Schiffer–Zalcman [3].

Setup. Fix a positive weight $\mu \in C^2(\overline{U_2})$, where $U_1 \subset U_0 \subset U_2$ are bounded C^2 domains with $\overline{U_1} \subset U_0$ and $\overline{U_0} \subset U_2$. For a bounded measurable domain $U \subset U_2$, define its weighted Cauchy transform by

$$g_U(z) := \int_U \frac{\mu(w)}{z - w}\, dA(w), \qquad z \in \mathbb{C} \setminus \overline{U}.$$

The function g_U is holomorphic on $\mathbb{C}\setminus\overline{U}$, continuous up to ∂U, and, in the sense of distributions on $\mathbb{C}$,

$$\partial_{\overline{z}}\, g_U \;=\; \pi\, \mu\, \mathbf{1}_U.$$

At infinity it admits the expansion

$$g_U(z) = \frac{1}{z} \int_U \mu(w)\, dA(w) + O(|z|^{-2}),$$

so in particular $\lim_{z\to\infty} g_U(z) = 0$ and $\lim_{z\to\infty} z g_U(z) = \int_U \mu\, dA > 0$.

As before, every domain U sufficiently C^2-close to U_0 may be represented as a normal graph over ∂U_0,

$$\partial U = \{x + \varphi(x)\, \nu(x) : x \in \partial U_0\},$$

with $\varphi \in C^2(\partial U_0)$ of small C^2 norm, where ν denotes the outward unit normal to ∂U_0. We denote the resulting domain by U_φ and write $\Phi_\varphi(x) := x + \varphi(x)\nu(x)$ for the boundary diffeomorphism, extended smoothly to a tubular neighborhood of ∂U_0 (and by the identity outside) so as to define a diffeomorphism of $\overline{U_2}$ onto itself.

Proposition 2.2 ([13])**.** *Let $U_1 \subset U_0 \subset U_2$ be bounded C^2 domains, and let $\mu \in C^2(\overline{U_2})$ satisfy*

$$\mu > 0 \qquad \text{on } \overline{U_2}.$$

Then there exist $\rho > 0$ and $\varepsilon > 0$ with the following property.

Suppose g is holomorphic on $\mathbb{C} \setminus \overline{U_1}$, continuous up to ∂U_1, and satisfies

$$\lim_{z\to\infty} g(z) = 0, \qquad \lim_{z\to\infty} z\, g(z) = c_0 > 0, \tag{2.16}$$

together with

$$\sup_{z\in\partial U_1} |g(z) - g_{U_0}(z)| < \varepsilon.$$

Then there exists a unique function $\varphi \in C^2(\partial U_0)$, with $\|\varphi\|_{C^2(\partial U_0)} < \rho$, such that the associated domain U_φ satisfies

$$U_1 \subset U_\varphi \subset U_2$$

and

$$g_{U_\varphi}(z) = g(z) \qquad \text{for all } z \in \mathbb{C} \setminus \overline{U_\varphi}.$$

Moreover, the assignment

$$g \mapsto \varphi(g)$$

is C^1 with respect to the $C^2(\partial U_1)$ topology on the trace data and the $C^2(\partial U_0)$ topology on the boundary graph.

Proof. The proof reduces matters to a boundary equation on ∂U_1 and applies the implicit function theorem in suitable Banach spaces.

Step 1: reduction to a boundary matching problem. Both g and g_{U_φ} are holomorphic on $\mathbb{C} \setminus \overline{U_\varphi}$ and vanish at infinity. As long as $U_1 \subset U_\varphi$, the curve ∂U_1 lies in the (connected) unbounded component of $\mathbb{C} \setminus \overline{U_\varphi}$. Suppose

$$g_{U_\varphi}(z) \;=\; g(z) \qquad \text{for all } z \in \partial U_1. \tag{2.17}$$

The difference $g_{U_\varphi} - g$ is then holomorphic on $\mathbb{C} \setminus \overline{U_\varphi}$, continuous up to the boundary, vanishes at infinity, and vanishes on the closed curve ∂U_1. Since the zero set of a non-trivial holomorphic function on a connected open set is discrete, vanishing on the one-dimensional set ∂U_1 forces $g_{U_\varphi} \equiv g$ on the unbounded component of $\mathbb{C} \setminus \overline{U_\varphi}$. As both functions are holomorphic on the full open set $\mathbb{C} \setminus \overline{U_\varphi}$, the identity theorem extends the equality to that whole open set.

Thus it suffices to solve (2.17). We define

$$\mathcal{G} : \; \varphi \;\longmapsto\; g_{U_\varphi}\big|_{\partial U_1},$$

and view $\mathcal{G}$ as a map between Banach spaces

$$\mathcal{G} \colon \; \mathcal{O} \subset C^2(\partial U_0) \;\longrightarrow\; C^2(\partial U_1),$$

defined on a small C^2 neighborhood $\mathcal{O}$ of 0, small enough that $\overline{U_1} \subset U_\varphi \subset U_2$ for all $\varphi \in \mathcal{O}$. Note that for such φ, ∂U_1 stays a positive distance from $\overline{U_\varphi}$, so the integral defining g_{U_φ} is non-singular for $z \in \partial U_1$.

Step 2: C^1 dependence of $\mathcal{G}$ on φ. Performing the change of variables $w = \Phi_\varphi(w')$ in the bulk integral yields

$$g_{U_\varphi}(z) = \int_{U_0} \frac{\mu(\Phi_\varphi(w'))\, J_\varphi(w')}{z - \Phi_\varphi(w')}\, dA(w'), \qquad z \in \partial U_1,$$

where $J_\varphi = |\det D\Phi_\varphi|$. Because $z \in \partial U_1$ keeps a uniform positive distance $d_0 > 0$ from $\overline{U_0}$ (and hence from $\overline{U_\varphi}$ for all φ in a sufficiently small neighborhood of zero), the integrand is a smooth function of (φ, w') and the differentiation in φ under the integral sign is fully justified by dominated convergence. Combined with the smoothness of μ and of the dependence $\varphi \mapsto \Phi_\varphi$, this gives that $\mathcal{G}$ is of class C^1 (in fact C^k whenever $\mu \in C^k$) from $\mathcal{O} \subset C^2(\partial U_0)$ to $C^2(\partial U_1)$.

Step 3: Hadamard shape derivative. We compute $D\mathcal{G}(0)$. Let $\psi \in C^2(\partial U_0)$ and set $\varphi = t\psi$ for small $t \in \mathbb{R}$. Writing $U_{t\psi} \setminus U_0$ as the set swept out by the normal flow with speed ψ in time t, one has, for $z \in \partial U_1$ (so that $z \notin \overline{U_{t\psi}}$ for $|t|$ small),

$$g_{U_{t\psi}}(z) - g_{U_0}(z) = \int_{U_{t\psi} \triangle U_0} \frac{\pm\mu(w)}{z - w}\, dA(w),$$

with the sign $+$ (resp. $-$) on $U_{t\psi} \setminus U_0$ where $\psi > 0$ (resp. on $U_0 \setminus U_{t\psi}$ where $\psi < 0$). Parametrizing the symmetric difference as a tubular shell of width $t\psi(x)$ over ∂U_0 and using the surface area element $d\sigma$ on ∂U_0, Fubini gives

$$g_{U_{t\psi}}(z) - g_{U_0}(z) = \int_{\partial U_0} \Big(\int_0^{t\psi(x)} \frac{\mu(x + s\nu(x))}{z - x - s\nu(x)}\, ds \Big)\, d\sigma(x),$$

which after dividing by t and letting $t \to 0$ yields the standard Hadamard variation formula

$$D\mathcal{G}(0)[\psi](z) = \int_{\partial U_0} \frac{\mu(w)\, \psi(w)}{z - w}\, d\sigma(w), \qquad z \in \partial U_1. \tag{2.18}$$

Thus the linearization is the (single-layer) Cauchy transform on ∂U_0 of the boundary density $\mu\psi$, restricted to ∂U_1. We denote it by L,

$$L\psi(z) := \int_{\partial U_0} \frac{\mu(w)\, \psi(w)}{z - w}\, d\sigma(w), \qquad z \in \partial U_1.$$

Step 4: invertibility of the linearization. We show that L is an isomorphism between $C^2(\partial U_0)$ and the natural trace space

$$\mathcal{T} := \Big\{ H\big|_{\partial U_1} : H \text{ holomorphic on } \mathbb{C} \setminus \overline{U_1},\ H(\infty) = 0 \Big\} \cap C^2(\partial U_1),$$

equipped with the $C^2(\partial U_1)$ norm.

Injectivity. Suppose $L\psi \equiv 0$ on ∂U_1. The Cauchy-type transform

$$F(z) := \int_{\partial U_0} \frac{\mu(w)\, \psi(w)}{z - w}\, d\sigma(w)$$

is holomorphic on $\mathbb{C} \setminus \partial U_0$ and vanishes at infinity. By hypothesis $F \equiv 0$ on ∂U_1; the same identity-theorem argument as in Step 1 (applied on the connected open set $\mathbb{C} \setminus \overline{U_0}$) shows that $F \equiv 0$ on $\mathbb{C} \setminus \overline{U_0}$. The Sokhotski–Plemelj jump relations for the Cauchy integral on the C^2 curve ∂U_0 [19, Ch. III] yield, for every $\zeta \in \partial U_0$ at which $\mu\psi$ is continuous,

$$F^{\mathrm{int}}(\zeta) - F^{\mathrm{ext}}(\zeta) \;=\; c_0\, \mu(\zeta)\psi(\zeta),$$

where F^{int} and F^{ext} denote the non-tangential limits from the interior and the exterior of U_0 respectively, and c_0 is a non-zero constant depending only on the chosen orientation conventions for the kernel $1/(z-w)\, d\sigma$ (in the standard convention with ∂U_0 oriented counterclockwise and $d\sigma$ the arclength measure, c_0 equals a non-zero scalar multiple of the integrand of $dw/d\sigma$ on ∂U_0; the precise value is irrelevant here). Since $F^{\mathrm{ext}} \equiv 0$ and $\mu > 0$ on ∂U_0, this forces $\psi \equiv 0$. Hence L is injective.

Surjectivity. Let $h \in \mathcal{T}$ and let H denote its unique holomorphic extension to $\mathbb{C} \setminus \overline{U_1}$ vanishing at infinity. The annular region $A := U_0 \setminus \overline{U_1}$ is contained in $\mathbb{C} \setminus \overline{U_1}$, so H is holomorphic in a neighborhood of $\overline{A}$; in particular the trace $H|_{\partial U_0}$ is well defined, lies in $C^2(\partial U_0)$, and depends continuously on h in the C^2 topology by the Cauchy integral formula on ∂U_1 (interior holomorphic regularity inside the annulus). We now seek a density $\sigma_0 \in C^2(\partial U_0)$ such that

$$H(z) \;=\; \int_{\partial U_0} \frac{\sigma_0(w)}{z-w}\, d\sigma(w) \qquad \text{for all } z \in \mathbb{C} \setminus \overline{U_0}.$$

Such a density is unique: if two densities σ_0 and σ_0' produced the same Cauchy integral on $\mathbb{C} \setminus \overline{U_0}$, then their difference $\delta := \sigma_0 - \sigma_0'$ generates a Cauchy integral F with exterior limit $F^{\mathrm{ext}} \equiv 0$. The standard Sokhotski–Plemelj jump relations [19, Ch. III] yield $F^{\mathrm{int}} - F^{\mathrm{ext}} = c_0\, \delta$ at every continuity point of δ, with the same non-zero kernel constant c_0 as above. With $F^{\mathrm{ext}} \equiv 0$, the function F extends holomorphically across ∂U_0 to the whole plane (and remains holomorphic) only if $F^{\mathrm{int}} \equiv 0$ as well; combined with the vanishing of F at infinity this would force $F \equiv 0$ and hence $\delta \equiv 0$. One then verifies the consistency by noting that the C^2 regularity of the curve and density guarantees that the non-tangential limits exist pointwise and the jump identity holds in the classical sense.

Existence of $\sigma_0 \in C^2(\partial U_0)$ is the classical exterior Cauchy inversion problem on the C^2 curve ∂U_0. The boundary integral operator

$$T\sigma_0(\zeta) \;:=\; \text{p.v.} \int_{\partial U_0} \frac{\sigma_0(w)}{\zeta - w}\, d\sigma(w), \qquad \zeta \in \partial U_0,$$

is a classical singular integral operator on the C^2 curve, bounded on $C^2(\partial U_0)$ by the standard mapping properties of the Cauchy transform on smooth curves [19, Ch. III]. Combining T with the Sokhotski–Plemelj jump identity gives, for the exterior boundary value of the Cauchy integral with density σ_0,

$$F^{\mathrm{ext}}(\zeta) \;=\; T\sigma_0(\zeta) - \tfrac{1}{2}\, c_0\, \sigma_0(\zeta),$$

so that the exterior matching equation $F^{\mathrm{ext}} = H|_{\partial U_0}$ becomes the integral equation

$$T\sigma_0 - \tfrac{1}{2}\, c_0\, \sigma_0 \;=\; H\big|_{\partial U_0} \qquad \text{on } \partial U_0.$$

The associated Fredholm theory (with T a compact perturbation of a multiple of the identity in the appropriate scale of Hölder spaces, see [19, Ch. III]) combined with the uniqueness statement just established yields, by the Fredholm alternative, a unique solution $\sigma_0 \in C^2(\partial U_0)$ depending continuously on $H|_{\partial U_0}$, and hence on h, in the C^2 topology.

Setting $\psi := \sigma_0/\mu \in C^2(\partial U_0)$ (which is admissible since $\mu \geq c > 0$ on $\overline{U_2}$ and $\mu \in C^2$), we obtain

$$L\psi(z) = \int_{\partial U_0} \frac{\mu(w)\psi(w)}{z-w}\, d\sigma(w) = \int_{\partial U_0} \frac{\sigma_0(w)}{z-w}\, d\sigma(w) = H(z) \qquad \text{for } z \in \partial U_1.$$

Hence L is surjective onto $\mathcal{T}$. Combining injectivity and surjectivity, L is a continuous bijection between the Banach spaces $C^2(\partial U_0)$ and $\mathcal{T}$; the open mapping theorem then guarantees that $L^{-1}\colon \mathcal{T} \to C^2(\partial U_0)$ is bounded.

Step 5: implicit function theorem. We now have all ingredients to invoke the implicit function theorem between the Banach spaces

$$X := C^2(\partial U_0), \qquad Y := \left\{ H\big|_{\partial U_1} : H \text{ holom. on } \mathbb{C} \setminus \overline{U_1},\ H(\infty) = 0 \right\} \cap C^2(\partial U_1) = \mathcal{T},$$

both equipped with the appropriate C^2 norms. By Step 2, the map $\mathcal{G}\colon \mathcal{O} \to Y$ is C^1 on a neighborhood $\mathcal{O} \subset X$ of 0, and by Step 4 its differential at 0 is the Banach isomorphism $D\mathcal{G}(0) = L : X \to Y$. The implicit function theorem (in Banach spaces) therefore furnishes $\rho_0 > 0$ and $\varepsilon_0 > 0$ such that, for every $\widetilde{h} \in Y$ with $\|\widetilde{h} - \mathcal{G}(0)\|_{C^2(\partial U_1)} < \varepsilon_0$, there is a unique $\varphi \in X$ with $\|\varphi\|_{C^2(\partial U_0)} < \rho_0$ solving $\mathcal{G}(\varphi) = \widetilde{h}$, and the assignment $\widetilde{h} \mapsto \varphi$ is of class C^1 from a neighborhood of $\mathcal{G}(0) = g_{U_0}|_{\partial U_1}$ in Y to X.

To match the hypotheses of the proposition, we observe that any g holomorphic on $\mathbb{C} \setminus \overline{U_1}$, continuous up to ∂U_1, and satisfying (2.16), with

$$\sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)| < \varepsilon,$$

is in particular bounded on ∂U_1; since $g - g_{U_0}$ is holomorphic on $\mathbb{C} \setminus \overline{U_1}$ and vanishes at infinity, the Cauchy integral formula on a slightly larger smooth contour surrounding ∂U_1 inside $U_0 \setminus \overline{U_1}$ upgrades the sup-norm bound on ∂U_1 to a C^2-norm bound on ∂U_1, namely

$$\|g - g_{U_0}\|_{C^2(\partial U_1)} \ \leq\ C \sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)|,$$

with a constant C depending only on U_0, U_1, and μ. Choosing ε so that $C\varepsilon < \varepsilon_0$, and setting $\rho := \rho_0$, the IFT yields the unique $\varphi \in C^2(\partial U_0)$ with the asserted properties; in particular $g \mapsto \varphi(g)$ is C^1 with respect to the $C^2(\partial U_1)$ norm on the trace data, as claimed.

By Step 1, the boundary equality $g_{U_\varphi} = g$ on ∂U_1 extends to all of $\mathbb{C} \setminus \overline{U_\varphi}$ by holomorphic continuation.

Finally, the geometric inclusion $U_1 \subset U_\varphi \subset U_2$ follows from a uniform tubular neighborhood argument. Set

$$d_1 := \operatorname{dist}(\partial U_1, \partial U_0) > 0, \qquad d_2 := \operatorname{dist}(\partial U_0, \partial U_2) > 0,$$

which are strictly positive by the standing assumption $\overline{U_1} \subset U_0$ and $\overline{U_0} \subset U_2$. For $\varphi \in C^2(\partial U_0)$ with $\|\varphi\|_{C^0(\partial U_0)} < \min(d_1, d_2)$, each boundary point of U_φ lies within distance $\|\varphi\|_{C^0}$ of ∂U_0, so

∂U_φ is contained in the open tubular neighborhood $\{x : \mathrm{dist}(x, \partial U_0) < \min(d_1, d_2)\}$, which is in turn contained in $U_2 \setminus \overline{U_1}$. Combined with the fact that U_φ is a normal-graph perturbation of U_0 (so its topological position relative to ∂U_0 is preserved for small φ), this forces $U_1 \subset U_\varphi \subset U_2$. Since the inclusion $C^2(\partial U_0) \hookrightarrow C^0(\partial U_0)$ is continuous, shrinking ρ if necessary ensures the smallness condition above. This proves the proposition. □

Remark 2.1. The only delicate point in Proposition 2.2 is the invertibility of the linearized map. The mechanism is that the Cauchy transform of a boundary density is determined by its trace on any inner contour (by holomorphic continuation through the annulus), and the Sokhotski–Plemelj jump relation then recovers the density itself. Positivity of μ is what allows one to pass from $\mu\psi$ back to ψ; the same proof would fail at any point where μ vanishes.

We postpone the proof of Theorem 2 until after the perturbative inverse theorem. The same inverse/free-boundary mechanism used to construct domains for prescribed eigenfunctions also recovers the classical Abreu–Dörfler disk theorem in the simply connected setting; see Section 4 below.

3. Proof of Theorem 1

Proof of Theorem 1. We again use the Bargmann transform. Let

$$P(z) = 1 + \sum_{k=1}^{N} b_k z^k,$$

where the coefficients b_k are chosen so that the Bargmann transform of

$$f_0 = h_0 + \sum_{k=1}^{N} a_k h_k$$

is exactly P. Then the condition that f_0 be an eigenfunction of $\mathcal{L}_U$ with eigenvalue λ is equivalent to

$$\int_U P(z) e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda P(w), \qquad w \in \mathbb{C}. \tag{3.1}$$

The proof proceeds in four steps. In Step 0 we apply a polynomial differential operator $P(\partial_w/\pi)$ to (3.1) in order to symmetrize the integrand and reduce matters to a moment identity for $|P|^2 e^{-\pi|z|^2}$. In Step 1 we construct an auxiliary radially symmetric domain U_0 for which the corresponding eigenvalue is the constant function λ; this is achieved by perturbing a Daubechies disk via the Isakov free-boundary scheme of Proposition 2.1. In Step 2 we use a Laplace transform in the variable w to convert the polynomial right-hand side $\lambda Q(w)$ into the rational right-hand side required by Proposition 2.2 and produce a domain U satisfying the symmetrized moment identity. Finally, in Step 3 we invert the differentiation of Step 0 by analyzing the entire solutions of the constant-coefficient ODE $P(\partial_w/\pi)H = 0$ and ruling out nontrivial exponential solutions through a growth estimate.

Step 0: a first reduction. For a polynomial

$$R(z) = \alpha_0 + \alpha_1 z + \cdots + \alpha_n z^n,$$

define the differential operator

$$R(\partial_w/\pi) := \sum_{k=0}^{n} \alpha_k \Big(\frac{\partial_w}{\pi}\Big)^k.$$

We apply $P(\partial_w/\pi)$ to both sides of (3.1). The right-hand side is the polynomial $\lambda P(w)$, on which the differential operator acts in the obvious way. For the left-hand side, write

$$F(w) := \int_U P(z) e^{-\pi|z|^2} e^{\pi \overline{z} w}\, dA(z).$$

Since U is bounded, the integrand is jointly continuous on $\overline{U} \times \mathbb{C}$, and for every compact set $K \subset \mathbb{C}$ and every nonnegative integer j one has the uniform bound

$$\big|\partial_w^j \big(P(z) e^{-\pi|z|^2} e^{\pi \overline{z} w}\big)\big| \le \big(\pi \sup_{z\in U} |z|\big)^j \sup_{z\in U}\big(|P(z)| e^{-\pi|z|^2}\big)\, e^{\pi(\sup_{z\in U}|z|)\sup_{w\in K}|w|},$$

valid for $(z,w) \in U \times K$. The right-hand side is integrable on the bounded set U, so dominated convergence justifies differentiating under the integral sign arbitrarily often, and F is entire. Combining this with the identity

$$\Big(\frac{\partial_w}{\pi}\Big)^j e^{\pi \overline{z} w} = \overline{z}^{\,j} e^{\pi \overline{z} w},$$

we obtain

$$P(\partial_w/\pi) F(w) = \int_U P(z)\, \overline{P(z)}\, e^{-\pi|z|^2} e^{\pi \overline{z} w}\, dA(z) = \int_U |P(z)|^2 e^{-\pi|z|^2} e^{\pi \overline{z} w}\, dA(z).$$

Hence (3.1) is equivalent to

$$\int_U |P(z)|^2 e^{-\pi|z|^2} e^{\pi \overline{z} w}\, dA(z) = \lambda Q(w), \qquad w \in \mathbb{C}, \tag{3.2}$$

where

$$Q(w) := P(\partial_w/\pi)\big(P(w)\big).$$

Strictly speaking, applying $P(\partial_w/\pi)$ only gives the implication (3.1) $\Rightarrow$ (3.2); the converse implication will be recovered in Step 3 by an ODE-theoretic argument. Our first task is therefore to construct a domain U satisfying (3.2).

Step 1: construction of an auxiliary domain. We begin by constructing a C^2 domain U_0, close to a fixed disk, for which

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi \overline{z} w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}. \tag{3.3}$$

Note that (3.3) is the special case of (3.2) in which the right-hand side is the constant λ; it will serve as the unperturbed reference for Step 2.

Let $\psi \in C_c^\infty(\mathbb{C})$ be a nonnegative radial function such that

$$\operatorname{supp} \psi \subset D_2(0) \setminus D_1(0), \qquad \int_{\mathbb{C}} \psi\, dA = 1,$$

and set

$$\psi_\delta(z) := \delta^{-2} \psi(z/\delta), \qquad 0 < \delta < 1.$$

Then ψ_δ has total mass one and $\operatorname{supp}\psi_\delta \subset D_{2\delta}(0) \setminus D_\delta(0)$. Since ψ_δ is radial, expanding $e^{\pi\bar{z}w} = \sum_{j\geq 0}(\pi\bar{z}w)^j/j!$ and integrating in polar coordinates kills every angular harmonic with $j \geq 1$, so that

$$\int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z) = c_\delta, \qquad w \in \mathbb{C},$$

where

$$c_\delta := \int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2}\, dA(z) > 0.$$

Note that $c_\delta \to 1$ as $\delta \to 0$. Hence (3.3) is equivalent to

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z) = \frac{\lambda}{c_\delta} \int_{\mathbb{C}} \psi_\delta(z) e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z). \tag{3.4}$$

We shall construct a compactly supported distributional solution of the Poisson equation

$$\Delta u_0 = -|P(z)|^2 e^{-\pi|z|^2} \mathbf{1}_{U_0} + \frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}. \tag{3.5}$$

Once such a solution has been constructed, the moment identity (3.3) follows by applying the same Fourier-analytic divisibility argument used in the proof of Theorem 2 to the corresponding compactly supported distribution

$$\omega_0 := |P(z)|^2 e^{-\pi|z|^2} \mathbf{1}_{U_0} - \frac{\lambda}{c_\delta} \psi_\delta(z) e^{-\pi|z|^2}.$$

Indeed, (3.5) says exactly that $\Delta u_0 = -\omega_0$, and since u_0 is compactly supported, the vanishing of $\widehat{\omega_0}$ on the two characteristic lines of the Bargmann transform is equivalent to (3.3).

Unperturbed reference. We first treat the unperturbed case $P \equiv 1$. By the theorem of Daubechies [15], there exists a disk D_0 centered at the origin such that

$$\int_{D_0} e^{-\pi|z|^2} e^{\pi\bar{z}w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}.$$

Equivalently, $\Delta u_* = -e^{-\pi|z|^2}\mathbf{1}_{D_0} + \lambda\,\delta_0$ admits a compactly supported radial solution u_*. This solution is unique among compactly supported distributional solutions: by radial symmetry, the Poisson equation reduces to the ODE

$$\frac{1}{r}\big(r u_*'(r)\big)' = -e^{-\pi r^2}\mathbf{1}_{(0,r_0)}(r), \qquad r > 0,$$

together with the asymptotic condition $u_*'(r) \sim \lambda/(2\pi r)$ as $r \to 0^+$ enforced by the distributional source $\lambda\delta_0$. Integrating twice with the natural boundary condition that u_* vanish for $r \geq r_0$ gives a unique radial profile. Explicitly,

$$u_*(r) = \frac{\lambda}{2\pi}\log\frac{r_0}{r} - \int_r^{r_0} \frac{1}{s}\int_0^s \rho\, e^{-\pi\rho^2}\, d\rho\, ds, \qquad 0 < r \leq r_0,$$

and $u_* \equiv 0$ for $r \geq r_0$, where r_0 is the radius of D_0. From this explicit formula one verifies directly:

- $-\Delta u_* = e^{-\pi|z|^2} > 0$ on $D_0 \setminus \{0\}$, so u_* is superharmonic on $D_0 \setminus \{0\}$;
- $u_* = 0$ on ∂D_0;
- the logarithmic singularity created by the mass $\lambda\delta_0$ has positive coefficient $\lambda/(2\pi)$, so $u_*(r) \to +\infty$ as $r \to 0^+$.

By the maximum principle for superharmonic functions, $u_* > 0$ throughout $D_0 \setminus \{0\}$, and in particular $u_* > 0$ on every compact subset of $D_0 \setminus \{0\}$.

For δ small enough that $\operatorname{supp}\psi_\delta \subset D_0$, the same argument yields a compactly supported radial solution $\widetilde{u}_0$ of

$$\Delta\widetilde{u}_0 = -e^{-\pi|z|^2}\mathbf{1}_{D_0} + \frac{\lambda}{c_\delta}\psi_\delta(z)e^{-\pi|z|^2}, \tag{3.6}$$

and $\widetilde{u}_0$ is uniquely determined among compactly supported radial distributional solutions by the same ODE reduction (the smooth radial source on the right-hand side replaces the Dirac mass, and the requirements of compact support and continuity through the origin pin down the integration constants).

We claim that, for δ sufficiently small,

$$\widetilde{u}_0 > 0 \qquad \text{in } D_0.$$

To see this, we first verify that

$$\frac{\lambda}{c_\delta}\,\psi_\delta(z)\,e^{-\pi|z|^2} \xrightarrow[\delta\to 0]{} \lambda\,\delta_0 \qquad \text{in } \mathcal{D}'(\mathbb{C}). \tag{3.7}$$

Indeed, for any $\varphi \in C_c^\infty(\mathbb{C})$,

$$\Big\langle \frac{\lambda}{c_\delta}\psi_\delta\, e^{-\pi|\cdot|^2}, \varphi\Big\rangle = \frac{\lambda}{c_\delta}\int_{\mathbb{C}} \psi_\delta(z)\, e^{-\pi|z|^2}\varphi(z)\, dA(z).$$

Decompose $\varphi(z)e^{-\pi|z|^2} = \varphi(0) + r(z)$ with r continuous and $r(0) = 0$; then

$$\frac{1}{c_\delta}\int_{\mathbb{C}} \psi_\delta(z)\, e^{-\pi|z|^2}\varphi(z)\, dA(z) = \varphi(0)\cdot\frac{1}{c_\delta}\int_{\mathbb{C}} \psi_\delta(z)e^{-\pi|z|^2}\, dA(z) + \frac{1}{c_\delta}\int_{\mathbb{C}} \psi_\delta(z)\, r(z)\, dA(z).$$

The first summand equals $\varphi(0)$. The second is bounded in absolute value by $c_\delta^{-1}\,\|\psi_\delta\|_{L^1}\,\sup_{|z|\le 2\delta}|r(z)| = c_\delta^{-1}\sup_{|z|\le 2\delta}|r(z)| \to 0$ as $\delta \to 0$, since r is continuous, $r(0) = 0$, and $c_\delta \to 1$. Multiplying by λ proves (3.7).

In particular, $\Delta\widetilde{u}_0 \to \Delta u_*$ in $\mathcal{D}'(\mathbb{C})$. To upgrade this to local uniform convergence, we use the Newtonian-potential representations of $\widetilde{u}_0$ and u_* together with the harmonic correction enforcing compact support: writing

$$\widetilde{u}_0(z) = \frac{1}{2\pi}\int_{\mathbb{C}} \log\frac{1}{|z-\zeta|}\Big(-e^{-\pi|\zeta|^2}\mathbf{1}_{D_0}(\zeta) + \frac{\lambda}{c_\delta}\psi_\delta(\zeta)e^{-\pi|\zeta|^2}\Big)dA(\zeta) + h_\delta(z),$$

$$u_*(z) = \frac{1}{2\pi}\int_{\mathbb{C}} \log\frac{1}{|z-\zeta|}\Big(-e^{-\pi|\zeta|^2}\mathbf{1}_{D_0}(\zeta)\Big)dA(\zeta) + \frac{\lambda}{2\pi}\log\frac{1}{|z|} + h_*(z),$$

where h_δ, h_* are harmonic functions chosen so as to enforce compact support, the contribution from $-e^{-\pi|\zeta|^2}\mathbf{1}_{D_0}$ cancels in the difference and one is left with

$$\widetilde{u}_0(z) - u_*(z) = \frac{\lambda}{2\pi}\Big(\int_{\mathbb{C}} \log\frac{1}{|z-\zeta|}\frac{1}{c_\delta}\psi_\delta(\zeta)e^{-\pi|\zeta|^2}\, dA(\zeta) - \log\frac{1}{|z|}\Big) + \big(h_\delta(z) - h_*(z)\big).$$

By dominated convergence applied to the logarithmic potential with kernel uniformly bounded on compact subsets of $\overline{D}_0 \setminus \{0\}$, together with (3.7), the displayed difference tends to zero locally uniformly on $\overline{D}_0 \setminus \{0\}$. Combined with interior elliptic regularity for $\Delta(\widetilde{u}_0 - u_*)$, which is harmonic on any compact subset of $D_0 \setminus \{0\}$ once δ is so small that $\operatorname{supp}\psi_\delta$ is contained in

any prescribed neighborhood of the origin, this gives

$$\widetilde{u}_0 \xrightarrow[\delta\to 0]{} u_* \qquad \text{locally uniformly on } \overline{D}_0 \setminus \{0\}.$$

Since u_* is strictly positive on every compact subset of $D_0 \setminus \{0\}$ and continuous up to ∂D_0 where it vanishes, this implies $\widetilde{u}_0 > 0$ on every fixed compact subset of $D_0 \setminus \{0\}$ for all δ small enough; the strict inequality $u_* > 0$ at interior points provides the room needed to absorb the local-uniform error. On the small annulus $\delta \le |z| \le 2\delta$, where ψ_δ is supported, the explicit profile near the origin shows that $\widetilde{u}_0$ is comparable to a truncated logarithm $-\log(|z|/r_0)$ shifted by an $O(1)$ smooth correction, and is therefore strictly positive. We conclude that $\widetilde{u}_0 > 0$ throughout D_0 for $\delta > 0$ small.

Perturbed problem via Isakov. We now invoke the perturbative free-boundary theorem of Isakov [25], given by Proposition 2.1, verifying its hypotheses carefully. Set

$$f_0(z) := e^{-\pi|z|^2}\Big(1 - \frac{\lambda}{c_\delta}\psi_\delta(z)\Big).$$

Since $\psi_\delta \in C_c^\infty(\mathbb{C})$, we have $f_0 \in C^\infty(\mathbb{R}^2)$, hence in particular $f_0 \in C^5(\mathbb{R}^2)$. If δ is chosen so small that $\operatorname{supp}\psi_\delta \subset D_0$, then on ∂D_0,

$$f_0(z) = e^{-\pi|z|^2} > 0.$$

Define

$$v_0 := -\widetilde{u}_0.$$

Then (3.6) becomes

$$-\Delta v_0 = f_0\,\mathbf{1}_{D_0} \qquad \text{in } \mathbb{C},$$

with $v_0 = 0$ on ∂D_0 and $v_0 < 0$ in D_0, by the positivity of $\widetilde{u}_0$ established above. Thus all hypotheses of Proposition 2.1 are satisfied with $\Omega_0 = D_0$.

For the perturbed source, define

$$f(z) := e^{-\pi|z|^2}\Big(|P(z)|^2 - \frac{\lambda}{c_\delta}\psi_\delta(z)\Big).$$

Since P is a polynomial, $f \in C^\infty(\mathbb{R}^2)$ as well. Moreover, for any fixed disk $\Omega' \supset D_0$, the $C^5(\Omega')$-norm of $f - f_0$ is controlled by the coefficients of $P - 1$: writing

$$|P(z)|^2 - 1 = \sum_{k=1}^{N} b_k z^k + \sum_{k=1}^{N} \overline{b_k}\,\overline{z}^{\,k} + \sum_{k,\ell=1}^{N} b_k\overline{b_\ell}\, z^k\overline{z}^{\,\ell},$$

which is a polynomial in $z, \overline{z}$ that vanishes when all $b_k = 0$ and depends polynomially on the coefficients, a standard estimate yields

$$\|f - f_0\|_{C^5(\Omega')} \le C(\Omega', N, \lambda) \max_{1\le k\le N} |b_k|.$$

Hence, if the coefficients b_k are sufficiently small, we may arrange that

$$\|f - f_0\|_{C^5(\Omega')} < \varepsilon_1,$$

where ε_1 is the smallness threshold furnished by Proposition 2.1. Applying that proposition, we obtain a C^2 domain U_0 and a function v such that

$$-\Delta v = f\,\mathbf{1}_{U_0} \qquad \text{in } \mathbb{C},$$

with $v = 0$ outside U_0 and $v < 0$ in U_0. Set

$$u_0 := -v.$$

Then

$$\Delta u_0 = f\,\mathbf{1}_{U_0}.$$

If the perturbation is sufficiently small, then U_0 is a small C^2 deformation of D_0, and since $\operatorname{supp}\psi_\delta \subset D_0$ we still have

$$\operatorname{supp}\psi_\delta \subset U_0.$$

Therefore

$$\Delta u_0 = -|P(z)|^2 e^{-\pi|z|^2}\mathbf{1}_{U_0} + \frac{\lambda}{c_\delta}\psi_\delta(z)e^{-\pi|z|^2},$$

so (3.5) holds. This proves (3.3), and completes Step 1: we have produced a C^2 domain U_0, lying in a small C^2-neighborhood of the Daubechies disk D_0, on which the symmetrized moment identity holds with constant right-hand side λ. We will use U_0 in Step 2 as the unperturbed reference for the application of Proposition 2.2.

Step 2: construction of a domain satisfying (3.2). With the auxiliary domain U_0 at our disposal, we now seek a C^2 domain U for which the right-hand side of the moment identity is the polynomial $\lambda Q(w)$ rather than the constant λ. The crucial observation is that integrating the parameter w along the positive real axis against an exponential weight converts (3.2) into an explicit identity for a weighted Cauchy transform; this is exactly the form to which Proposition 2.2 of Cherednichenko [13] applies. The reformulation we are about to perform is closely related to the quadrature-domain framework developed in [23, 41, 26, 40].

Restrict (3.2) to real values $w = t > 0$, multiply by $e^{-\pi\zeta t}$, and integrate in $t > 0$. If

$$\zeta > \sup_{z\in U}|z|,$$

the integrand of the resulting double integral is absolutely integrable: indeed, for real $\zeta > \sup_{z\in U}|z|$,

$$\Re(\overline{z} - \zeta) \le |z| - \zeta < 0 \qquad \text{for every } z \in U,$$

so

$$\int_U \int_0^\infty \left|e^{\pi(\overline{z}-\zeta)t}\right| dt\, |P(z)|^2 e^{-\pi|z|^2}\, dA(z) \le \int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\pi(\zeta - |z|)}\, dA(z) < \infty,$$

because U is bounded. Fubini's theorem therefore applies and yields

$$\int_U \left(\int_0^\infty e^{\pi(\overline{z}-\zeta)t}\, dt\right) |P(z)|^2 e^{-\pi|z|^2}\, dA(z) = \lambda \int_0^\infty e^{-\pi\zeta t} Q(t)\, dt.$$

Writing

$$Q(w) = \alpha_0 + \alpha_1 w + \cdots + \alpha_n w^n,$$

we obtain

$$\frac{1}{\pi}\int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta-\overline{z}}\,dA(z) = \lambda\sum_{k=0}^{n}\alpha_k\int_0^\infty e^{-\pi\zeta t}t^k\,dt.$$

Since

$$\int_0^\infty e^{-\pi\zeta t}t^k\,dt = \frac{\Gamma(k+1)}{(\pi\zeta)^{k+1}},$$

it follows that

$$\frac{1}{\pi}\int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta-\overline{z}}\,dA(z) = \lambda\sum_{k=0}^{n}\alpha_k\frac{\Gamma(k+1)}{\pi^{k+1}\zeta^{k+1}}.$$

To put this into the form of a weighted Cauchy transform, let

$$V := \overline{U}, \qquad \mu(w) := |P(\overline{w})|^2 e^{-\pi|w|^2}.$$

The change of variables $w = \overline{z}$ yields

$$\int_U \frac{|P(z)|^2 e^{-\pi|z|^2}}{\zeta-\overline{z}}\,dA(z) = \int_V \frac{\mu(w)}{\zeta-w}\,dA(w) = g_V(\zeta).$$

Hence

$$g_V(\zeta) = \lambda\sum_{k=0}^{n}\alpha_k\frac{\Gamma(k+1)}{\pi^k\zeta^{k+1}}, \qquad \zeta\in\mathbb{R}_+,\ \zeta\gg 1. \tag{3.8}$$

Since both sides are holomorphic for large ζ, analytic continuation gives the desired identity on $\mathbb{C}\setminus V$.

Conversely, suppose a domain V satisfies (3.8). Expanding $1/(\zeta-w) = \sum_{k\geq 0} w^k/\zeta^{k+1}$ for $|\zeta|$ large and matching coefficients yields the moments

$$\int_V w^k\mu(w)\,dA(w) = \lambda\,\alpha_k\,\frac{\Gamma(k+1)}{\pi^k}, \qquad 0\leq k\leq n,$$

together with the vanishing of the corresponding moments for $k > n$ (since Q has degree n). These moment identities recover (3.2) for the conjugate domain $U = \overline{V}$ via the Laplace-transform calculation read in reverse, in conjunction with the identity theorem for entire functions. Thus the conjugation $V = \overline{U}$ is merely a bookkeeping device that places the kernel in the form $1/(\zeta-w)$ required by the weighted Cauchy transform formalism.

Let g denote the rational function on the right-hand side of (3.8). We now verify the hypotheses of Proposition 2.2. Since

$$Q(w) = P(\partial_w/\pi)\big(P(w)\big),$$

and $P(0) = 1$, the constant term of Q is

$$Q(0) = 1 + \sum_{k=1}^{N}\frac{k!}{\pi^k}b_k^2.$$

In particular, if the coefficients b_k are sufficiently small, then

$$Q(0) > 0,$$

so the leading term in the expansion of g at infinity is

$$\frac{\lambda Q(0)}{z},$$

and consequently

$$g(z) \to 0, \qquad z\,g(z) \to \lambda Q(0) > 0 \qquad \text{as } z \to \infty.$$

Thus (2.16) holds, with $c_0 = \lambda Q(0)$.

We next verify the closeness condition on ∂U_1. Since U_0 was obtained in Step 1 as a small perturbation of the Daubechies disk D_0, we choose $U_1 \subset U_0 \subset U_2$ to be concentric disks (centered at the center of D_0) with radii, say, $(1-10^{-3})$ and $(1+10^{-3})$ times the radius of D_0. By Step 1,

$$\int_{U_0} |P(z)|^2 e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}.$$

Repeating the Laplace-transform argument above, with the constant λ in place of the polynomial $\lambda Q(w)$, gives

$$g_{U_0}(z) = \frac{\lambda}{z} \qquad \text{for } z \in \mathbb{C} \setminus \overline{U_0},$$

where g_{U_0} denotes the weighted Cauchy transform of the conjugate domain $\overline{U_0}$ with weight $\mu(w) = |P(\overline{w})|^2 e^{-\pi|w|^2}$. Consequently, on ∂U_1,

$$g(z) - g_{U_0}(z) = \lambda \sum_{k=1}^{n} \alpha_k \frac{\Gamma(k+1)}{\pi^k z^{k+1}} + \lambda\,(Q(0)-1)\,\frac{1}{z}.$$

Since ∂U_1 is a fixed circle staying a positive distance from the origin, it follows that

$$\sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)| \le C(U_1, \lambda)\Big(|Q(0)-1| + \sum_{k=1}^{n} |\alpha_k|\Big).$$

Each coefficient α_k of $Q-1$ depends polynomially on the coefficients of $P-1$, with no constant term, and therefore tends to zero with $\max_{1\le j\le N} |b_j|$. Hence, by choosing the coefficients sufficiently small, we may ensure that

$$\sup_{z \in \partial U_1} |g(z) - g_{U_0}(z)| < \varepsilon_0,$$

where ε_0 is the smallness threshold provided by Proposition 2.2. Applying that proposition, we obtain a C^2 domain $\widetilde{U}$ such that

$$g_{\widetilde{U}}(z) = g(z), \qquad z \in \mathbb{C} \setminus \widetilde{U}.$$

Replacing $\widetilde{U}$ by its conjugate if necessary, we obtain a domain U satisfying (3.2).

Finally, since $U \subset U_2$ and U_2 is fixed once λ is fixed, by shrinking the admissible size of the coefficients of $P-1$ once more we may ensure that

$$P(z) \neq 0 \qquad \text{for all } z \in U_2.$$

In particular,

$$P(z) \neq 0 \qquad \text{for all } z \in U.$$

This non-vanishing of P on U will play no direct role until Step 3, where it is needed to push the zeros of the characteristic polynomial $\widetilde{P}(\zeta) = P(\zeta/\pi)$ to large modulus.

Step 3: recovery of the original eigenvalue equation. It remains to upgrade the symmetrized identity (3.2) to the original equation (3.1). Define

$$G(w) := \int_U P(z)e^{-\pi|z|^2}e^{\pi\bar{z}w}\,dA(z), \qquad w \in \mathbb{C}.$$

Then G is entire (the integrand is entire in w, and the dominated-convergence justification of differentiation under the integral sign carried out in Step 0 applies verbatim with P in place of $|P|^2$). By construction, (3.2) holds, so applying $P(\partial_w/\pi)$ to G yields, by the same argument as in Step 0,

$$P(\partial_w/\pi)G(w) = \int_U |P(z)|^2 e^{-\pi|z|^2}e^{\pi\bar{z}w}\,dA(z) = \lambda Q(w) = P(\partial_w/\pi)(\lambda P)(w), \qquad w \in \mathbb{C}.$$

Hence

$$H := G - \lambda P$$

is an entire function satisfying the constant-coefficient linear differential equation

$$P(\partial_w/\pi)H = 0.$$

Introduce the characteristic polynomial of this operator,

$$\widetilde{P}(\zeta) := P(\zeta/\pi),$$

and let $\zeta_1, \dots, \zeta_\ell$ be its zeros, with multiplicities $m_1, \dots, m_\ell$. By the structure theorem for entire solutions of constant-coefficient linear differential equations of finite exponential type (see e.g. [24, Ch. XII]), every entire solution of

$$P(\partial_w/\pi)H = 0$$

of at most exponential type is of the form

$$H(w) = \sum_{j=1}^{\ell} R_j(w)\, e^{\zeta_j w}, \tag{3.9}$$

where each R_j is a polynomial of degree strictly less than m_j.

We will reach a contradiction by showing that all R_j must vanish. First, since P has no zeros on U_2, and the zeros of $\widetilde{P}(\zeta) = P(\zeta/\pi)$ are obtained by scaling those of P by the factor π, by choosing the coefficients of $P - 1$ sufficiently small we may guarantee that all zeros of $\widetilde{P}$ stay at distance at least $10\sup_{z\in U}|z|$ from the origin:

$$|\zeta_j| > 10\sup_{z\in U}|z| =: 10M, \qquad 1 \le j \le \ell.$$

(Indeed, the roots of $\widetilde{P}$ depend continuously on its coefficients, and at $b = 0$ the polynomial $\widetilde{P}$ degenerates to the constant 1, which has no roots; for small b the roots therefore lie in any prescribed neighborhood of infinity.)

Next, we establish an upper bound on H from its definition. Since P is a fixed polynomial of degree $\le N$ and U is bounded with $M = \sup_{z\in U}|z|$,

$$|G(w)| \le \int_U |P(z)|e^{-\pi|z|^2}e^{\pi|z||w|}\,dA(z) \le \Big(\sup_{z\in U}|P(z)|e^{-\pi|z|^2}\Big)\,|U|\,e^{\pi M|w|} \le C(1+|w|)^N e^{\pi M|w|}, \qquad w \in \mathbb{C},$$

where the first inequality uses $|e^{\pi\bar{z}w}| \leq e^{\pi|z||w|}$ and in the last inequality we have absorbed $\sup_{z\in U}|P(z)| \leq C_0(1+M)^N$ and the area $|U|$ into the constant C. Since every polynomial is dominated by $e^{\varepsilon|w|}$ for arbitrary $\varepsilon > 0$, it follows that for every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$|G(w)| \leq C_\varepsilon e^{(\pi+\varepsilon)M|w|}, \qquad w \in \mathbb{C}.$$

As $\lambda P(w)$ grows only polynomially in $|w|$, the same estimate holds for H:

$$|H(w)| \leq C'_\varepsilon e^{(\pi+\varepsilon)M|w|}.$$

In particular, H has finite exponential type at most πM, which justifies the use of (3.9).

Suppose, for contradiction, that some R_j is not identically zero. By a small generic rotation of the coordinate w if necessary, we may assume that the zeros ζ_j are arranged so that there is a unique ζ_1 realizing

$$|\zeta_1| = \max_j |\zeta_j| \qquad \text{and} \qquad R_1 \not\equiv 0,$$

and that, for every other zero ζ_j with $j \neq 1$,

$$\Re(\zeta_j\overline{\zeta_1}) < |\zeta_1|^2.$$

(The strict inequality $\Re(\zeta_j\overline{\zeta_1}) \leq |\zeta_j||\zeta_1| < |\zeta_1|^2$ holds automatically when $|\zeta_j| < |\zeta_1|$; ties at maximal modulus are broken by the rotation.) Then, along the ray

$$w = t\,\overline{\zeta_1}, \qquad t > 0,$$

the term $R_1(w)e^{\zeta_1 w}$ has modulus $|R_1(t\overline{\zeta_1})|e^{t|\zeta_1|^2}$, while every other term $R_j(w)e^{\zeta_j w}$ has modulus bounded by a polynomial in t times $e^{t\,\Re(\zeta_j\overline{\zeta_1})}$, with $\Re(\zeta_j\overline{\zeta_1}) < |\zeta_1|^2$. Hence the dominant exponential term yields

$$|H(t\overline{\zeta_1})| \geq c\,e^{t|\zeta_1|^2}$$

for all sufficiently large t, with $c > 0$.

Evaluating the upper bound on H at $w = t\overline{\zeta_1}$ gives

$$|H(t\overline{\zeta_1})| \leq C'_\varepsilon e^{(\pi+\varepsilon)Mt|\zeta_1|}.$$

Comparing with the lower bound $|H(t\overline{\zeta_1})| \geq c\,e^{t|\zeta_1|^2}$ and letting $t \to \infty$, we obtain

$$|\zeta_1|^2 \leq (\pi+\varepsilon)M|\zeta_1|, \qquad \text{i.e.,} \qquad |\zeta_1| \leq (\pi+\varepsilon)M.$$

Since $\varepsilon > 0$ is arbitrary, this gives

$$|\zeta_1| \leq \pi M,$$

contradicting our choice $|\zeta_1| > 10M$ (recall that $\pi < 10$). Hence every R_j vanishes identically, so

$$H \equiv 0.$$

Equivalently,

$$G = \lambda P,$$

which is exactly (3.1). This proves that P, and therefore f_0, is an eigenfunction of the localization operator associated with the domain U with eigenvalue λ. □

4. A Classical Rigidity Corollary

The localization operator $\mathcal{L}_U$ on the Bargmann–Fock space is diagonalized by Hermite functions whenever U is a disk centered at the origin: this is an immediate consequence of the rotational symmetry of U together with the radial structure of the Hermite weight. The converse question, raised by Abreu and Dörfler, asks whether eigenfunction status of even a *single* Hermite function already forces the underlying domain to be a disk. The next theorem records a positive answer in the simply connected C^2 regime.

For convenience we recall the statement of Theorem 2 from the introduction: *let $U \subset \mathbb{C}$ be a simply connected bounded domain with C^2 boundary, and let h_k be a Hermite function ($k \geq 0$). If h_k is an eigenfunction of the localization operator $\mathcal{L}_U$, then U is a disk.* (The original Abreu–Dörfler theorem [1] dispenses with the boundary regularity assumption; we will recover this stronger form in the second proof below.)

We will give two proofs of Theorem 2. The first, presented below, leverages the Fourier-analytic machinery developed earlier in the paper, reducing the question to a free boundary problem for a logarithmic potential. The second, in the next subsection, proceeds along more classical complex-analytic lines.

4.1. **First proof.**

First proof under the additional assumption that ∂U is C^2. The Bargmann transform of the Hermite function h_k is a nonzero scalar multiple of the monomial z^k. Thus, arguing as in [1], the assumption that h_k is an eigenfunction of the localization operator $\mathcal{L}_U$ is equivalent to the identity

$$\int_U z^k\, e^{\pi w \overline{z}}\, e^{-\pi|z|^2}\, dA(z) = \lambda\, w^k, \qquad w \in \mathbb{C}.$$

Both sides are entire functions of w. Since U is bounded and the weight $e^{-\pi|z|^2}$ is integrable, the dominated convergence theorem applied to difference quotients shows that for each multi-index α in w, the partial derivative ∂_w^α commutes with the integral on the left-hand side; the absolute integrability of $|z|^{|\alpha|+k} e^{\pi|w||z|} e^{-\pi|z|^2} \mathbf{1}_U$ for fixed w provides the required local majorant. Iterating k times, we obtain

$$\frac{1}{(\pi)^k}\, \partial_w^k \Big[\int_U z^k\, e^{\pi w \overline{z}}\, e^{-\pi|z|^2}\, dA(z) \Big] = \int_U z^k \overline{z}^k\, e^{\pi w \overline{z}}\, e^{-\pi|z|^2}\, dA(z) = \int_U |z|^{2k}\, e^{\pi w \overline{z}}\, e^{-\pi|z|^2}\, dA(z).$$

On the right-hand side, $\partial_w^k(\lambda w^k) = \lambda\, k!$. Setting $c(\lambda, k) := \lambda\, k!/\pi^k \in \mathbb{R}$ and using $\mathbf{1}_U$ to extend the integration to $\mathbb{C}$, we obtain

$$\int_{\mathbb{C}} \mathbf{1}_U(z)\, |z|^{2k}\, e^{-\pi|z|^2}\, e^{\pi \overline{z}\, w}\, dA(z) = c(\lambda, k), \tag{4.1}$$

for some real constant $c(\lambda, k)$.

Consider now the compactly supported distribution

$$\eta := |z|^{2k} e^{-\pi|z|^2} \mathbf{1}_U(z) - c(\lambda, k)\, \delta_0,$$

which has compact support in any closed ball containing $U \cup \{0\}$. Writing $z = x+iy$ and $w \in \mathbb{C}$, we may rewrite the exponential factor in (4.1) as

$$e^{\pi \overline{z}\, w} = e^{\pi(x-iy)w} = e^{\pi x w}\, e^{-i\pi y w}.$$

Hence (4.1) implies that the Fourier transform $\widehat{\eta}$, viewed as an entire function on $\mathbb{C}^2$ via Paley–Wiener, satisfies

$$\widehat{\eta}(w, -iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Since $c(\lambda, k) \in \mathbb{R}$, taking complex conjugates in (4.1) after replacing w by $\overline{w}$ also yields

$$\widehat{\eta}(w, iw) = 0, \qquad \forall w \in \mathbb{C}.$$

Applying Lemma 2.6 to the distribution η, we obtain a function $u \in L^2(\mathbb{R}^2)$ such that

$$\Delta u = f(|z|)\, \mathbf{1}_U - c(\lambda, k)\, \delta_0 \tag{4.2}$$

in the sense of distributions, where

$$f(|z|) = |z|^{2k} e^{-\pi |z|^2}.$$

Writing u explicitly as a logarithmic potential of the right-hand side of (4.2), we conclude that U satisfies

$$\int_U \log|z - z'|\, f(|z'|)\, dA(z') = c(\lambda, k)\, \log|z|, \qquad z \in \mathbb{C} \setminus (U \cup \{0\}). \tag{4.3}$$

Reduction to $0 \notin U$. Suppose first that $0 \in U$, and pick $\rho > 0$ small enough that $\overline{D_\rho(0)} \subset U$. The radial measure $f(|z|)\, \mathbf{1}_{D_\rho(0)}\, dA$ generates a logarithmic potential

$$\Phi_\rho(z) := \int_{D_\rho(0)} \log|z - z'|\, f(|z'|)\, dA(z'),$$

which, by Newton's shell theorem in the plane, is radial; outside $D_\rho(0)$ it equals

$$\Phi_\rho(z) = \Big(\int_{D_\rho(0)} f(|z'|)\, dA(z') \Big) \log|z| + \text{const}.$$

Subtracting $D_\rho(0)$ from U thus preserves identity (4.3) on the complement of the new set, at the cost of replacing the constant $c(\lambda, k)$ by another real constant. Hence we may assume $0 \notin U$. Moreover, $0 \notin \partial U$, because the left-hand side of (4.3) stays bounded as $z \to 0$, whereas the right-hand side diverges to $-\infty$ if $c(\lambda, k) \neq 0$. The case $c(\lambda, k) = 0$ is incompatible with the integral being positive at $z = 0$, so it is also excluded.

Define the logarithmic potential

$$\Psi_U(z) := \int_U \log|z - z'|\, f(|z'|)\, dA(z').$$

We claim that Ψ_U is *locally radial*: for each $z_0 \in U$, there exists a ball $B(z_0) \subset U$ such that Ψ_U agrees on $B(z_0)$ with a radial function of $|z|$.

To prove this, let $B \supset U$ be a ball centered at 0, and define

$$\Phi(z) := \int_B \log|z - z'|\, f(|z'|)\, dA(z').$$

Since both B and the weight $f(|z'|)$ are radial, Φ is radial; in particular, $\nabla\Phi(z)$ is parallel to z for every $z \in \mathbb{R}^2$. Furthermore, on the interior of B,

$$\Delta\Phi(z) = 2\pi\, f(|z|).$$

Now Ψ_U satisfies (4.3) on the unbounded component of $\mathbb{C} \setminus (U \cup \{0\})$. Differentiating that identity in z for $z \notin \overline{U} \cup \{0\}$ gives

$$\nabla\Psi_U(z) = c(\lambda, k)\, \frac{z}{|z|^2}, \qquad z \in \mathbb{C} \setminus (\overline{U} \cup \{0\}).$$

The classical jump relations for the single-layer / volume logarithmic potential of an L^∞ density on a C^2 domain (see, e.g., [23]) imply that $\nabla\Psi_U$ extends continuously across ∂U from *both* sides, and the boundary values agree. Hence on ∂U,

$$\nabla\Psi_U(z) = c(\lambda, k)\, \frac{z}{|z|^2}, \qquad z \in \partial U,$$

which is, in particular, parallel to z. Therefore the difference

$$v := \Psi_U - \Phi$$

satisfies $\Delta v = 2\pi f(|z|)\mathbf{1}_U - 2\pi f(|z|)\mathbf{1}_B = -2\pi f(|z|)\mathbf{1}_{B\setminus U}$ in the sense of distributions on B, so v is harmonic in U and in $B \setminus \overline{U}$, and on ∂U satisfies

$$\nabla v(x) \text{ is parallel to } x.$$

The next lemma converts this gradient condition into local radiality of v.

Lemma 4.1. *Let $V \subset \mathbb{R}^2$ be a domain, and suppose there exists a harmonic function $v \in C^1(\overline{V})$ such that $\nabla v(x)$ is parallel to x for every $x \in \partial V$. Then $\nabla v(x)$ is parallel to x for every $x \in V$. In particular, on every circle $S = \{|x| = r\} \subset \mathbb{R}^2$, the function v is constant on every connected component of $S \cap V$.*

Proof. Consider the angular derivative

$$g(x) := x_1\, \partial_2 v(x) - x_2\, \partial_1 v(x) = \partial_\theta v(x),$$

which equals the tangential derivative of v along the circle through x centered at 0. A direct computation gives

$$\Delta g = x_1\, \Delta(\partial_2 v) - x_2\, \Delta(\partial_1 v) + 2\big(\partial_1\partial_2 v - \partial_2\partial_1 v\big) = 0,$$

since v is harmonic and partial derivatives of harmonic functions are themselves harmonic. By the boundary hypothesis, $\nabla v(x)$ is parallel to x on ∂V, so $g \equiv 0$ on ∂V. Together with $g \in C(\overline{V})$, the maximum principle for harmonic functions yields $g \equiv 0$ in V. Hence $\nabla v(x)$ is parallel to x at every interior point as well, and on any circle $S = \{|x| = r\}$ the tangential derivative $\partial_\theta v$ vanishes on $S \cap V$. Therefore v is constant on each connected component of $S \cap V$. □

Applying Lemma 4.1 on small subdomains $V \subset U$ where v is harmonic and C^1 up to the boundary, we conclude that v is locally radial on U. Since Φ is globally radial, so is $\Psi_U = \Phi + v$, locally on U.

Lemma 4.2. *Let $W \subset \mathbb{R}^2 \setminus \{0\}$ be a connected open set, and let $u \in C(W)$. Suppose that for every $x \in W$ there exist $r_x > 0$ and a scalar function ϕ_x such that*

$$u(y) = \phi_x(|y|) \qquad \text{for all } y \in B(x, r_x) \subset W.$$

Then there exists a single scalar function ϕ, defined on the interval

$$I_W := \{|y| : y \in W\},$$

such that

$$u(y) = \phi(|y|) \qquad \text{for all } y \in W.$$

Proof. For each $x \in W$, choose a ball $B_x := B(x, r_x) \subset W$ on which u is a radial function of $|y|$, with profile ϕ_x. We first note that whenever $B_x \cap B_{x'} \neq \emptyset$,

$$\phi_x(|y|) = u(y) = \phi_{x'}(|y|) \qquad \text{for every } y \in B_x \cap B_{x'}.$$

Since $B_x \cap B_{x'}$ is open and avoids the origin, the map $y \mapsto |y|$ is a submersion on it, so the set of radii it attains contains a nontrivial open interval on which ϕ_x and $\phi_{x'}$ agree.

Fix $x_* \in W$ and define a candidate function ϕ on I_W as follows. Given $y \in W$, connect x_* to y by a polygonal path in W, which exists because W is open and connected. Its image is compact, hence covered by finitely many balls $B_{x_1}, \dots, B_{x_m}$ with successive nonempty overlaps and $x_1 = x_*$, $y \in B_{x_m}$. The previous paragraph implies ϕ_{x_j} and $\phi_{x_{j+1}}$ agree on a nontrivial interval of radii, so by analytic-continuation along the chain we may unambiguously set $\phi(|y|) := \phi_{x_m}(|y|)$. Independence of the chain follows from the same overlap argument: any two chains from x_* to y can be interleaved into a single chain, and successive overlaps force the final value to coincide. Hence ϕ is well-defined on I_W and satisfies $u(y) = \phi(|y|)$ for every $y \in W$. □

We now conclude the proof. Since U is simply connected and bounded, it suffices to rule out the possibility that U fails to be a disk centered at the origin. Suppose therefore, for contradiction, that U is not such a disk. Then ∂U is not a single circle centered at 0, so there exist radii $0 < r_1 < r_2$ such that the annulus

$$A := \{z \in \mathbb{R}^2 : \ r_1 < |z| < r_2\}$$

meets both U and ∂U in a non-trivial way; in particular, we may pick a point

$$x_0 \in \partial U \cap A.$$

Let

$$\widetilde{U}_0 := U \cap B_r(x_0),$$

for $r > 0$ sufficiently small that $\widetilde{U}_0$ is connected and $0 \notin \overline{\widetilde{U}_0}$. Since Ψ_U is locally radial in U, Lemma 4.2 applied to $W = \widetilde{U}_0$ yields a scalar function ψ_U on $I_{\widetilde{U}_0}$ such that

$$\Psi_U(x) = \psi_U(|x|), \qquad x \in \widetilde{U}_0.$$

On the other hand, by (4.3) and standard regularity for logarithmic potentials with C^2 boundary, the gradient $\nabla\Psi_U$ extends continuously across ∂U, and for $z \in \partial U$ we have

$$\nabla\Psi_U(z) = c\,\frac{z}{|z|^2}$$

for the same constant $c = c(\lambda, k) \in \mathbb{R}$ as in (4.3). Combining this with $\Psi_U(x) = \psi_U(|x|)$ on $\widetilde{U}_0$ and passing to $z \in \partial\widetilde{U}_0 \cap \partial U$ from the inside,

$$\psi_U'(|z|)\,\frac{z}{|z|} = c\,\frac{z}{|z|^2}, \qquad z \in \partial U \cap \overline{B_r(x_0)},$$

hence

$$\psi_U'(r) = \frac{c}{r}$$

for every r in the set of radii attained by points of $\partial U \cap \overline{B_r(x_0)}$. Since ∂U is a C^2 curve, we may, after shrinking r once more, parametrize $\partial U \cap B_r(x_0)$ by a C^2 embedding $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}^2$ with $\gamma(0) = x_0$. Because ∂U is not a circle centered at 0, the function $t \mapsto |\gamma(t)|$ is not constant on any subinterval; by choosing $x_0 \in \partial U \cap A$ at which $|\gamma|$ is non-constant on every neighbourhood (such points form an open dense subset of $\partial U \cap A$ when ∂U is not a circle centered at 0), we may further assume that the derivative $\frac{d}{dt}|\gamma(t)|\big|_{t=0} \neq 0$. Hence by the open-mapping property for C^1 functions of one variable with nonzero derivative, the image of $|\gamma|$ contains a nontrivial open interval $I \subset (r_1, r_2)$. Therefore

$$\psi_U'(r) = \frac{c}{r} \qquad \text{for every } r \in I,$$

whence

$$\psi_U(r) = c\log r + C_0, \qquad r \in I,$$

for some real constant C_0.

Let

$$W_I := \{x \in \widetilde{U}_0 : \ |x| \in I\}.$$

Then W_I is a nonempty open subset of U, and on W_I,

$$\Psi_U(x) = c\log|x| + C_0.$$

Hence

$$\Delta\Psi_U \equiv 0 \qquad \text{in } W_I,$$

since the origin is excluded from $\widetilde{U}_0$. On the other hand, distribution-theoretically,

$$\Delta\Psi_U = 2\pi\, f(|z|)\,\mathbf{1}_U - c\,\delta_0$$

in $\mathbb{C}$, so on the open set $W_I \subset U \setminus \{0\}$ we have

$$\Delta\Psi_U = 2\pi\, f(|z|) = 2\pi\,|z|^{2k} e^{-\pi|z|^2}.$$

The right-hand side vanishes only at the origin, which lies outside W_I, while the left-hand side just shown to be 0 on W_I — a contradiction. This rules out the case in which U is not a disk centered at 0, completing the proof. □

Remark 4.1. The structure of the argument above is in the spirit of [3], where logarithmic-potential rigidity statements were derived from Schiffer-style overdetermined boundary conditions, and of the quadrature-domain literature [23, 41, 40, 26], in which the regularity of free boundaries and the matching of an interior potential with the Newtonian potential of a radially symmetric density play a central role. Concretely, identity (4.3) can be read as saying that U is a *generalized quadrature domain* for the radial weight $f(|z|)$, with quadrature data concentrated at the origin. From this perspective, the conclusion that U is a disk is a (sharp) instance of the classical principle that radial quadrature data forces a radial domain. The role of the C^2 boundary regularity is concentrated in the boundary extension of $\nabla\Psi_U$ and in the open-interval-of-radii argument that ends the proof; both can be relaxed at the cost of additional technicalities, which we will revisit in the second proof.

4.2. **Second proof.**

Second proof of Theorem 2. We keep the notation

$$f(|z|) := |z|^{2k}e^{-\pi|z|^2}, \qquad c := c(\lambda, k),$$

from the first proof. The argument below is in the spirit of the holomorphic moving-plane/quadrature-domain methods of [3, 23, 26], in that we exploit the existence of a holomorphic function on U with prescribed real boundary values to recover full radial symmetry. Starting again from (4.1) and applying Lemma 2.6, we obtain a compactly supported distributional solution of

$$\Delta u = f(|z|)\mathbf{1}_U - c\,\delta_0 \qquad \text{in } \mathcal{D}'(\mathbb{R}^2). \tag{4.4}$$

Using the planar fundamental solution $(2\pi)^{-1}\log|z|$, we may write

$$u(z) := \frac{1}{2\pi}\int_U \log|z-z'|\, f(|z'|)\, dA(z') - \frac{c}{2\pi}\log|z|. \tag{4.5}$$

By (4.3), this function satisfies

$$u \equiv 0 \qquad \text{on } \mathbb{R}^2 \setminus (U \cup \{0\}).$$

We first show that $0 \in U$. Since the singular term in (4.4) is $-c\delta_0$, one must have $0 \in \overline{U}$. If $0 \in \partial U$, then there exist exterior points $z_j \to 0$ with

$$u(z_j) = 0.$$

On the other hand, the first term in (4.5) remains finite as $z \to 0$, because $f \in L^\infty(U)$ and $\log|z-z'|$ is locally integrable, whereas the second term equals $-\frac{c}{2\pi}\log|z| \to +\infty$. This contradicts the exterior identity $u(z_j) = 0$. Hence $0 \notin \partial U$, and therefore

$$0 \in U.$$

We next show directly from (4.5) that

$$u \in C^1(\mathbb{R}^2 \setminus \{0\}).$$

Indeed, if we write

$$g(z) := f(|z|)\mathbf{1}_U(z) \in L^\infty_c(\mathbb{R}^2),$$

then

$$u(z) = \frac{1}{2\pi} \int_{\mathbb{R}^2} \log|z - z'|\, g(z')\, dA(z') - \frac{c}{2\pi} \log|z|.$$

Since the kernel $K(\zeta) = \zeta/|\zeta|^2$ belongs to $L^1_{\mathrm{loc}}(\mathbb{R}^2)$ (note that $|K(\zeta)| = |\zeta|^{-1}$, which is integrable on bounded subsets of $\mathbb{R}^2$), the translation map $z \mapsto K(z - \cdot)$ is continuous in $L^1_{\mathrm{loc}}(\mathbb{R}^2)$ by the standard density argument: continuous compactly supported functions are dense in L^1_{loc}, and for each such function continuity of translations is immediate. Together with $g \in L^\infty_c(\mathbb{R}^2)$, this implies that the convolution

$$(K * g)(z) := \int_{\mathbb{R}^2} \frac{z - z'}{|z - z'|^2}\, g(z')\, dA(z')$$

is a continuous function of $z \in \mathbb{R}^2$. On the other hand, a direct distributional computation shows

$$\nabla\left(\int_{\mathbb{R}^2} \log|z - z'|\, g(z')\, dA(z')\right) = K * g \qquad \text{in } \mathcal{D}'(\mathbb{R}^2),$$

so the gradient of the convolution $\log|\cdot| * g$ coincides almost everywhere with the continuous function $K * g$. It follows that $\log|\cdot| * g$ is of class C^1 on $\mathbb{R}^2$, and hence $u \in C^1(\mathbb{R}^2 \setminus \{0\})$ (the only obstruction to global C^1 regularity being the explicit logarithmic singularity at the origin coming from the $-\frac{c}{2\pi}\log|z|$ term).

Now let $z_* \in \partial U$. Since $0 \in U$, the boundary point z_* is bounded away from 0; fix a small ball $B_\rho(z_*) \Subset \mathbb{R}^2 \setminus \{0\}$. On the exterior open set $\mathbb{R}^2 \setminus \overline{U}$ we have $u \equiv 0$, so $\nabla u \equiv 0$ there. As ∂U is contained in the boundary of the exterior, every $z_* \in \partial U$ is a limit of exterior points $z_j \to z_*$ along which $\nabla u(z_j) = 0$. By the C^1 regularity of u on $B_\rho(z_*)$, the gradient ∇u is continuous at z_*, and we conclude $\nabla u(z_*) = 0$. In complex notation $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$, this yields

$$\partial_z u = 0 \qquad \text{on } \partial U. \tag{4.6}$$

We emphasize that *no* boundary regularity of ∂U has been used at this point, in contrast with the first proof: the vanishing of ∇u on ∂U is forced solely by exterior continuity, which in turn is supplied by the explicit formula (4.5).

Define

$$\Phi(t) := \int_0^t s^k e^{-\pi s}\, ds \qquad \text{and} \qquad \Psi(z) := \frac{\Phi(|z|^2)}{z}, \quad z \neq 0.$$

A direct computation, using $|z|^2 = z\bar{z}$ and $\partial_{\bar{z}}(z\bar{z}) = z$, gives, for $z \neq 0$,

$$\partial_{\bar{z}}\Psi(z) = \frac{\Phi'(|z|^2)\, \partial_{\bar{z}}(|z|^2)}{z} = \frac{|z|^{2k} e^{-\pi|z|^2} \cdot z}{z} = |z|^{2k} e^{-\pi|z|^2} = f(|z|).$$

At the origin, the apparent singularity $1/z$ is integrable against the vanishing factor $\Phi(|z|^2)$: since $k \geq 0$ and $\Phi(t) = \int_0^t s^k e^{-\pi s}\, ds = O(t^{k+1})$ as $t \to 0$, we have $\Psi(z) = O(|z|^{2k+1})$ near $z = 0$, so $\Psi \in L^\infty_{\mathrm{loc}}$ (in fact continuous, with $\Psi(0) = 0$). Consequently, the identity $\partial_{\bar{z}}\Psi = f(|z|)$ persists in the distributional sense on all of U; no extra δ_0 contribution arises at the origin.

Recall also the standard identity

$$\partial_{\bar{z}}\left(\frac{1}{z}\right) = \pi\delta_0 \qquad \text{in } \mathcal{D}'(\mathbb{R}^2),$$

which is equivalent to the fact that $\frac{1}{\pi z}$ is a fundamental solution of $\partial_{\bar z}$; equivalently, $\Delta \log|z| = 4\partial_z\partial_{\bar z}\log|z| = 2\pi\delta_0$ combined with $\partial_z \log|z| = \frac{1}{2z}$. Now set

$$F(z) := 4\partial_z u(z) - \Psi(z) + \frac{c}{\pi z}, \qquad z \in U \setminus \{0\}.$$

Using $\Delta = 4\partial_z\partial_{\bar z}$, the relation $\partial_{\bar z}(1/z) = \pi\delta_0$, the computation of $\partial_{\bar z}\Psi$ above, and (4.4), we compute, in $\mathcal{D}'(U)$,

$$\partial_{\bar z}F = 4\partial_z\partial_{\bar z}u - \partial_{\bar z}\Psi + \frac{c}{\pi}\partial_{\bar z}\left(\frac{1}{z}\right) = \Delta u - f(|z|) + c\,\delta_0 = \big(f(|z|) - c\delta_0\big) - f(|z|) + c\delta_0 = 0.$$

Hence F is a distributional solution of the Cauchy–Riemann equation on U, and by Weyl's lemma (or, equivalently, hypoellipticity of $\partial_{\bar z}$) it is represented by a genuine holomorphic function on U; the apparent pole of Ψ and of $c/(\pi z)$ at the origin is exactly cancelled by the contribution of the singular term $-c\delta_0$ in (4.4) arriving through $4\partial_z\partial_{\bar z}u$. Thus F is holomorphic on all of U.

Define

$$W(z) := \pi z\, F(z).$$

Then W is holomorphic on U and $W(0) = 0$. Since $\partial_z u$ is continuous up to ∂U, so is W, and by (4.6),

$$W|_{\partial U} = \pi z\left(4\partial_z u - \frac{\Phi(|z|^2)}{z} + \frac{c}{\pi z}\right)\Big|_{\partial U} = c - \pi\Phi(|z|^2).$$

In particular, $W|_{\partial U}$ is real-valued. The imaginary part of W is therefore harmonic in U, continuous up to ∂U, and vanishes on ∂U. By the maximum principle,

$$\Im W \equiv 0 \qquad \text{in } U.$$

Since a holomorphic function with zero imaginary part is constant, there exists $\kappa \in \mathbb{R}$ such that

$$W \equiv \kappa \qquad \text{in } U.$$

Restricting to the boundary, and using that W is continuous up to ∂U with boundary value $c - \pi\Phi(|z|^2)$, we obtain

$$c - \pi\Phi(|z|^2) = \kappa \qquad \text{for all } z \in \partial U.$$

The function $t \mapsto \Phi(t) = \int_0^t s^k e^{-\pi s}\,ds$ has derivative $\Phi'(t) = t^k e^{-\pi t} > 0$ for $t > 0$, hence is strictly increasing on $[0,\infty)$. The identity above therefore forces $|z|^2$ to take a single value on ∂U, i.e. $|z|$ is constant on ∂U. Consequently ∂U is contained in a circle centered at the origin. Let $R > 0$ be such that $\partial U \subset \{|z| = R\}$. Since U is open and connected, it cannot meet both $\{|z| < R\}$ and $\{|z| > R\}$, because these are disjoint open sets and $U \cap \{|z| = R\} = \varnothing$. As U is bounded, we must therefore have $U \subset \{|z| < R\}$. If this inclusion were strict, then $\{|z| < R\} \setminus U$ would contain a point, and hence, being open, would contribute a boundary point of U lying strictly inside the disk $\{|z| < R\}$, contradicting $\partial U \subset \{|z| = R\}$. Therefore

$$U = \{|z| < R\}$$

that is, U is a disk. □

Remark 4.2. The two proofs use the boundary assumptions in rather different ways, and it is instructive to keep careful track of the regularity bookkeeping.

In the first proof, once the logarithmic potential Ψ_U is written explicitly, no elliptic boundary regularity is needed to conclude that $\Psi_U \in C^1(\mathbb{R}^2 \setminus \{0\})$: this follows directly from convolution with the locally integrable kernel $\nabla \log |z|$, exactly as in the L^1_{loc}-translation argument used here for (4.5). Thus the local radiality argument itself only uses continuity of the gradient up to the boundary. The genuine extra input appears only in the final geometric step, where one needs that, near a suitable point $x_0 \in \partial U$, the set $\partial U \cap B_r(x_0)$ contains a nontrivial boundary arc, so that the radii attained there form an interval. Accordingly, the first proof does not really need C^2 regularity; it works under any hypothesis guaranteeing such a local arc structure, for instance when U is a Jordan domain. In particular, any Lipschitz, C^1, or C^2 boundary is more than sufficient.

By contrast, the second proof does not use any boundary regularity at all. The explicit formula (4.5) already yields $u \in C^1(\mathbb{R}^2 \setminus \{0\})$, and since $u \equiv 0$ on the exterior open set $\mathbb{R}^2 \setminus \overline{U}$, continuity implies that both u and ∇u vanish on ∂U – precisely the boundary information packaged in (4.6). This is exactly the input needed to construct the holomorphic function $W = \pi z F$ and conclude that $|z|$ is constant on ∂U. Hence the second proof recovers the theorem under the original assumptions of Abreu and Dörfler [1], namely for arbitrary bounded simply connected domains, with no boundary smoothness hypothesis. The mechanism is also conceptually closer to the classical quadrature-domain rigidity results of [3, 23, 26], where the existence of a holomorphic function on U with real boundary values forces the boundary to lie on a level set of a real-analytic function, here the elementary radial function $z \mapsto |z|^2$.

5. Proof of Theorem 3

We prove that the quantitative stability estimate in Corollary 1.4 of [20], which builds on the rigidity result of [33] characterizing the disk as the unique optimizer of the first localization eigenvalue, is sharp. The proof proceeds by constructing a family of domains which are perturbations of the disk, whose first localization eigenvalue has the same first-order behaviour as the disk, but whose distance to the class of disks is of order ε.

Proof of Theorem 3. Fix

$$\lambda_0 := 1 - e^{-\pi},$$

which is the eigenvalue of the localization operator of the unit disk D_1 associated with the ground state h_0. For ε small, define

$$P_\varepsilon(w) := 1 + \varepsilon w^2.$$

By Theorem 1, applied with $\lambda = \lambda_0$ and $P_\varepsilon(w) = 1 + \varepsilon w^2$, for all sufficiently small $|\varepsilon|$ there exists a domain U_ε such that

$$\int_{U_\varepsilon} P_\varepsilon(z) e^{\pi \bar{z} w} e^{-\pi |z|^2}\, dA(z) = \lambda_0 P_\varepsilon(w), \qquad w \in \mathbb{C}. \tag{5.1}$$

Moreover, by construction, U_ε is a C^2 perturbation of the unit disk. Hence, after possibly shrinking the range of admissible ε, its boundary can be written as

$$\partial U_\varepsilon = \{(1 + \varphi_\varepsilon(y))y : \ y \in \mathbb{S}^1\}, \tag{5.2}$$

where

$$\varphi_\varepsilon \in C^2(\mathbb{S}^1), \qquad \|\varphi_\varepsilon\|_{C^2(\mathbb{S}^1)} \lesssim |\varepsilon|.$$

The map

$$\varepsilon \mapsto \varphi_\varepsilon$$

may in fact be chosen C^1 as a map from a neighborhood of 0 into $C^1(\mathbb{S}^1)$. To see this, we briefly retrace the proof of Theorem 1. The construction proceeds in two steps: first, an Isakov-type step which converts the weighted quadrature identity into a free boundary problem of obstacle type, and second, the regularity/uniqueness step which produces φ_ε via an implicit-function-theorem argument. In both steps, the unknown is the solution of an equation of the form $F(\varepsilon, \varphi) = 0$, where

$$F : (-\varepsilon_0, \varepsilon_0) \times \mathcal{U} \to \mathcal{V}$$

is a C^1 map between open subsets of suitable Banach spaces, with $F(0,0) = 0$ and $\partial_\varphi F(0,0)$ invertible (the invertibility being the linearization computed at the disk, which is a Fredholm operator of index zero whose kernel is removed by the normalization fixing the centroid and the total area at first order). Since the polynomial $P_\varepsilon(z) = 1 + \varepsilon z^2$ depends polynomially on ε, the dependence of F on ε is C^∞. The C^1 version of the implicit function theorem in Banach spaces then yields a C^1 branch $\varepsilon \mapsto \varphi_\varepsilon$ into $C^1(\mathbb{S}^1)$. We therefore write

$$\dot\varphi_0(y) := \partial_\varepsilon|_{\varepsilon=0}\, \varphi_\varepsilon(y).$$

Step 1: first variation of the eigenvalue identity. We differentiate (5.1) at $\varepsilon = 0$. Since $P_\varepsilon \to 1$ in C^∞ and $\varepsilon \mapsto \varphi_\varepsilon$ is C^1 into $C^1(\mathbb{S}^1)$, all terms in (5.1) are differentiable at $\varepsilon = 0$. Since U_ε is given by a normal graph over $\mathbb{S}^1$, we apply Reynolds' transport formula. Concretely, in polar coordinates $z = re^{it}$, for any smooth test function H we have

$$\int_{U_\varepsilon} H(z)\, dA(z) = \int_0^{2\pi} \int_0^{1+\varphi_\varepsilon(e^{it})} H(re^{it})\, r\, dr\, dt.$$

Differentiating in ε at $\varepsilon = 0$ and using the fundamental theorem of calculus,

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \int_{U_\varepsilon} H(z)\, dA(z) = \int_0^{2\pi} \dot\varphi_0(e^{it}) H(e^{it})\, dt = \int_{\mathbb{S}^1} \dot\varphi_0(z) H(z)\, d\mathcal{H}^1(z),$$

where the factor $r = 1$ on the unit circle has been absorbed into the arc-length measure. This is precisely Reynolds' transport identity:

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \int_{U_\varepsilon} H(z)\, dA(z) = \int_{\mathbb{S}^1} \dot\varphi_0(z) H(z)\, d\mathcal{H}^1(z).$$

Applying this to

$$H(z) = e^{\pi \bar z w} e^{-\pi|z|^2},$$

and differentiating the factor $P_\varepsilon(z) = 1 + \varepsilon z^2$, we obtain

$$\int_{\mathbb{S}^1} \dot\varphi_0(z) e^{\pi \bar z w} e^{-\pi|z|^2}\, d\mathcal{H}^1(z) + \int_{D_1} z^2 e^{\pi \bar z w} e^{-\pi|z|^2}\, dA(z) = \lambda_0 w^2. \tag{5.3}$$

Since D_1 is radial, the monomial z^2 is an eigenfunction of the localization operator of D_1; therefore there exists $\lambda_2 > 0$ such that

$$\int_{D_1} z^2 e^{\pi \overline{z} w} e^{-\pi |z|^2}\, dA(z) = \lambda_2 w^2.$$

Hence (5.3) becomes

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \overline{z} w} e^{-\pi |z|^2}\, d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \tag{5.4}$$

In fact, λ_2 can be computed explicitly. Expanding $e^{\pi \overline{z} w} = \sum_{n \geq 0} \frac{\pi^n}{n!} \overline{z}^n w^n$ and using orthogonality of the monomials $\overline{z}^n$ over the disk, the only term contributing to w^2 is the one with $n = 2$:

$$\int_{D_1} z^2 e^{\pi \overline{z} w} e^{-\pi |z|^2}\, dA(z) = \frac{\pi^2}{2} w^2 \int_{D_1} z^2 \overline{z}^2 e^{-\pi |z|^2}\, dA(z) = \frac{\pi^2}{2} w^2 \int_{D_1} |z|^4 e^{-\pi |z|^2}\, dA(z).$$

Passing to polar coordinates and integrating by parts twice,

$$\int_{D_1} |z|^4 e^{-\pi |z|^2}\, dA(z) = 2\pi \int_0^1 r^4 e^{-\pi r^2}\, r\, dr = \pi \int_0^1 s^2 e^{-\pi s}\, ds$$

where we substituted $s = r^2$. A direct computation, integrating by parts twice, yields

$$\pi \int_0^1 s^2 e^{-\pi s}\, ds = \frac{2}{\pi^2} \Big(1 - \Big(1 + \pi + \frac{\pi^2}{2} \Big) e^{-\pi} \Big).$$

Multiplying by $\frac{\pi^2}{2}$, we obtain

$$\lambda_2 = \frac{\pi^2}{2} \int_{D_1} |z|^4 e^{-\pi |z|^2}\, dA(z) = 1 - \Big(1 + \pi + \frac{\pi^2}{2} \Big) e^{-\pi}.$$

Therefore

$$\lambda_0 - \lambda_2 = (1 - e^{-\pi}) - \Big(1 - \Big(1 + \pi + \frac{\pi^2}{2} \Big) e^{-\pi} \Big) = \pi \Big(1 + \frac{\pi}{2} \Big) e^{-\pi} > 0,$$

so the coefficient of w^2 on the right-hand side is indeed nonzero. In particular, since $|z| = 1$ on $\mathbb{S}^1$, we may rewrite this as

$$e^{-\pi} \int_{\mathbb{S}^1} \dot{\varphi}_0(z) e^{\pi \overline{z} w}\, d\mathcal{H}^1(z) = (\lambda_0 - \lambda_2) w^2. \tag{5.5}$$

Setting $w = 0$ gives

$$\int_{\mathbb{S}^1} \dot{\varphi}_0\, d\mathcal{H}^1 = 0. \tag{5.6}$$

Step 2: identification of the leading mode. Expanding

$$e^{\pi \overline{z} w} = \sum_{n \geq 0} \frac{\pi^n}{n!} \overline{z}^n w^n,$$

and comparing coefficients in (5.5), we find

$$\int_{\mathbb{S}^1} \dot{\varphi}_0(z) \overline{z}^n\, d\mathcal{H}^1(z) = 0 \qquad \text{for all } n \neq 2,$$

whereas the coefficient corresponding to $n = 2$ is nonzero. Since $\dot{\varphi}_0$ is real-valued, it follows that

$$\dot{\varphi}_0(e^{it}) = a \cos(2t) + b \sin(2t) \tag{5.7}$$

for some pair $(a,b) \neq (0,0)$. In particular,

$$\int_{\mathbb{S}^1} |\dot{\varphi}_0| \, d\mathcal{H}^1 > 0. \tag{5.8}$$

Step 3: normalization of the area. The area of U_ε is given by

$$|U_\varepsilon| = \frac{1}{2} \int_{\mathbb{S}^1} (1 + \varphi_\varepsilon(y))^2 \, d\mathcal{H}^1(y). \tag{5.9}$$

Since

$$\varphi_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1),$$

and (5.6) gives the vanishing of the first-order term, we obtain from (5.9)

$$|U_\varepsilon| = \pi + O(\varepsilon^2). \tag{5.10}$$

Let

$$r_\varepsilon := \sqrt{\frac{\pi}{|U_\varepsilon|}}, \qquad \widetilde{U}_\varepsilon := r_\varepsilon U_\varepsilon.$$

Then $|\widetilde{U}_\varepsilon| = \pi$, and by (5.10),

$$r_\varepsilon = 1 + O(\varepsilon^2).$$

Hence $\widetilde{U}_\varepsilon$ still admits a normal graph representation

$$\partial \widetilde{U}_\varepsilon = \{(1 + \widetilde{\varphi}_\varepsilon(y))y : \ y \in \mathbb{S}^1\},$$

with

$$\widetilde{\varphi}_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1).$$

Thus the area normalization does not alter the first-order deformation.

Step 4: distance from the centered disk. Since both $\widetilde{U}_\varepsilon$ and D_1 are star-shaped with respect to the origin, the symmetric difference may be computed in polar coordinates:

$$|\widetilde{U}_\varepsilon \triangle D_1| = \frac{1}{2} \int_{\mathbb{S}^1} \left|(1 + \widetilde{\varphi}_\varepsilon(y))^2 - 1\right| \, d\mathcal{H}^1(y).$$

Since $\widetilde{\varphi}_\varepsilon = \varepsilon \dot{\varphi}_0 + O(\varepsilon^2)$ in $C^1(\mathbb{S}^1)$, we have in particular $\|\widetilde{\varphi}_\varepsilon\|_{L^\infty(\mathbb{S}^1)} \lesssim |\varepsilon|$, whence $\widetilde{\varphi}_\varepsilon^2 = O(\varepsilon^2)$ in $L^\infty(\mathbb{S}^1)$. Hence

$$(1 + \widetilde{\varphi}_\varepsilon)^2 - 1 = 2\widetilde{\varphi}_\varepsilon + \widetilde{\varphi}_\varepsilon^2 = 2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2) \quad \text{in } L^\infty(\mathbb{S}^1),$$

where the implied constant depends only on the C^1-norm bound on $\widetilde{\varphi}_\varepsilon$. Therefore

$$|\widetilde{U}_\varepsilon \triangle D_1| = \frac{1}{2} \int_{\mathbb{S}^1} \left|2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)\right| \, d\mathcal{H}^1.$$

Set $\alpha := \int_{\mathbb{S}^1} |\dot{\varphi}_0| \, d\mathcal{H}^1$. By (5.8), $\alpha > 0$. The reverse triangle inequality in L^1 and the $O(\varepsilon^2)$ remainder give

$$\int_{\mathbb{S}^1} \left|2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)\right| \, d\mathcal{H}^1 \geq 2|\varepsilon|\alpha - O(\varepsilon^2)$$

and similarly the upper bound

$$\int_{\mathbb{S}^1} \left|2\varepsilon \dot{\varphi}_0 + O(\varepsilon^2)\right| \, d\mathcal{H}^1 \leq 2|\varepsilon|\alpha + O(\varepsilon^2).$$

Consequently there exist constants $c, C > 0$ such that

$$c|\varepsilon| \le |\widetilde{U}_\varepsilon \triangle D_1| \le C|\varepsilon| \tag{5.11}$$

for all sufficiently small ε.

Step 5: comparison with arbitrary disks. Let D be any disk of area π. Then $D = D_1 + a$ for some $a \in \mathbb{C}$. We first consider disks whose centers are not too close to the origin. Since

$$|D_1 \triangle (D_1 + a)| \gtrsim \min\{|a|, 1\},$$

the triangle inequality and (5.11) imply that there exists $K \gg 1$ such that, if $|a| \ge K|\varepsilon|$, then

$$|\widetilde{U}_\varepsilon \triangle D| \ge |D_1 \triangle D| - |\widetilde{U}_\varepsilon \triangle D_1| \gtrsim |a| - C|\varepsilon| \gtrsim |\varepsilon|. \tag{5.12}$$

It remains to consider disks whose centers satisfy $|a| \le K|\varepsilon|$. Write

$$a = \eta\varepsilon e^{i\theta}, \qquad |\eta| \le K, \quad \theta \in [0, 2\pi).$$

We claim that, for ε sufficiently small, the translated domain

$$\widetilde{U}_\varepsilon - a$$

is still a normal graph over $\mathbb{S}^1$:

$$\partial(\widetilde{U}_\varepsilon - a) = \{(1 + \psi_\varepsilon^{\theta,\eta}(y))y : \ y \in \mathbb{S}^1\},$$

with

$$\psi_\varepsilon^{\theta,\eta} \in C^1(\mathbb{S}^1), \qquad \|\psi_\varepsilon^{\theta,\eta}\|_{C^1} \lesssim |\varepsilon|,$$

uniformly in (θ, η) with $|\eta| \le K$. This follows because $\widetilde{U}_\varepsilon$ is already an $O(\varepsilon)$-C^1 perturbation of D_1, and the translation vector a is also of order ε. We now compute the first-order term of $\psi_\varepsilon^{\theta,\eta}$. A translation by a contributes, to first order in ε, the normal displacement

$$-\eta\Re(e^{-i\theta}y).$$

Indeed, for $y \in \mathbb{S}^1$,

$$|y - a| = \big(1 - 2\Re(\overline{y}a) + |a|^2\big)^{1/2} = 1 - \Re(\overline{y}a) + O(|a|^2),$$

uniformly in y, and here $a = \eta\varepsilon e^{i\theta}$. Therefore

$$\psi_\varepsilon^{\theta,\eta}(y) = \varepsilon\Big(\dot{\varphi}_0(y) - \eta\Re(e^{-i\theta}y)\Big) + O(\varepsilon^2) \quad \text{in } C^1(\mathbb{S}^1), \tag{5.13}$$

uniformly for $|\eta| \le K$.

Define

$$g^{\theta,\eta}(y) := \dot{\varphi}_0(y) - \eta\Re(e^{-i\theta}y).$$

Using (5.7), we see that $g^{\theta,\eta}$ belongs to the finite-dimensional space spanned by the first and second harmonics. More precisely,

$$g^{\theta,\eta}(e^{it}) = a\cos(2t) + b\sin(2t) - \eta\cos(t - \theta).$$

The second-harmonic part is independent of (θ,η) and nonzero, because $(a,b)\neq(0,0)$. To extract a uniform L^1 lower bound, note that $g^{\theta,\eta}$ lies in the 4-dimensional space

$$V := \operatorname{span}\{\cos t, \sin t, \cos 2t, \sin 2t\} \subset L^1(\mathbb{S}^1).$$

On this finite-dimensional space, all norms are equivalent; in particular, the $L^1(\mathbb{S}^1)$ and $L^\infty(\mathbb{S}^1)$ norms are equivalent, and convergence in coefficients is equivalent to convergence in any of these norms. The map

$$(\theta,\eta)\in[0,2\pi)\times[-K,K] \longmapsto g^{\theta,\eta}\in V$$

is continuous, and its image

$$\mathcal{G}_K := \{g^{\theta,\eta} :\ \theta\in[0,2\pi),\ |\eta|\le K\}$$

is the continuous image of a compact set, hence compact in V. Moreover, the $(\cos 2t, \sin 2t)$-coefficients of every $g^{\theta,\eta}$ are exactly $(a,b)\neq(0,0)$, independent of (θ,η); thus $0\notin\mathcal{G}_K$. The function $g\mapsto\|g\|_{L^1(\mathbb{S}^1)}$ is continuous and strictly positive on $V\setminus\{0\}$, so it attains a positive minimum on the compact set $\mathcal{G}_K$. It follows that there exists $\delta_K>0$ such that

$$\int_{\mathbb{S}^1} |g^{\theta,\eta}(y)|\, d\mathcal{H}^1(y) \ge \delta_K \qquad \text{for all } \theta\in[0,2\pi),\ |\eta|\le K. \tag{5.14}$$

Now,

$$|\widetilde{U}_\varepsilon \triangle D| = |(\widetilde{U}_\varepsilon - a)\triangle D_1|.$$

Since $\widetilde{U}_\varepsilon - a$ is star-shaped with radial graph $1+\psi_\varepsilon^{\theta,\eta}$, the same polar-coordinate computation as in Step 4 gives

$$|(\widetilde{U}_\varepsilon - a)\triangle D_1| = \frac{1}{2}\int_{\mathbb{S}^1} \left|(1+\psi_\varepsilon^{\theta,\eta}(y))^2 - 1\right| d\mathcal{H}^1(y).$$

Using (5.13),

$$(1+\psi_\varepsilon^{\theta,\eta})^2 - 1 = 2\varepsilon g^{\theta,\eta} + O(\varepsilon^2) \quad \text{in } L^\infty(\mathbb{S}^1),$$

uniformly for $|\eta|\le K$. Thus, by (5.14), there exists $c_K>0$ such that

$$|\widetilde{U}_\varepsilon \triangle D| = |(\widetilde{U}_\varepsilon - a)\triangle D_1| \ge c_K|\varepsilon| \tag{5.15}$$

whenever $|a|\le K|\varepsilon|$. Combining (5.12) and (5.15), we conclude that

$$\inf_{\substack{D \text{ disk}\\ |D|=\pi}} |\widetilde{U}_\varepsilon \triangle D| \gtrsim |\varepsilon|.$$

Since $r_\varepsilon = 1+O(\varepsilon^2)$, the dilation from U_ε to $\widetilde{U}_\varepsilon$ changes the set only by

$$|\widetilde{U}_\varepsilon \triangle U_\varepsilon| = O(\varepsilon^2),$$

and therefore does not affect the first-order distance estimate. Thus the area-normalized family $\widetilde{U}_\varepsilon$ remains at symmetric-difference distance $\gtrsim|\varepsilon|$ from the class of disks.

Step 6: square-root sharpness of the deficit estimate. It remains to translate the distance estimate into the corresponding deficit estimate. Recall that, by construction via Theorem 1, the first localization eigenvalue of U_ε equals λ_0 *exactly*, since (5.1) states that P_ε is an eigenfunction with eigenvalue λ_0 for the localization operator on U_ε, and a standard variational argument shows that this is the principal eigenvalue. By (5.10), $|U_\varepsilon| = \pi + O(\varepsilon^2)$, and rescaling by

$r_\varepsilon = 1 + O(\varepsilon^2)$ to area π changes the eigenvalue only at order $O(\varepsilon^2)$ (since the localization operator scales analytically in the dilation parameter, and the disk is a critical point for this scaling). It follows that the eigenvalue deficit of $\widetilde{U}_\varepsilon$ with respect to the disk satisfies

$$\lambda_0 - \lambda(\widetilde{U}_\varepsilon) = O(\varepsilon^2),$$

where $\lambda(\widetilde{U}_\varepsilon)$ denotes the first localization eigenvalue of $\widetilde{U}_\varepsilon$. On the other hand, by Step 5, the symmetric-difference asymmetry satisfies

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\widetilde{U}_\varepsilon \triangle D| \gtrsim |\varepsilon|.$$

Comparing the two,

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\widetilde{U}_\varepsilon \triangle D| \gtrsim |\varepsilon| \simeq \sqrt{\lambda_0 - \lambda(\widetilde{U}_\varepsilon)}.$$

This shows that the inequality

$$\inf_{\substack{D \text{ disk} \\ |D|=\pi}} |\Omega \triangle D| \lesssim \sqrt{\lambda_0 - \lambda(\Omega)}$$

of Corollary 1.4 in [20] is square-root-tight along the family $\widetilde{U}_\varepsilon$: the deficit is of order ε^2, the asymmetry is of order ε, and so the square root of the deficit and the asymmetry are of comparable size. This exhibits the optimality of the exponent $1/2$ and concludes the proof of the sharpness of the estimate in Corollary 1.4 of [20], complementing the rigidity result of [33]. □

6. Local maximizers for the Faber–Krahn problem in the Gaussian STFT setting

The purpose of this section is to establish a local rigidity statement for the Faber–Krahn problem associated with the Gaussian STFT. The global problem, solved in [33], asserts that, among measurable subsets of $\mathbb{C}$ of fixed measure, the centered disk uniquely maximizes the first eigenvalue of the Gaussian STFT localization operator. Here we show that this rigidity is already visible at the *local* level: any set whose first eigenvalue cannot be increased by small symmetric-difference perturbations must, up to a translation, be a disk. This complements the quantitative stability result of [20] and provides the variational counterpart of the inverse-spectral analysis carried out in the previous sections.

We work throughout in the Bargmann–Fock representation; we refer the reader to [4, 17, 22] for the relevant background. Thus, if $F \in \mathcal{F}^2(\mathbb{C})$ is the Bargmann transform of $f \in L^2(\mathbb{R})$, then

$$|\mathcal{V}_\varphi f(z)|^2 = |F(z)|^2 e^{-\pi|z|^2},$$

up to the normalization already fixed in the previous sections. Accordingly, for $F \in \mathcal{F}^2(\mathbb{C})$ we set

$$u_F(z) := |F(z)|^2 e^{-\pi|z|^2}.$$

For a measurable set $\Omega \subset \mathbb{C}$, let

$$\lambda_{1,\varphi}(\Omega) := \sup_{\|f\|_2=1} \int_\Omega |\mathcal{V}_\varphi f(z)|^2 \, dz.$$

We say that Ω is a *local maximizer* for $\lambda_{1,\varphi}$ under the measure constraint if there exists $\delta_0 > 0$ such that, whenever $E \subset \mathbb{C}$ is measurable, $|E| = |\Omega|$, and

$$\int_{\mathbb{C}} |\mathbf{1}_\Omega - \mathbf{1}_E| \, e^{-\pi|z|^2} \, dz < \delta_0,$$

one has

$$\lambda_{1,\varphi}(\Omega) \geq \lambda_{1,\varphi}(E).$$

We now prove the local rigidity statement announced in the introduction. The proof proceeds in six steps. Steps 1–3 are essentially the analogues, in our setting, of standard arguments in the Faber–Krahn theory: Σ is forced to be a level set of u_F, the corresponding eigenvalue is simple, and a first-variation identity ties F to a divergence-free vector field built from holomorphic data. The crucial new ingredient lies in Step 4, where we show that the *derivative* F' is again a first eigenfunction. Combined with the simplicity from Step 2, this forces F to satisfy a first-order linear ODE, and hence to be a Gaussian. Once this is in place, Steps 5–6 use the Weyl invariance of the Bargmann–Fock setting and a one-dimensional rearrangement argument to identify Σ with a centered disk.

Proof of the local-maximizer theorem. Let $\Sigma \subset \mathbb{C}$ be a local maximizer for $\lambda_{1,\varphi}$ under the measure constraint, and let $F \in \mathcal{F}^2(\mathbb{C})$ be a normalized first eigenfunction for the localization operator associated with Σ. Equivalently,

$$\int_\Sigma H(z)\overline{F(z)} \, d\mu(z) = \lambda_{1,\varphi}(\Sigma) \int_{\mathbb{C}} H(z)\overline{F(z)} \, d\mu(z) \tag{6.1}$$

for every $H \in \mathcal{F}^2(\mathbb{C})$, where

$$d\mu(z) := e^{-\pi|z|^2} \, dA(z).$$

Step 1: Σ is a superlevel set of u_F.

We claim that there exists $t_0 \geq 0$ such that, up to sets of measure zero,

$$\Sigma = \{u_F > t_0\}. \tag{6.2}$$

The strategy is a bathtub-style perturbation argument: if Σ were not a superlevel set of u_F, one could move a small piece of Σ where u_F is small to a region where u_F is larger, increasing the localization integral while preserving Lebesgue measure and remaining within the local-maximizer neighborhood of Σ.

Recall that F is admissible in the variational problem defining $\lambda_{1,\varphi}(\Sigma')$ for any measurable set Σ' of finite measure. In particular,

$$\lambda_{1,\varphi}(\Sigma') \geq \int_{\Sigma'} u_F \, dA, \qquad \lambda_{1,\varphi}(\Sigma) = \int_\Sigma u_F \, dA,$$

the latter equality holding because F is the actual eigenfunction for Σ. It therefore suffices to find a competitor Σ' with $|\Sigma'| = |\Sigma|$, close to Σ in the sense of weighted symmetric difference, and with $\int_{\Sigma'} u_F \, dA > \int_\Sigma u_F \, dA$; that would violate local maximality.

Suppose, for the sake of contradiction, that (6.2) fails. Choose, by the layer-cake representation, the unique $t_0 \geq 0$ such that

$$A := \{u_F > t_0\} \quad \text{satisfies} \quad |A| = |\Sigma|;$$

then A and Σ do not coincide a.e., so both $A \setminus \Sigma$ and $\Sigma \setminus A$ have positive Lebesgue measure (and in fact positive weighted measure, since $e^{-\pi|z|^2} > 0$ everywhere).

We now perform the bathtub construction explicitly. By inner regularity of the Lebesgue measure, choose an open subset $S' \subset A \setminus \Sigma$ and an open subset $T' \subset \Sigma \setminus A$ with $|S'|, |T'| > 0$. Replacing each by a sufficiently small open subset, we may further assume $\int_{S'} d\mu$, $\int_{T'} d\mu < \delta_0/4$. Since the Lebesgue measure restricted to a fixed open set has the intermediate-value property, we may choose subsets $S \subset S'$ and $T \subset T'$ with the common Lebesgue measure

$$|S| = |T| = \tfrac{1}{2} \min(|S'|, |T'|) > 0.$$

Then

$$\int_{\mathbb{C}} (\mathbf{1}_S + \mathbf{1}_T)\, e^{-\pi|z|^2}\, dz < \tfrac{\delta_0}{2} < \delta_0.$$

Define

$$\Sigma' := (\Sigma \setminus T) \cup S.$$

Then $|\Sigma'| = |\Sigma|$ and

$$\int_{\mathbb{C}} |\mathbf{1}_\Sigma - \mathbf{1}_{\Sigma'}| e^{-\pi|z|^2}\, dz = \int_{\mathbb{C}} (\mathbf{1}_S + \mathbf{1}_T) e^{-\pi|z|^2}\, dz < \delta_0.$$

On the other hand, by construction,

$$u_F(z) > t_0 \quad \text{on } S \subset A, \qquad u_F(z) \leq t_0 \quad \text{on } T \subset \Sigma \setminus A.$$

Moreover, since u_F is continuous and $S \subset A = \{u_F > t_0\}$ is open with positive measure, we may by shrinking S further assume that there exists $\eta > 0$ with $u_F \geq t_0 + \eta$ on S. It then follows that

$$\int_S u_F\, dA \geq (t_0 + \eta)\, |S|, \qquad \int_T u_F\, dA \leq t_0\, |T| = t_0\, |S|,$$

and hence

$$\int_{\Sigma'} u_F\, dA = \int_\Sigma u_F\, dA + \int_S u_F\, dA - \int_T u_F\, dA \geq \int_\Sigma u_F\, dA + \eta\, |S| > \int_\Sigma u_F\, dA.$$

Since F is admissible for the variational problem defining $\lambda_{1,\varphi}(\Sigma')$,

$$\lambda_{1,\varphi}(\Sigma') \geq \int_{\Sigma'} u_F\, dA > \int_\Sigma u_F\, dA = \lambda_{1,\varphi}(\Sigma),$$

contradicting the local maximality of Σ. This proves (6.2).

In particular, since u_F is real-analytic outside the (discrete) zero set of F, u_F is constant on $\partial\Sigma$ in the sense that there exists $c_\Sigma \geq 0$ with

$$u_F(z) = c_\Sigma \qquad \text{for } z \in \partial\Sigma. \tag{6.3}$$

We will see in Step 2 that F does not vanish on $\overline{\Sigma}$, and in particular $c_\Sigma > 0$.

Step 2: simplicity of the first eigenvalue.

We claim that the first eigenvalue $\lambda_{1,\varphi}(\Sigma)$ is simple. The argument uses a modular-deformation principle: any two first eigenfunctions have the same modulus on $\partial\Sigma$ up to a constant, and a holomorphic function with constant modulus on a connected open set must itself be constant. This kind of argument plays a role also in the stability analysis of [20].

Suppose, for contradiction, that $G \in \mathcal{F}^2(\mathbb{C})$ is another first eigenfunction, linearly independent of F. Step 1 applied to G shows that Σ is also a superlevel set of u_G; hence there exists $c'_\Sigma \geq 0$ such that

$$u_G(z) = c'_\Sigma \qquad \text{for } z \in \partial\Sigma.$$

Equivalently,

$$|F(z)|^2 e^{-\pi|z|^2} = c_\Sigma, \qquad |G(z)|^2 e^{-\pi|z|^2} = c'_\Sigma, \qquad z \in \partial\Sigma.$$

We next observe that first eigenfunctions are zero-free in the interior of Σ. Indeed, by Step 1 we have $u_F > c_\Sigma$ on the (open) interior of Σ. Since

$$u_F = |F|^2 e^{-\pi|z|^2}$$

and the Gaussian factor $e^{-\pi|z|^2}$ is strictly positive everywhere, this yields $|F| > 0$ throughout the interior of Σ; the same applies to $|G|$. In particular, $c_\Sigma > 0$ and $c'_\Sigma > 0$. Moreover, since F, G are entire and continuous, $|F| > 0$ on a neighborhood of any compact subset of Σ°; by Step 1, we may take a slight enlargement of Σ (still a superlevel set of u_F) on which $|F|, |G| > 0$, so we may work on a connected open set $U \Supset \Sigma$ where both F and G are zero-free.

The ratio F/G is holomorphic and nowhere zero on U, and the function

$$\log\left|\frac{F}{G}\right| = \frac{1}{2}\log\frac{|F|^2}{|G|^2} = \frac{1}{2}\log\frac{u_F}{u_G}$$

is harmonic on U: it is the real part of any local branch of $\log(F/G)$, which is well defined on simply connected subsets, and harmonic in any case as the logarithm of the modulus of a zero-free holomorphic function. Note that the Gaussian factors cancel inside the logarithm, so on $\partial\Sigma$ by (6.3) this function equals the constant $\frac{1}{2}\log(c_\Sigma/c'_\Sigma)$. By the maximum principle applied separately on each connected component of Σ, the function $\log|F/G|$ is constant on each component. Hence

$$\left|\frac{F}{G}\right| \equiv \text{constant} \qquad \text{on each connected component of } \Sigma.$$

A nonconstant holomorphic function is open, so a holomorphic function with constant modulus on a connected open set must itself be constant. Therefore

$$F = \gamma_j G \qquad \text{on each connected component } \Sigma_j \text{ of } \Sigma.$$

The relation $F = \gamma_j G$ is an identity between two entire functions on a set with an accumulation point; the analytic continuation principle gives $F = \gamma_j G$ on all of $\mathbb{C}$ for any single j. In particular all γ_j coincide, and F and G are linearly dependent, contradicting our assumption. This proves simplicity.

Step 3: a first-variation identity.

By Step 2, the eigenfunction F is determined up to a unimodular constant. We now derive a first-variation identity satisfied by F, based on the fact that Σ is a level set of u_F and that holomorphic vector fields are divergence-free.

The function $u_F = |F|^2 e^{-\pi|z|^2}$ is real-analytic on $\mathbb{C} \setminus \{F = 0\}$, since F is entire and the exponential factor is real-analytic. By Step 2, F does not vanish in the interior of Σ, and by the argument of Step 4 below it does not vanish on $\overline{\Sigma}$ either; so u_F is in fact real-analytic on a neighborhood of $\overline{\Sigma}$. The boundary $\partial\Sigma$ is therefore a level set of a non-constant real-analytic function and so is the union of finitely many real-analytic arcs, plus a possible finite set of singular points where ∇u_F vanishes (see, e.g., the standard local structure of real-analytic varieties). In particular, $\partial\Sigma$ is Lipschitz, and C^1 away from the critical set of u_F; the divergence theorem applies to Σ for any C^1-vector field on a neighborhood of $\overline{\Sigma}$, the singular boundary points forming a set of one-dimensional Hausdorff measure zero.

Let ψ be holomorphic on an open neighborhood V of $\overline{\Sigma}$, and write

$$\psi = u + iv$$

with u, v real-valued. Consider the real vector field

$$X_\psi := (u, -v).$$

Since ψ is holomorphic, the Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x,$$

imply

$$\operatorname{div} X_\psi = u_x + (-v)_y = u_x - v_y = 0 \qquad \text{in } V.$$

Because $u_F \equiv c_\Sigma$ on $\partial\Sigma$, the divergence theorem on Σ gives

$$\int_\Sigma \langle \nabla u_F, X_\psi \rangle \, dA = \int_\Sigma \operatorname{div}(u_F X_\psi) \, dA = \int_{\partial\Sigma} u_F \, X_\psi \cdot \nu \, d\mathcal{H}^1 = c_\Sigma \int_{\partial\Sigma} X_\psi \cdot \nu \, d\mathcal{H}^1 = 0,$$

where the last equality uses $\int_{\partial\Sigma} X_\psi \cdot \nu \, d\mathcal{H}^1 = \int_\Sigma \operatorname{div} X_\psi \, dA = 0$ by Cauchy–Riemann. Since $u_F = |F|^2 e^{-\pi|z|^2}$ and $\nabla(e^{-\pi|z|^2}) = -2\pi e^{-\pi|z|^2}(x, y)$, this becomes

$$0 = \int_\Sigma \big(\langle \nabla |F|^2, X_\psi \rangle - 2\pi |F|^2 \langle z, X_\psi \rangle \big) \, d\mu. \tag{6.4}$$

We now express (6.4) in complex notation. A direct computation, using $\partial_z |F|^2 = F'\overline{F}$ and $\partial_{\bar z} |F|^2 = F\overline{F'}$, gives

$$\partial_x |F|^2 = F'\overline{F} + \overline{F'}F = 2\Re(F'\overline{F}), \qquad \partial_y |F|^2 = i(F'\overline{F} - \overline{F'}F) = -2\Im(F'\overline{F}).$$

Writing $\psi = u + iv$ and recalling $X_\psi = (u, -v)$,

$$\langle \nabla |F|^2, X_\psi \rangle = 2u \, \Re(F'\overline{F}) + 2v \, \Im(F'\overline{F}) = 2 \, \Re\big(\overline{\psi} \, F'\overline{F}\big) = 2 \, \Re\big(\psi \, \overline{F'}F\big),$$

where in the last equality we used $\Re w = \Re \overline{w}$. Likewise, writing $z = x + iy$,

$$\langle z, X_\psi \rangle = xu + y(-v) = xu - yv = \Re\big((x + iy)(u + iv)\big) = \Re(z\psi).$$

Substituting these expressions into (6.4), we obtain

$$0 = \Re \int_\Sigma \psi(z)\Big(\overline{F'(z)}\, F(z) - \pi z\, |F(z)|^2\Big)\, d\mu(z).$$

Replacing ψ by $i\psi$ (which is also holomorphic on V) gives the analogous identity for the imaginary part:

$$0 = \Im \int_\Sigma \psi(z)\Big(\overline{F'(z)}\, F(z) - \pi z\, |F(z)|^2\Big)\, d\mu(z).$$

Combining the two,

$$0 = \int_\Sigma \psi(z)\, F(z)\Big(\overline{F'(z)} - \pi z\, \overline{F(z)}\Big)\, d\mu(z), \tag{6.5}$$

which holds for every holomorphic ψ on a neighborhood of $\overline{\Sigma}$. Equivalently, after factoring out $F(z)$ inside the integrand,

$$0 = \int_\Sigma \psi(z)\Big(\overline{F'(z)}\, F(z) - \pi z\, |F(z)|^2\Big)\, d\mu(z).$$

The form (6.5) is the one used in Step 4.

Step 4: F' is again a first eigenfunction.

This step is the crux of the argument and is the main new ingredient compared to classical Faber–Krahn rigidity proofs. Rather than working with general test functions in the eigenvalue equation, we exploit the divergence identity (6.5) together with a clever choice of test function involving the reproducing kernel of the Fock space. The end result is that *the derivative F'* of any first eigenfunction is itself a first eigenfunction; since the eigenspace is one-dimensional by Step 2, this forces a first-order ODE on F, and hence pins down F up to translation as a Gaussian.

For every $w \in \mathbb{C}$, set

$$K_w(z) := e^{\pi \overline{w} z};$$

this is the reproducing kernel of $\mathcal{F}^2(\mathbb{C})$ at the point w, in the sense that

$$\int_{\mathbb{C}} H(z)\, e^{\pi w \overline{z}}\, d\mu(z) = H(w), \qquad H \in \mathcal{F}^2(\mathbb{C}); \tag{6.6}$$

see, e.g., [17, Ch. 1] or [4]. Equivalently, $\langle H, K_w \rangle_{\mathcal{F}^2} = H(w)$.

By Step 1, after replacing Σ by the corresponding superlevel-set representative if necessary, we may suppose that

$$\Sigma = \{u_F > c_\Sigma\}.$$

In particular, $u_F \geq c_\Sigma > 0$ on $\overline{\Sigma}$. Since $u_F(z) = |F(z)|^2 e^{-\pi |z|^2}$, it follows that F has no zeros on $\overline{\Sigma}$. By continuity of F and the compactness of $\overline{\Sigma}$, there exists an open neighborhood $V \supset \overline{\Sigma}$ on which F has no zeros. Hence, for every $w \in \mathbb{C}$, the function

$$\psi_w(z) := \frac{K_w(z)}{F(z)} = \frac{e^{\pi \overline{w} z}}{F(z)}$$

is holomorphic on V. Applying (6.5) with $\psi = \psi_w$, so that $\psi_w(z) F(z) = e^{\pi \overline{w} z}$, we obtain

$$0 = \int_\Sigma e^{\pi \overline{w} z}\Big(\overline{F'(z)} - \pi z\, \overline{F(z)}\Big)\, d\mu(z),$$

or, separating the two terms,

$$\int_\Sigma e^{\pi\overline{w}z}\overline{F'(z)}\,d\mu(z) = \int_\Sigma \pi z\, e^{\pi\overline{w}z}\overline{F(z)}\,d\mu(z). \tag{6.7}$$

We now relate the right-hand side of (6.7) to $\overline{F'(w)}$ using the eigenvalue equation (6.1) together with the reproducing identity (6.6). Choosing

$$H(z) := z\, e^{\pi\overline{w}z} \in \mathcal{F}^2(\mathbb{C})$$

(the product of the kernel K_w by the polynomial z; both factors preserve membership in $\mathcal{F}^2$) in (6.1), we obtain

$$\int_\Sigma z\, e^{\pi\overline{w}z}\,\overline{F(z)}\,d\mu(z) = \lambda_{1,\varphi}(\Sigma)\int_{\mathbb{C}} z\, e^{\pi\overline{w}z}\,\overline{F(z)}\,d\mu(z). \tag{6.8}$$

On the other hand, taking complex conjugates in (6.6) applied to F at the point w, one finds

$$\int_{\mathbb{C}} e^{\pi\overline{w}z}\,\overline{F(z)}\,d\mu(z) = \overline{F(w)}.$$

Both sides depend on w through the antiholomorphic variable $\overline{w}$; we differentiate with respect to $\overline{w}$, exchanging differentiation and integration on the left. The interchange is legitimate by the dominated convergence theorem: locally in w, the integrand $\partial_{\overline{w}}(e^{\pi\overline{w}z}\overline{F(z)}) = \pi z\, e^{\pi\overline{w}z}\overline{F(z)}$ is dominated by an $L^1(d\mu)$ function on $\mathbb{C}$, as a consequence of the Cauchy–Schwarz inequality and the fact that $F \in \mathcal{F}^2(\mathbb{C})$ and the Fock-space norm of zK_w is locally bounded. We conclude

$$\pi\int_{\mathbb{C}} z\, e^{\pi\overline{w}z}\,\overline{F(z)}\,d\mu(z) = \overline{F'(w)}.$$

Multiplying (6.8) by π and substituting,

$$\pi\int_\Sigma z\, e^{\pi\overline{w}z}\,\overline{F(z)}\,d\mu(z) = \lambda_{1,\varphi}(\Sigma)\,\overline{F'(w)}.$$

Combining this with (6.7),

$$\int_\Sigma e^{\pi\overline{w}z}\,\overline{F'(z)}\,d\mu(z) = \lambda_{1,\varphi}(\Sigma)\,\overline{F'(w)}, \qquad w \in \mathbb{C}.$$

Taking complex conjugates, this is exactly the eigenvalue equation (6.1) applied with F replaced by F' (and an arbitrary K_w playing the role of the test function). Since the linear span of $\{K_w\}_{w\in\mathbb{C}}$ is dense in $\mathcal{F}^2(\mathbb{C})$, the identity extends to every $H \in \mathcal{F}^2(\mathbb{C})$, and we conclude that F' is a first eigenfunction of the localization operator on Σ, with eigenvalue $\lambda_{1,\varphi}(\Sigma)$.

By the simplicity established in Step 2,

$$F' = \theta F$$

for some constant $\theta \in \mathbb{C}$. Solving this ODE gives

$$F(z) = Ce^{\theta z},$$

for some nonzero constant C.

Step 5: reduction to the constant eigenfunction.

We now use the Weyl invariance of the Bargmann–Fock setting to reduce the analysis to the case $F \equiv 1$. Recall from (2.5) that, for $a \in \mathbb{C}$, the Weyl translation

$$(T_a F)(z) = e^{\pi a z - \frac{\pi}{2}|a|^2} F(z-a)$$

is unitary on $\mathcal{F}^2(\mathbb{C})$ and satisfies $u_{T_a F}(z) = u_F(z-a)$. In particular, T_a maps eigenfunctions of the localization operator on Σ to eigenfunctions of the localization operator on the translated set $\Sigma + a$, with the same eigenvalue, and the quantity $\int_\Sigma u_F \, dA$ is invariant under the simultaneous substitutions $F \mapsto T_a F$ and $\Sigma \mapsto \Sigma + a$.

We claim that we may choose $a \in \mathbb{C}$ so that $T_{-a}F$ is a constant function. Indeed, by Step 4, $F(z) = Ce^{\theta z}$ for some $C, \theta \in \mathbb{C}$ with $C \neq 0$. Compute, using the identity $u_{T_{-a}F}(z) = u_F(z+a)$:

$$u_{T_{-a}F}(z) = |F(z+a)|^2 e^{-\pi|z+a|^2} = |C|^2 e^{2\Re(\theta(z+a))} \, e^{-\pi|z+a|^2}.$$

Expanding,

$$2\Re(\theta(z+a)) - \pi|z+a|^2 = 2\Re(\theta z) + 2\Re(\theta a) - \pi|z|^2 - 2\pi\Re(\overline{a}z) - \pi|a|^2.$$

The linear terms in z on the right cancel if and only if $2\Re(\theta z) = 2\pi\Re(\overline{a}z)$ for every $z \in \mathbb{C}$, which is the condition $\overline{a} = \theta/\pi$, i.e.

$$a := \frac{\overline{\theta}}{\pi}.$$

With this choice,

$$u_{T_{-a}F}(z) = |C'|^2 e^{-\pi|z|^2}, \qquad |C'|^2 := |C|^2 e^{2\Re(\theta a) - \pi|a|^2} = |C|^2 e^{|\theta|^2/\pi}.$$

Since $T_{-a}F$ is entire and $|T_{-a}F(z)|^2 e^{-\pi|z|^2} = |C'|^2 e^{-\pi|z|^2}$, it follows that $|T_{-a}F| \equiv |C'|$ is constant; a holomorphic function with constant modulus is constant, hence $T_{-a}F \equiv C''$ for some $C'' \in \mathbb{C}$ with $|C''| = |C'|$. Normalizing, we may absorb the unimodular constant into the eigenfunction and assume

$$F \equiv 1$$

after the change $\Sigma \rightsquigarrow \Sigma - a$ (and the corresponding unitary substitution $F \rightsquigarrow T_{-a}F$).

In this normalization the ground state h_0 (corresponding to $F \equiv 1$ in Bargmann–Fock coordinates) is an eigenfunction of the localization operator of Σ. Since Σ is a level set of $u_F(z) = e^{-\pi|z|^2}$, and the latter is a strictly radial function of $|z|$, the set Σ is itself radial; in particular, it is either a centered disk or a finite union of centered annuli. The next step rules out the annular configurations and identifies Σ with the disk.

Step 6: exclusion of annuli and conclusion.

It remains to verify that, among radial sets of prescribed measure, the only local maximizer is the centered disk. Although this is essentially automatic after Step 5, we record the standard rearrangement argument since it makes explicit *why* no annular local maximizer can exist; this provides the endpoint matching the global Faber–Krahn theorem of [33].

If Σ is radial, write

$$\widetilde{\Sigma} := \{r > 0 : \ re^{it} \in \Sigma \text{ for some } t \in [0, 2\pi)\}.$$

By Step 1 and the Lipschitz regularity established in Step 3, $\widetilde{\Sigma}$ is open and is a finite union of disjoint open intervals:

$$\widetilde{\Sigma} = \bigcup_{j=1}^{N} (a_j, b_j), \qquad 0 \le a_1 < b_1 \le a_2 < b_2 \le \cdots \le a_N < b_N.$$

Then, using $\int r e^{-\pi r^2}\, dr = -\frac{1}{2\pi} e^{-\pi r^2}$,

$$\int_{\Sigma} e^{-\pi |z|^2}\, dA(z) = 2\pi \int_{\widetilde{\Sigma}} r e^{-\pi r^2}\, dr = \sum_{j=1}^{N} \left(e^{-\pi a_j^2} - e^{-\pi b_j^2} \right),$$

under the constraint

$$|\Sigma| = 2\pi \int_{\widetilde{\Sigma}} r\, dr = \pi \sum_{j=1}^{N} (b_j^2 - a_j^2).$$

We are therefore led to the optimization problem

$$\text{maximize} \qquad \sum_{j=1}^{N} \left(e^{-\pi a_j^2} - e^{-\pi b_j^2} \right) \qquad \text{subject to} \qquad \sum_{j=1}^{N} (b_j^2 - a_j^2) = c, \tag{6.9}$$

where $c = |\Sigma|/\pi$ is fixed. Substituting $s_j := a_j^2$, $t_j := b_j^2$ (with $0 \le s_1 < t_1 \le s_2 < \cdots$), and writing $\Phi(x) := e^{-\pi x}$, (6.9) becomes

$$\text{maximize} \qquad \sum_{j=1}^{N} \big(\Phi(s_j) - \Phi(t_j) \big) \qquad \text{subject to} \qquad \sum_{j=1}^{N} (t_j - s_j) = c.$$

The function Φ is strictly convex and strictly decreasing on $[0, \infty)$; its derivative $-\Phi'(\sigma) = \pi e^{-\pi\sigma}$ is strictly decreasing in σ. For any $t > s \ge 0$,

$$\Phi(s) - \Phi(t) = \int_s^t (-\Phi'(\sigma))\, d\sigma,$$

so

$$\sum_{j=1}^{N} \big(\Phi(s_j) - \Phi(t_j) \big) = \int_{\widetilde{\Sigma}_2} (-\Phi'(\sigma))\, d\sigma, \qquad \widetilde{\Sigma}_2 := \bigcup_{j=1}^{N} (s_j, t_j).$$

The set $\widetilde{\Sigma}_2 \subset [0, \infty)$ has Lebesgue measure equal to c, and the integrand $-\Phi'$ is strictly decreasing in σ. The bathtub principle (or, equivalently, a direct rearrangement argument) shows that, among all measurable subsets of $[0, \infty)$ of total Lebesgue measure c, the integral $\int_{\widetilde{\Sigma}_2} (-\Phi'(\sigma))\, d\sigma$ is strictly maximized by the leftmost interval $(0, c)$. This corresponds to $N = 1$, $s_1 = 0$, $t_1 = c$, i.e.

$$\widetilde{\Sigma} = (0, R), \qquad R := \sqrt{c},$$

and equivalently Σ is the centered disk D_R with $|D_R| = |\Sigma|$. Any annular configuration ($N \ge 2$, or $N = 1$ with $s_1 > 0$) produces a strictly smaller value of $\int_{\Sigma} u_F\, dA$, and is therefore not a local maximizer.

We conclude that every local maximizer of $\lambda_{1,\varphi}$ under the measure constraint is, up to translation, a disk. $\square$

7. Global maximizers for the Faber–Krahn problem in the Gaussian STFT setting

In this section we prove existence of global maximizers for the Gaussian time-frequency concentration problem under a measure constraint, and then combine that existence result with the local rigidity theorem from the previous section to recover the disk as the unique maximizer, up to translation. The strategy is by now classical in time-frequency analysis: a profile decomposition in the spirit of P.-L. Lions' concentration-compactness, transplanted to the Fock space via the Bargmann transform, dichotomizes the loss of compactness of a maximizing sequence into the action of the Weyl translation group. A vanishing remainder is then ruled out by an iterative extraction whose maximality forces all the energy to remain captured by the extracted profiles. Closely related arguments have appeared in the time-frequency setting in [32, 31, 18, 39], and the disk-rigidity result we recover is the celebrated theorem of Nicola–Tilli [33, 34].

Since the Bargmann transform is unitary, the variational problem may be written on the Fock space side as

$$M(s) := \sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega) = \sup_{\|F\|_{\mathcal{F}^2}=1} I_F(s).$$

Here, and throughout the section, we exploit the translation invariance (2.8) together with the pointwise identity (2.7) (cf. also (2.6)), the basic regularity properties recorded in Lemma 2.5, and the elementary properties of the Fock space gathered in Lemma 2.1.

We begin with a Weyl-orthogonality lemma that captures the asymptotic decoupling of two functions whose translation parameters drift apart. This plays the role of *Pythagorean orthogonality at infinity* on the Fock space.

Lemma 7.1. *Let $F, G \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k\} \subset \mathbb{C}$ satisfy $|a_k| \to \infty$. Then*

$$T_{a_k} G \rightharpoonup 0 \qquad \text{weakly in } \mathcal{F}^2(\mathbb{C}),$$

and in particular

$$\langle F, T_{a_k} G\rangle_{\mathcal{F}^2} \to 0.$$

Consequently,

$$\|F + T_{a_k} G\|^2_{\mathcal{F}^2} = \|F\|^2_{\mathcal{F}^2} + \|G\|^2_{\mathcal{F}^2} + o(1).$$

More generally, if $|a_k - b_k| \to \infty$, then

$$\langle T_{a_k} F, T_{b_k} G\rangle_{\mathcal{F}^2} \to 0.$$

Proof. Using unitarity of the Weyl operators T_a on $\mathcal{F}^2(\mathbb{C})$ (cf. (2.5)),

$$\langle F, T_{a_k} G\rangle_{\mathcal{F}^2} = \langle T_{-a_k} F, G\rangle_{\mathcal{F}^2}.$$

Now $T_{-a_k} F$ is bounded in $\mathcal{F}^2$ by unitarity, and the identity (2.7) (equivalently (2.6)) yields

$$u_{T_{-a_k} F}(z) = u_F(z + a_k) \qquad (z \in \mathbb{C}),$$

so that the densities $u_{T_{-a_k}F}$ drift uniformly to infinity. Equivalently, since $u_F \in C_0(\mathbb{C})$ by Lemma 2.1(iii), for every compact $K \subset \mathbb{C}$,

$$\sup_{z\in K} |T_{-a_k}F(z)|e^{-\pi|z|^2/2} = \sup_{z\in K} u_{T_{-a_k}F}(z)^{1/2} = \sup_{z\in K} u_F(z+a_k)^{1/2} \to 0.$$

Hence $T_{-a_k}F \to 0$ locally uniformly. Combining this with the boundedness of $\{T_{-a_k}F\}$ in $\mathcal{F}^2(\mathbb{C})$, and using the fact that the linear span of the reproducing kernels $\{K_w\}_{w\in\mathbb{C}}$ is dense in $\mathcal{F}^2$, we conclude that $T_{-a_k}F \rightharpoonup 0$ weakly in $\mathcal{F}^2$; in particular $\langle T_{-a_k}F, G\rangle \to 0$. This proves the first claim.

For the norm decoupling, expand the square:

$$\|F + T_{a_k}G\|_{\mathcal{F}^2}^2 = \|F\|_{\mathcal{F}^2}^2 + \|T_{a_k}G\|_{\mathcal{F}^2}^2 + 2\Re\langle F, T_{a_k}G\rangle_{\mathcal{F}^2} = \|F\|_{\mathcal{F}^2}^2 + \|G\|_{\mathcal{F}^2}^2 + o(1),$$

where we used unitarity of T_{a_k} and the cross-term decay just proved.

Finally, the general case reduces to the first by writing

$$\langle T_{a_k}F, T_{b_k}G\rangle_{\mathcal{F}^2} = \langle F, T_{-a_k}T_{b_k}G\rangle_{\mathcal{F}^2} = e^{i\theta_k}\langle F, T_{b_k-a_k}G\rangle_{\mathcal{F}^2},$$

for some real phase θ_k coming from the cocycle of the Weyl representation; since this phase factor has modulus one, the conclusion follows from $|b_k - a_k| \to \infty$. □

The next lemma compares the density of a finite sum of widely separated translates with the density of each individual translate. The precise statement is what makes the profile decomposition compatible with the variational quantities $I_F(s)$, since these depend only on the densities and not on the phases of the underlying functions.

Lemma 7.2. *Fix $J \geq 1$, let $G^{(1)}, \dots, G^{(J)} \in \mathcal{F}^2(\mathbb{C})$, and let $\{a_k^{(j)}\}_{k\geq 1} \subset \mathbb{C}$ for $1 \leq j \leq J$ satisfy*

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad \text{whenever } j \neq \ell.$$

For each k, set

$$S_k := \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)}.$$

Then the following hold.

(i) For every $R > 0$ and every $1 \leq j \leq J$,

$$\sup_{z\in D_R(a_k^{(j)})} \left|u_{S_k}(z) - u_{T_{a_k^{(j)}}G^{(j)}}(z)\right| \to 0 \qquad (k \to \infty).$$

(ii) For every $R > 0$, if

$$E_{k,R} := \mathbb{C} \setminus \bigcup_{j=1}^{J} D_R(a_k^{(j)}),$$

then

$$\limsup_{k\to\infty} \sup_{z\in E_{k,R}} u_{S_k}(z) \leq \left(\sum_{j=1}^{J} \sup_{\zeta\in\mathbb{C}\setminus D_R} u_{G^{(j)}}(\zeta)^{1/2}\right)^2.$$

In particular, given $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that for every $R \geq R_\varepsilon$,

$$\sup_{z \in E_{k,R}} u_{S_k}(z) \leq \varepsilon$$

for all sufficiently large k.

Proof. Throughout the proof we use repeatedly the identity

$$u_{T_a F}(z) = u_F(z-a), \qquad z, a \in \mathbb{C},$$

which is (2.7) (equivalently (2.6)); together with the elementary subadditivity

$$u_{F+G}(z)^{1/2} \leq u_F(z)^{1/2} + u_G(z)^{1/2}, \qquad z \in \mathbb{C},$$

which follows from the definition $u_F = |F|^2 e^{-\pi|z|^2}$ and the triangle inequality applied to $|F+G| \leq |F| + |G|$.

Proof of (i). Fix $R > 0$ and $j \in \{1, \dots, J\}$. Write

$$S_k = T_{a_k^{(j)}} G^{(j)} + H_{k,j}, \qquad H_{k,j} := \sum_{\ell \neq j} T_{a_k^{(\ell)}} G^{(\ell)}.$$

For every $z \in \mathbb{C}$, the elementary inequality $\big||a+b|^2 - |a|^2\big| \leq 2|a|\,|b| + |b|^2$ yields

$$\begin{aligned} \big|u_{S_k}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z)\big| &= e^{-\pi|z|^2} \big||T_{a_k^{(j)}} G^{(j)}(z) + H_{k,j}(z)|^2 - |T_{a_k^{(j)}} G^{(j)}(z)|^2\big| \\ &\leq 2\, u_{T_{a_k^{(j)}} G^{(j)}}(z)^{1/2} u_{H_{k,j}}(z)^{1/2} + u_{H_{k,j}}(z). \end{aligned}$$

By Lemma 2.1(ii),

$$u_{T_{a_k^{(j)}} G^{(j)}}(z) \leq \|G^{(j)}\|^2_{\mathcal{F}^2} \qquad \text{for every } z \in \mathbb{C} \text{ and every } k.$$

Thus, to prove (i) it is enough to show that

$$\sup_{z \in D_R(a_k^{(j)})} u_{H_{k,j}}(z) \to 0.$$

For $z \in D_R(a_k^{(j)})$ and $\ell \neq j$,

$$z - a_k^{(\ell)} \in D_R(a_k^{(j)} - a_k^{(\ell)}).$$

Since $|a_k^{(j)} - a_k^{(\ell)}| \to \infty$,

$$\operatorname{dist}\big(D_R(a_k^{(j)} - a_k^{(\ell)}), 0\big) \geq |a_k^{(j)} - a_k^{(\ell)}| - R \to \infty.$$

Because $u_{G^{(\ell)}} \in C_0(\mathbb{C})$ by Lemma 2.1(iii),

$$\sup_{z \in D_R(a_k^{(j)})} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z) = \sup_{\zeta \in D_R(a_k^{(j)} - a_k^{(\ell)})} u_{G^{(\ell)}}(\zeta) \xrightarrow[k \to \infty]{} 0.$$

The subadditivity of $u_F^{1/2}$ yields

$$u_{H_{k,j}}(z)^{1/2} \leq \sum_{\ell \neq j} u_{T_{a_k^{(\ell)}} G^{(\ell)}}(z)^{1/2},$$

so the preceding convergence implies

$$\sup_{z\in D_R(a_k^{(j)})} u_{H_{k,j}}(z)^{1/2} \to 0.$$

This proves (i).

Proof of (ii). Fix $R > 0$ and $z \in E_{k,R}$. By definition, $z \notin D_R(a_k^{(j)})$ for every j, and hence $|z - a_k^{(j)}| \geq R$. Therefore

$$u_{T_{a_k^{(j)}}G^{(j)}}(z) = u_{G^{(j)}}(z - a_k^{(j)}) \leq \sup_{\zeta\in\mathbb{C}\setminus D_R} u_{G^{(j)}}(\zeta).$$

Using again the subadditivity of $u_F^{1/2}$,

$$u_{S_k}(z)^{1/2} \leq \sum_{j=1}^{J} u_{T_{a_k^{(j)}}G^{(j)}}(z)^{1/2} \leq \sum_{j=1}^{J} \sup_{\zeta\in\mathbb{C}\setminus D_R} u_{G^{(j)}}(\zeta)^{1/2}.$$

Squaring yields

$$u_{S_k}(z) \leq \left(\sum_{j=1}^{J} \sup_{\zeta\in\mathbb{C}\setminus D_R} u_{G^{(j)}}(\zeta)^{1/2}\right)^2.$$

Taking the supremum over $z \in E_{k,R}$ proves the displayed estimate in (ii). Since each $u_{G^{(j)}} \in C_0(\mathbb{C})$, the right-hand side tends to 0 as $R \to \infty$; the final claim follows by choosing R_ε so that the displayed bracket is at most $\sqrt{\varepsilon}$. □

The next lemma is a robust uniform-density perturbation estimate. Concretely, it says that adding to a bounded sequence in $\mathcal{F}^2(\mathbb{C})$ a residue whose density vanishes uniformly leaves the density essentially unchanged in L^∞. This will be applied with S_k the truncated profile sum and R_k the residual of the profile decomposition.

Lemma 7.3. *Let $\{S_k\} \subset \mathcal{F}^2(\mathbb{C})$ be bounded and let $\{R_k\} \subset \mathcal{F}^2(\mathbb{C})$ satisfy*

$$\|u_{R_k}\|_{L^\infty(\mathbb{C})} \to 0.$$

Then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \to 0.$$

More precisely, if $\sup_k \|S_k\|_{\mathcal{F}^2} \leq M$, then

$$\|u_{S_k+R_k} - u_{S_k}\|_{L^\infty(\mathbb{C})} \leq 2M\,\|u_{R_k}\|_{L^\infty(\mathbb{C})}^{1/2} + \|u_{R_k}\|_{L^\infty(\mathbb{C})}.$$

Proof. For every $z \in \mathbb{C}$,

$$\begin{aligned}\left|u_{S_k+R_k}(z) - u_{S_k}(z)\right| &= e^{-\pi|z|^2}\left||S_k(z) + R_k(z)|^2 - |S_k(z)|^2\right| \\ &\leq 2\,u_{S_k}(z)^{1/2}u_{R_k}(z)^{1/2} + u_{R_k}(z),\end{aligned}$$

again by the elementary inequality $\left||a+b|^2 - |a|^2\right| \leq 2|a|\,|b| + |b|^2$. By Lemma 2.1(ii),

$$u_{S_k}(z) \leq \|S_k\|_{\mathcal{F}^2}^2 \leq M^2 \qquad (z \in \mathbb{C}).$$

Hence

$$\left|u_{S_k+R_k}(z) - u_{S_k}(z)\right| \leq 2M\,\|u_{R_k}\|_{L^\infty}^{1/2} + \|u_{R_k}\|_{L^\infty}.$$

Taking the supremum over z yields the claim. □

The next lemma supplies the soft input on the function $s \mapsto I_F(s)$ that will be used at the end of the variational argument. It is essentially a consequence of the bathtub principle and the basic regularity of u_F, already noted in Lemma 2.5.

Lemma 7.4. *Let $F \in \mathcal{F}^2(\mathbb{C})$. Then the function*

$$s \mapsto I_F(s)$$

is finite, nondecreasing, and continuous on $[0, \infty)$.

Proof. By Lemma 2.5, $u_F \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$, so $I_F(s)$ is finite for every $s \geq 0$ and bounded above by $\|F\|^2_{\mathcal{F}^2}$. Monotonicity is immediate from the definition of $I_F(s)$ as a supremum over $|\Omega| = s$: if $s_1 \leq s_2$, every admissible set for s_1 can be enlarged to a set of measure s_2 without decreasing the integral, since $u_F \geq 0$.

To prove continuity, write $u := u_F$ and let u^* denote the decreasing rearrangement of u on $[0, \infty)$. Since $u \in C_0(\mathbb{C}) \cap L^1(\mathbb{C})$, $u^* \in L^1([0, \infty))$ and is equimeasurable with u. The bathtub principle yields

$$I_F(s) = \int_0^s u^*(\tau)\, d\tau, \qquad s \geq 0.$$

The right-hand side is the integral of an $L^1_{\text{loc}}([0, \infty))$ function over $[0, s]$ and is therefore absolutely continuous, hence continuous, in s. □

The next result is the profile decomposition adapted to the present setting, in the spirit of P.-L. Lions' concentration-compactness method. Although the overall scheme is by now standard, we spell out the points used in the sequel; the version stated here is a Fock-space variant of profile decompositions used in time-frequency analysis, see e.g. [32, 31, 18, 39].

Proposition 7.1 (Profile decomposition in the Fock space)**.** *Let $\{F_k\}_{k \geq 1} \subset \mathcal{F}^2(\mathbb{C})$ be a bounded sequence. Then, after passing to a subsequence, there exist profiles*

$$G^{(j)} \in \mathcal{F}^2(\mathbb{C}), \qquad j \geq 1,$$

and sequences of points

$$a_k^{(j)} \in \mathbb{C}, \qquad j \geq 1,\ k \geq 1,$$

such that:

(i) for each $j \neq \ell$,

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad (k \to \infty);$$

(ii) for every $J \geq 1$, if

$$R_k^{(J)} := F_k - \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)},$$

then

$$T_{-a_k^{(j)}} R_k^{(J)} \to 0 \qquad \textit{locally uniformly on } \mathbb{C}$$

for every $1 \leq j \leq J$;

(iii) for every $J \geq 1$,

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1);$$

(iv)

$$\lim_{J\to\infty} \left(\limsup_{k\to\infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} \right) = 0.$$

Proof. We argue by the standard iterative extraction scheme, but spell out the points used in the rest of the section. Throughout, we say that a function in $\mathcal{F}^2(\mathbb{C})$ is *nontrivial* if it is not identically zero.

Extraction of the first profile. If

$$\limsup_{k\to\infty} \|u_{F_k}\|_{L^\infty(\mathbb{C})} = 0,$$

there is nothing to prove: take all profiles equal to zero, all centers arbitrary (say zero), and observe that (i)–(iii) hold trivially while (iv) holds by hypothesis. Otherwise, after passing to a subsequence, choose $a_k^{(1)} \in \mathbb{C}$ such that

$$u_{F_k}(a_k^{(1)}) \geq \tfrac{1}{2}\|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

By (2.8) and the density-translation identity (2.7), the sequence $T_{-a_k^{(1)}}F_k$ has the same Fock norm and the corresponding density translated; relabelling, we may assume $a_k^{(1)} = 0$. Since $\{F_k\}$ is bounded in $\mathcal{F}^2(\mathbb{C})$, the reproducing-kernel bound $|F_k(z)| \leq \|F_k\|_{\mathcal{F}^2}\, e^{\pi|z|^2/2}$ (Lemma 2.3(ii)) makes $\{F_k\}$ a normal family of entire functions on every compact subset of $\mathbb{C}$. Passing to a subsequence and applying Montel's theorem, $F_k \to G^{(1)}$ locally uniformly on $\mathbb{C}$, with $G^{(1)} \in \mathcal{F}^2(\mathbb{C})$ by Fatou's lemma. By the choice of $a_k^{(1)} = 0$,

$$u_{G^{(1)}}(0) = \lim_{k\to\infty} u_{F_k}(0) \geq \tfrac{1}{2} \limsup_{k\to\infty} \|u_{F_k}\|_{L^\infty(\mathbb{C})} > 0,$$

so $G^{(1)}$ is nontrivial. Set

$$R_k^{(1)} := F_k - G^{(1)}.$$

Iteration. Suppose profiles $G^{(1)}, \dots, G^{(J-1)}$ and centers $a_k^{(1)}, \dots, a_k^{(J-1)}$ have been extracted, with remainder $R_k^{(J-1)}$. If

$$\limsup_{k\to\infty} \|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})} = 0,$$

we stop, set $G^{(j)} = 0$ and $a_k^{(j)} = 0$ for all $j \geq J$, and properties (i)–(iv) are easily seen to hold. Otherwise choose $a_k^{(J)} \in \mathbb{C}$ so that

$$u_{R_k^{(J-1)}}(a_k^{(J)}) \geq \tfrac{1}{2}\|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})}.$$

Property (i) at this stage (proved below) forces $|a_k^{(J)} - a_k^{(m)}| \to \infty$ for every $m < J$. Granted this, the sequence $\{T_{-a_k^{(J)}} R_k^{(J-1)}\}$ is bounded in $\mathcal{F}^2$ (by unitarity of $T_{-a_k^{(J)}}$ together with (iii) at stage $J-1$), so after passing to a further subsequence and applying Montel's theorem, it

converges locally uniformly to some $G^{(J)} \in \mathcal{F}^2(\mathbb{C})$. By the choice of $a_k^{(J)}$,

$$u_{G^{(J)}}(0) \geq \tfrac{1}{2} \limsup_{k\to\infty} \|u_{R_k^{(J-1)}}\|_{L^\infty(\mathbb{C})} > 0,$$

so $G^{(J)}$ is nontrivial. Set

$$R_k^{(J)} := R_k^{(J-1)} - T_{a_k^{(J)}} G^{(J)}.$$

Iterating either terminates at some finite step (in which case (iv) is immediate) or produces an infinite sequence of nontrivial profiles. By a standard diagonal extraction we obtain a single subsequence (still denoted $\{F_k\}$) for which all the partial decompositions are valid.

Proof of (i). Fix $j < \ell$, and suppose for contradiction that $|a_k^{(\ell)} - a_k^{(j)}|$ does not tend to infinity. After passing to a subsequence, $|a_k^{(\ell)} - a_k^{(j)}| \leq C$ for some constant C. At the ℓ-th step, by construction

$$T_{-a_k^{(\ell)}} R_k^{(\ell-1)} \to G^{(\ell)} \qquad \text{locally uniformly on } \mathbb{C},$$

with $G^{(\ell)}$ nontrivial. On the other hand, property (ii) verified up to stage $\ell - 1$ (proved below) gives

$$T_{-a_k^{(j)}} R_k^{(\ell-1)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}.$$

The Weyl identity $T_{-a_k^{(\ell)}} = e^{i\theta_k} T_{-a_k^{(j)}} T_{a_k^{(j)} - a_k^{(\ell)}}$ (for an appropriate real phase θ_k coming from the Weyl co-cycle, of modulus one) shows that

$$T_{-a_k^{(\ell)}} R_k^{(\ell-1)}(z) = e^{i\theta_k} \big(T_{-a_k^{(j)}} R_k^{(\ell-1)}\big)\big(z - (a_k^{(\ell)} - a_k^{(j)})\big).$$

Since $|a_k^{(\ell)} - a_k^{(j)}| \leq C$, the points $z - (a_k^{(\ell)} - a_k^{(j)})$ remain in a fixed compact set as z varies over a compact set; so the local uniform convergence of $T_{-a_k^{(j)}} R_k^{(\ell-1)}$ to 0 forces $T_{-a_k^{(\ell)}} R_k^{(\ell-1)} \to 0$ locally uniformly as well. This contradicts the nontriviality of $G^{(\ell)}$. Hence

$$|a_k^{(j)} - a_k^{(\ell)}| \to \infty \qquad \text{whenever } j \neq \ell.$$

Proof of (ii). Fix $J \geq 1$ and $1 \leq j \leq J$. By construction, at the j-th extraction step,

$$T_{-a_k^{(j)}} R_k^{(j-1)} \to G^{(j)} \qquad \text{locally uniformly on } \mathbb{C}.$$

Subtracting the j-th extracted profile,

$$T_{-a_k^{(j)}} R_k^{(j)} = T_{-a_k^{(j)}} R_k^{(j-1)} - G^{(j)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}.$$

For $J > j$, since

$$R_k^{(J)} = R_k^{(j)} - \sum_{\ell=j+1}^{J} T_{a_k^{(\ell)}} G^{(\ell)},$$

the linearity of the Weyl operators yields

$$T_{-a_k^{(j)}} R_k^{(J)} = T_{-a_k^{(j)}} R_k^{(j)} - \sum_{\ell=j+1}^{J} e^{i\theta_{k,\ell}}\, T_{a_k^{(\ell)} - a_k^{(j)}} G^{(\ell)}$$

for some real phases $\theta_{k,\ell}$ coming from the Weyl co-cycle. The first term tends to 0 locally uniformly by the previous paragraph. For each fixed $\ell > j$, part (i) gives $|a_k^{(\ell)} - a_k^{(j)}| \to \infty$, and

the identity $u_{T_bG}(z) = u_G(z-b)$ (i.e. (2.7)) implies that $T_{a_k^{(\ell)}-a_k^{(j)}}G^{(\ell)} \to 0$ locally uniformly in modulus (its density is $u_{G^{(\ell)}}(z-(a_k^{(\ell)}-a_k^{(j)})) \to 0$ for z in a compact set, since $u_{G^{(\ell)}} \in C_0(\mathbb{C})$). Thus

$$T_{-a_k^{(j)}}R_k^{(J)} \to 0 \qquad \text{locally uniformly on } \mathbb{C},$$

which is exactly (ii).

Proof of (iii). For fixed $J \geq 1$,

$$F_k = \sum_{j=1}^{J} T_{a_k^{(j)}}G^{(j)} + R_k^{(J)}.$$

Expanding the squared $\mathcal{F}^2$-norm and using Lemma 7.1 together with the pairwise divergence of the translation parameters from (i),

$$\Big\| \sum_{j=1}^{J} T_{a_k^{(j)}}G^{(j)} \Big\|_{\mathcal{F}^2}^2 = \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 + o_k(1).$$

Moreover, since

$$T_{-a_k^{(m)}}R_k^{(J)} \to 0 \qquad \text{locally uniformly on } \mathbb{C}$$

for every $1 \leq m \leq J$ by (ii), and the sequence $\{T_{-a_k^{(m)}}R_k^{(J)}\}_k$ is bounded in $\mathcal{F}^2$, the same weak-convergence argument as in Lemma 7.1 yields

$$\langle T_{a_k^{(m)}}G^{(m)}, R_k^{(J)}\rangle_{\mathcal{F}^2} = \langle G^{(m)}, T_{-a_k^{(m)}}R_k^{(J)}\rangle_{\mathcal{F}^2} \to 0,$$

for every fixed m. Expanding $\|F_k\|_{\mathcal{F}^2}^2$ and collecting terms,

$$\|F_k\|_{\mathcal{F}^2}^2 = \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 + \|R_k^{(J)}\|_{\mathcal{F}^2}^2 + o_k(1).$$

In particular, since $\|F_k\|_{\mathcal{F}^2}$ is bounded and the terms on the right are nonnegative, summing the profile-norm contributions over j yields the energy bound

$$\sum_{j\geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 \leq \limsup_{k\to\infty} \|F_k\|_{\mathcal{F}^2}^2. \tag{7.1}$$

This is the key compactness ingredient that controls the iteration and ensures (iv).

Proof of (iv). Set

$$\eta_J := \limsup_{k\to\infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})},$$

which is monotone nonincreasing in J by construction (since $R_k^{(J+1)}$ is obtained from $R_k^{(J)}$ by extracting one further nontrivial profile). Let $\eta_\infty := \lim_{J\to\infty} \eta_J$; we must show $\eta_\infty = 0$. Suppose, for contradiction, that $\eta_\infty \geq \delta > 0$.

By the iteration, at each stage J such that $\eta_J \geq \delta$, the extracted profile $G^{(J+1)}$ satisfies $u_{G^{(J+1)}}(0) \geq \delta/2$ by construction. By Lemma 2.1(ii),

$$\|G^{(J+1)}\|_{\mathcal{F}^2}^2 \geq u_{G^{(J+1)}}(0) \geq \delta/2.$$

If the iteration produced infinitely many such stages (which is the case under the assumption $\eta_\infty \geq \delta$), then

$$\sum_{j\geq 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = \infty,$$

contradicting (7.1). Hence $\eta_\infty = 0$, proving (iv). □

We now turn to the global problem. The proof of existence consists of combining the profile decomposition with the elementary continuity properties of $I_F(s)$, and using the homogeneity $I_{cF}(s) = |c|^2 I_F(s)$ recorded in Lemma 2.5(ii) to redistribute mass to a single nontrivial profile.

Theorem 6. *For every $s > 0$, there exists $F_* \in \mathcal{F}^2(\mathbb{C})$ with $\|F_*\|_{\mathcal{F}^2} = 1$ such that*

$$I_{F_*}(s) = M(s).$$

Equivalently, there exists a measurable set $\Omega_* \subset \mathbb{C}$ with $|\Omega_*| = s$ such that

$$\lambda_{1,\varphi}(\Omega_*) = M(s).$$

Proof. Let $\{F_k\} \subset \mathcal{F}^2(\mathbb{C})$ be a maximizing sequence:

$$\|F_k\|_{\mathcal{F}^2} = 1, \qquad I_{F_k}(s) \to M(s).$$

By replacing F_k with a Weyl translate if necessary, and using the translation invariance (2.8) together with (2.7), we may suppose that

$$u_{F_k}(0) = \|u_{F_k}\|_{L^\infty(\mathbb{C})}.$$

Note that $M(s) > 0$: indeed, taking $F \equiv 1$, one has $F \in \mathcal{F}^2(\mathbb{C})$ with $\|F\|_{\mathcal{F}^2}^2 = 1$ and density $u_F(z) = e^{-\pi|z|^2} > 0$ everywhere, so for any $s > 0$, $I_F(s) > 0$. Consequently, $\|u_{F_k}\|_{L^\infty(\mathbb{C})}$ is bounded below by a positive constant uniformly in k; otherwise, for any $\eta > 0$ we would have $\|u_{F_k}\|_{L^\infty} < \eta$ eventually, hence $I_{F_k}(s) \leq \eta s$, contradicting $I_{F_k}(s) \to M(s) > 0$.

Applying Proposition 7.1, after passing to a subsequence we may write

$$F_k = \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)} + R_k^{(J)}$$

for every fixed $J \geq 1$, with the conclusions of that proposition.

Step 1: an upper bound for the limit. Fix $J \geq 1$ and let $\Omega_k^{(J)}$ be a measurable set of measure s such that

$$I_{F_k}(s) = \int_{\Omega_k^{(J)}} u_{F_k}\, dA;$$

by Lemma 2.5(i) we may take $\Omega_k^{(J)}$ to be a superlevel set of u_{F_k}. Let $\varepsilon > 0$. Since $u_{G^{(j)}} \in L^1(\mathbb{C})$ for every j by Lemma 2.5, choose $R > 0$ so large that

$$\int_{\mathbb{C}\setminus D_R} u_{G^{(j)}}\, dA < \varepsilon \qquad \text{for all } 1 \leq j \leq J,$$

and large enough that the conclusion of Lemma 7.2(ii) holds with this choice of R. Since the translation parameters are pairwise divergent by Proposition 7.1(i), for k large enough the disks

$$D_R(a_k^{(1)}), \ldots, D_R(a_k^{(J)})$$

are pairwise disjoint. Write

$$B_{k,j} := D_R(a_k^{(j)}), \qquad E_{k,j} := \Omega_k^{(J)} \cap B_{k,j}, \qquad E_{k,0} := \Omega_k^{(J)} \setminus \bigcup_{j=1}^{J} B_{k,j}.$$

Set

$$s_{k,j} := |E_{k,j}|, \qquad 0 \le j \le J.$$

Then, since the $\{B_{k,j}\}_{j=1}^{J}$ are disjoint and $\Omega_k^{(J)}$ has measure s,

$$\sum_{j=0}^{J} s_{k,j} = s.$$

Set

$$S_k^{(J)} := \sum_{j=1}^{J} T_{a_k^{(j)}} G^{(j)}.$$

By Proposition 7.1(iii), the sequence $\{S_k^{(J)}\}_k$ is bounded in $\mathcal{F}^2(\mathbb{C})$, with bound $\le 1 + o_k(1)$. By Proposition 7.1(iv), if J is taken sufficiently large then

$$\limsup_{k\to\infty} \|u_{R_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon^2.$$

Lemma 7.3 therefore gives

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} \to 0 \qquad (k \to \infty),$$

and in particular, after taking k large enough,

$$\|u_{F_k} - u_{S_k^{(J)}}\|_{L^\infty(\mathbb{C})} < \varepsilon. \tag{7.2}$$

Now

$$E_{k,0} = \Omega_k^{(J)} \setminus \bigcup_{j=1}^{J} B_{k,j} \subset \mathbb{C} \setminus \bigcup_{j=1}^{J} D_R(a_k^{(j)}).$$

Lemma 7.2(i) gives, for each fixed $j \in \{1, \dots, J\}$,

$$\sup_{z \in B_{k,j}} \left| u_{S_k^{(J)}}(z) - u_{T_{a_k^{(j)}} G^{(j)}}(z) \right| \to 0 \qquad (k \to \infty),$$

while Lemma 7.2(ii) yields, for our choice of R and for k sufficiently large,

$$\sup_{z \in E_{k,0}} u_{S_k^{(J)}}(z) \le \varepsilon.$$

Combining these bounds with (7.2), we conclude that for all sufficiently large k,

(a) on each $B_{k,j}$,

$$u_{F_k}(z) \le u_{T_{a_k^{(j)}} G^{(j)}}(z) + 2\varepsilon;$$

(b) on $E_{k,0}$,

$$u_{F_k}(z) \le 2\varepsilon.$$

Therefore, decomposing the integral defining $I_{F_k}(s)$ according to the partition $\Omega_k^{(J)} = E_{k,0} \sqcup \bigsqcup_{j=1}^J E_{k,j}$,

$$\begin{aligned} I_{F_k}(s) &= \int_{\Omega_k^{(J)}} u_{F_k}\, dA \\ &\le \sum_{j=1}^J \int_{E_{k,j}} u_{T_{a_k^{(j)}} G^{(j)}}\, dA + 2\varepsilon \sum_{j=0}^J |E_{k,j}| \\ &= \sum_{j=1}^J \int_{E_{k,j} - a_k^{(j)}} u_{G^{(j)}}\, dA + 2\varepsilon s \\ &\le \sum_{j=1}^J I_{G^{(j)}}(s_{k,j}) + 2\varepsilon s, \end{aligned}$$

where in the next-to-last step we used the change of variables $z \mapsto z - a_k^{(j)}$ together with (2.7), and in the last step the definition of $I_{G^{(j)}}(s_{k,j})$ as the supremum of $\int_E u_{G^{(j)}}\, dA$ over sets of measure $s_{k,j} = |E_{k,j}| = |E_{k,j} - a_k^{(j)}|$.

Passing to a further subsequence if necessary, we may suppose

$$s_{k,j} \to \sigma_j \in [0, s] \qquad (k \to \infty)$$

for each $1 \le j \le J$. By the continuity of $I_{G^{(j)}}$ in the measure parameter, given by Lemma 7.4,

$$M(s) = \lim_{k\to\infty} I_{F_k}(s) \le \sum_{j=1}^J I_{G^{(j)}}(\sigma_j) + 2\varepsilon s.$$

Since $\varepsilon > 0$ was arbitrary,

$$M(s) \le \sum_{j=1}^J I_{G^{(j)}}(\sigma_j). \tag{7.3}$$

Step 2: redistribution of mass and saturation of the bound. We now use Lemma 2.5(ii). For each j with $G^{(j)} \neq 0$, set

$$\widehat{G}^{(j)} := \frac{G^{(j)}}{\|G^{(j)}\|_{\mathcal{F}^2}}, \qquad \|\widehat{G}^{(j)}\|_{\mathcal{F}^2} = 1,$$

and use the homogeneity $I_{cF}(\sigma) = |c|^2 I_F(\sigma)$ (with $c = \|G^{(j)}\|_{\mathcal{F}^2}$) to obtain

$$I_{G^{(j)}}(\sigma_j) = \|G^{(j)}\|_{\mathcal{F}^2}^2\, I_{\widehat{G}^{(j)}}(\sigma_j).$$

Since $\widehat{G}^{(j)}$ has unit $\mathcal{F}^2$-norm, by definition of M and the monotonicity of $I_{\widehat{G}^{(j)}}$ given by Lemma 7.4 (using $\sigma_j \le s$),

$$I_{\widehat{G}^{(j)}}(\sigma_j) \le I_{\widehat{G}^{(j)}}(s) \le M(s).$$

Hence

$$I_{G^{(j)}}(\sigma_j) \le \|G^{(j)}\|_{\mathcal{F}^2}^2\, M(s).$$

For j with $G^{(j)} = 0$ the inequality is trivial. Summing over $1 \le j \le J$ and using Proposition 7.1(iii) together with the normalization $\|F_k\|_{\mathcal{F}^2}^2 = 1$,

$$\sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j) \le M(s) \sum_{j=1}^{J} \|G^{(j)}\|_{\mathcal{F}^2}^2 \le M(s).$$

Combined with (7.3), this yields

$$M(s) \le \sum_{j=1}^{J} I_{G^{(j)}}(\sigma_j) \le M(s)$$

for every J. Therefore equality holds throughout for every J, which forces

$$\sum_{j\ge 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = 1$$

(so the residual $R_k^{(J)}$ carries no $\mathcal{F}^2$-mass in the limit), and moreover, for every j with $G^{(j)} \neq 0$,

$$I_{\widehat{G}^{(j)}}(\sigma_j) = M(s).$$

By monotonicity of $I_{\widehat{G}^{(j)}}$, $I_{\widehat{G}^{(j)}}(s) \ge I_{\widehat{G}^{(j)}}(\sigma_j) = M(s)$; since $M(s)$ is the supremum over unit-norm functions, we conclude

$$I_{\widehat{G}^{(j)}}(s) = M(s).$$

Thus every nonzero normalized profile is a global maximizer.

Step 3: producing a maximizing set. Choose any j_0 with $G^{(j_0)} \neq 0$ (such j_0 exists since $\sum_{j\ge 1} \|G^{(j)}\|_{\mathcal{F}^2}^2 = 1$) and set $F_* := \widehat{G}^{(j_0)}$, so that

$$\|F_*\|_{\mathcal{F}^2} = 1, \qquad I_{F_*}(s) = M(s).$$

By Lemma 2.5(i), there exists a superlevel set Ω_* of u_{F_*} with $|\Omega_*| = s$ and

$$\int_{\Omega_*} u_{F_*}\, dA = I_{F_*}(s) = M(s).$$

Then

$$\lambda_{1,\varphi}(\Omega_*) \ge \int_{\Omega_*} u_{F_*}\, dA = M(s),$$

while by definition of $M(s)$ we have $\lambda_{1,\varphi}(\Omega_*) \le M(s)$. Hence

$$\lambda_{1,\varphi}(\Omega_*) = M(s),$$

and Ω_* is a global maximizer of measure s. □

We may now identify the global maximizers, by combining the existence theorem above with the local-rigidity theorem of the previous section.

Theorem 7. *For every $s > 0$, every global maximizer for*

$$\sup_{|\Omega|=s} \lambda_{1,\varphi}(\Omega)$$

is, up to translation, a disk.

Proof. Let Ω_* be a global maximizer of measure s (which exists by Theorem 6). Then Ω_* is, in particular, a local maximizer of $\lambda_{1,\varphi}$ under the same measure constraint: every sufficiently small smooth deformation of Ω_* preserving the measure $|\Omega| = s$ yields a competitor whose value of $\lambda_{1,\varphi}$ is at most $M(s) = \lambda_{1,\varphi}(\Omega_*)$. The local-rigidity theorem proved in the previous section therefore applies: every such local maximizer is, up to translation, a disk. Since the problem is invariant under Weyl translations, this characterization is sharp. □

As an immediate consequence, we recover the classical Nicola–Tilli theorem [33, 34].

Corollary 7.1. *Among all measurable sets of prescribed measure, the disk maximizes Gaussian time-frequency concentration:*

$$\lambda_{1,\varphi}(\Omega) \leq \lambda_{1,\varphi}(D)$$

whenever $|\Omega| = |D|$ *and* D *is a disk.*

Proof. Existence of a global maximizer is given by Theorem 6, and Theorem 7 shows that every such maximizer is a disk, up to translation. Since translation by an element of $\mathbb{C}$ preserves both Lebesgue measure and the value of $\lambda_{1,\varphi}$, the claim follows. □

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